I wish to begin by acknowledging the Wurundjeri people of the Kulin nations, the Traditional Owners of the land on which we are gathered today. We pay our respects to the local people for allowing us to have our gathering on their land and to their Elders: past, present and future

To

MAF306
International Finance and Investment

* The Bachelor of Commerce at Deakin University is internationally [EPAS accredited]

AACSB ACCREDITED EQUIS ACCREDITED DEAKIN BUSINESS SCHOOL

DEAKIN BUSINESS SCHOOL

Academic staff

Dr Sohel Azad (unit chair, lecturer and tutor)

- Phone: W: 9244 6873; Office: LB4.108
- Email: s.azad@deakin.edu.au

Queries and consultation

- ## Academic queries
  - Academic queries should, in the first instance, be posted on Cloud Deakin. Teaching staff will use their best endeavours to respond within two business days of the query being received.
  - **If you don't receive a reply, please email me; after all, I am a human being!!**
- ## Consultation Times
  - Regular consultation: 10am-11am (Wednesday; LB4.108/Zoom)
  - Other time by appointment (can be run through zoom on request)

Seminar details

Day	Time	Location
Wednesday	12pm	MA2.103
Wednesday	1pm	LC6.108
Thursday	2pm	HE3.001

Online seminar: I will run an **online seminar** on **Wednesday 5pm through zoom (https://deakin.zoom.us/j/82003695285?pwd=ZmpTYk9FK1lqRUlQNHhlMUJudEZPdz09)**. While this seminar is mainly for online students, on campus students are welcome to join but we have to see how it goes with the attendance in the first few weeks.

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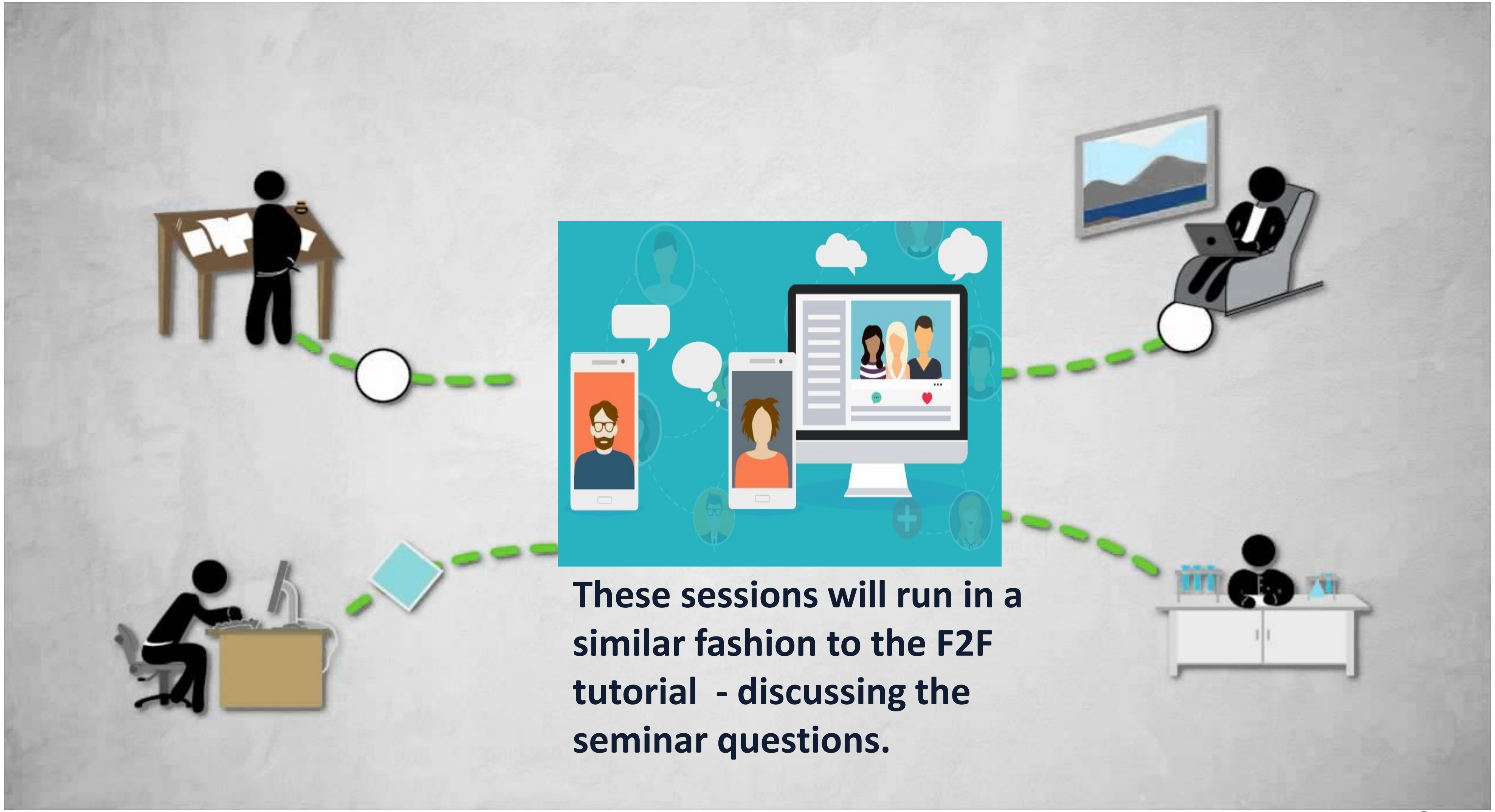
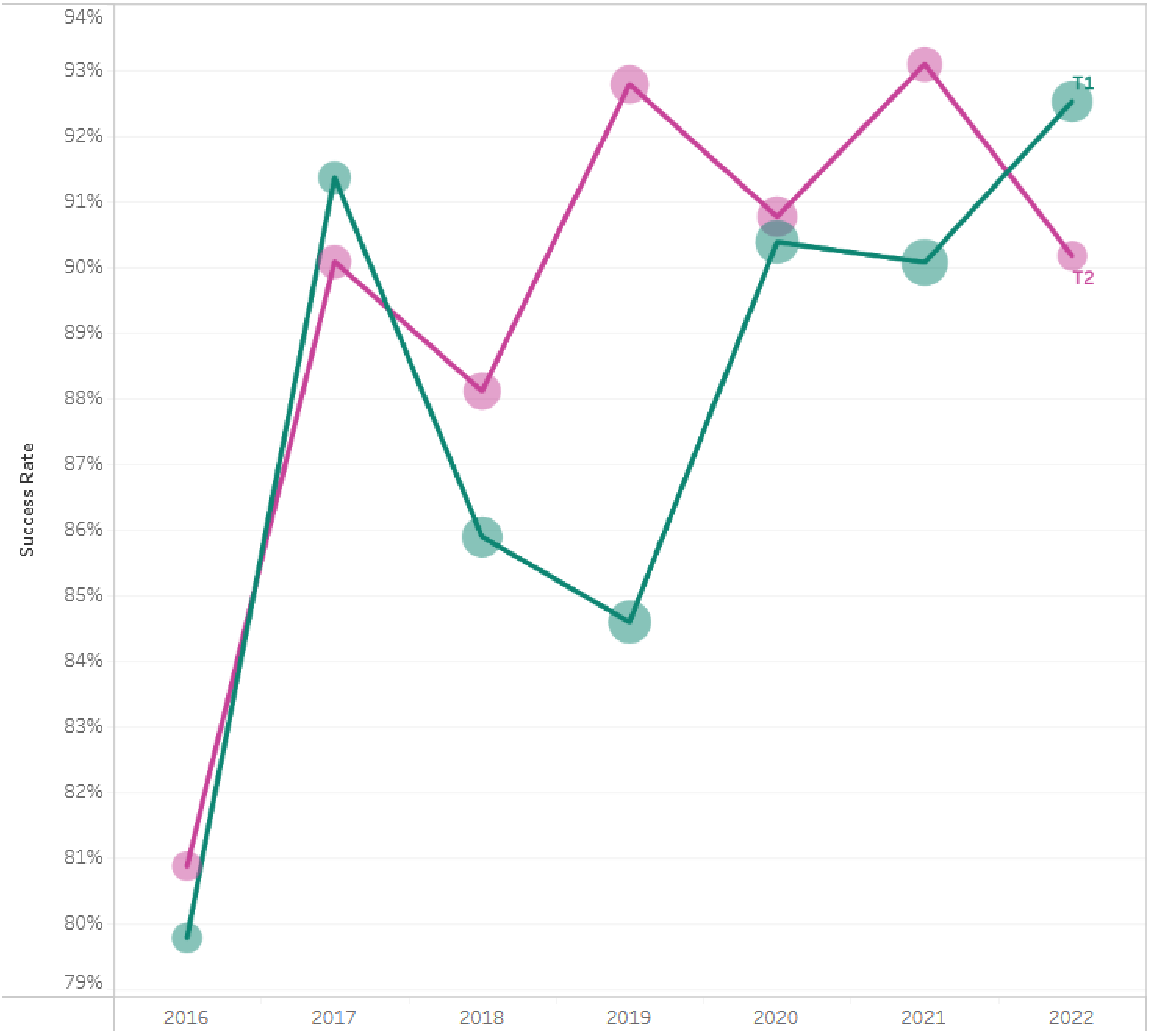
Online Collaborate Sessions



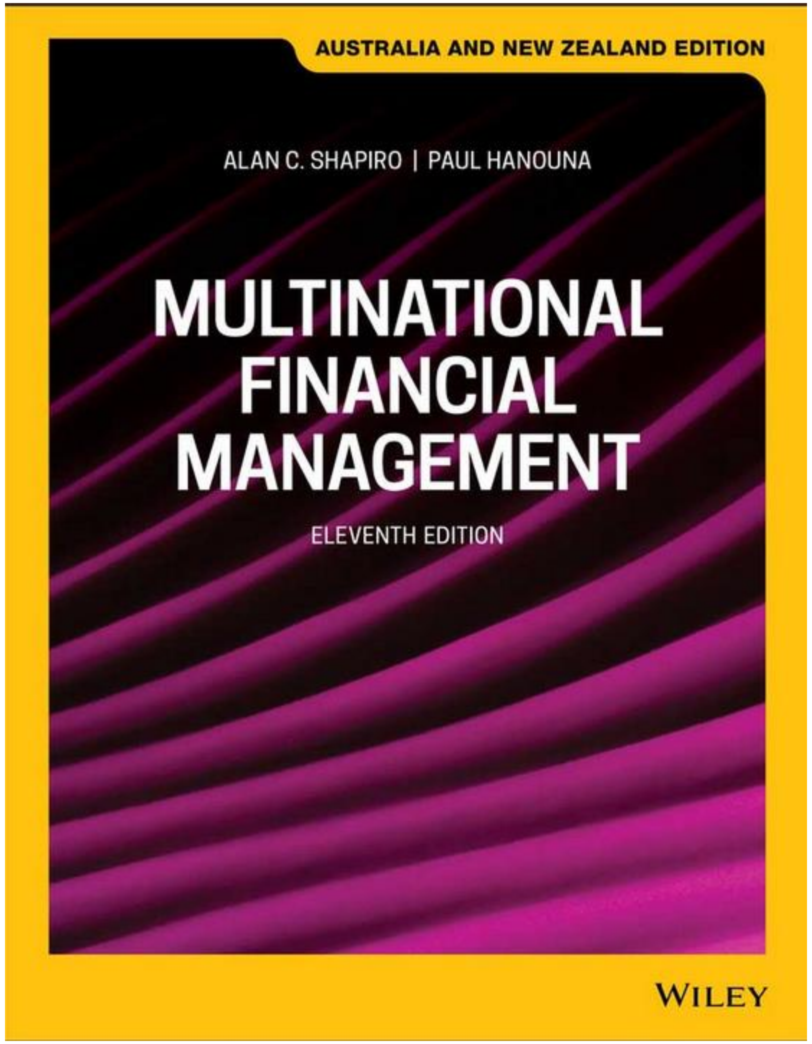
These sessions will run in a similar fashion to the F2F tutorial - discussing the seminar questions.

Unit's success rate over the years

Unit's success rate over the years

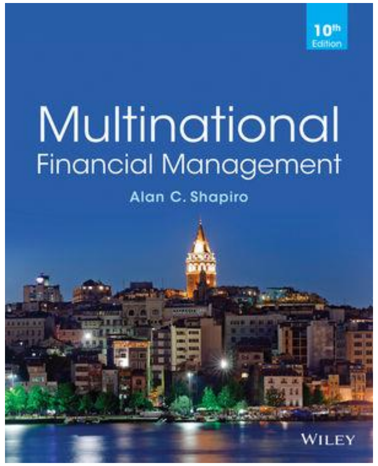


The prescribed text for MAF 306 :
Shapiro and Hanouna/Multinational Financial Management 11^{th} edition (Also called 13^{th} ANZ edition)



Link:
https://www.wileydirect.com.au/buy/multinational-financial-management-11th-australia-and-new-zealand-edition/

*No major difference of contents between 10^{th} and 11^{th} Edition except page numbers.



Assessment

Assessment name	Weight	Due date	Brief description	Linked Deakin Graduate Learning Outcomes
Assignment	**40%** **Part-A:** **30%** for Business Report (max 3000 words) **Part-B:** **10%** for individual reflective statement (max 500 words).	**Group assignment - one copy per group to be submitted into CloudDeakin site by AEST 8pm, 12 May 2024** via Cloud Deakin. **No hardcopy is required.** **You must form a group on Cloud Deakin by 24 March 2024. Members from different background are encouraged**	The assignment will consist of a business report to be written by a group of 3 students that analyses and compares contemporary International finance and investment issue(s). The report should be between 2,500 – 3,000 words (with a maximum of 1,000 words per student). Each student needs to specify which section(s) s/he has contributed individually and/or jointly to the assignment as the teamwork component will be assessed. Individual reflective statement must not exceed 500 words	ULO 1, 2, 3, 4 GLO 01 GLO 04 GLO 05 GLO 07
Multiple Choice Test(s)	**10%**	**MCQ** Opens 10:00 am on Thursday **May 16, 2024** and closes 8 pm Friday, **May 17, 2024**. Examines topics Week 1-10	**There will be an online multiple Choice tests with 20-40 questions with approximately 2-4 questions per topic. Test duration: 1 hour**	ULO 1, 2 GLO 01 GLO 04
Examination	**50%**	**Exam period** Exact date, time and location for the examination available from Student Connect.	**Two-hour online** **All Topics will be examined** **Any topics not included will be duly informed**	ULO 1, 2, 3 GLO 1, 4, 5

Graduate Learning Outcomes (GLO) for MAF306

GLO	Detail	Comment
1	**Discipline-specific knowledge and capabilities:**	Appropriate to the level of study related to a discipline or profession
4	**Critical thinking:**	Evaluating information using critical and analytical thinking and judgment
5	**Problem solving:**	Creating solutions to authentic (real world and ill-defined) problems
7	**Teamwork:**	Working and learning with others from different disciplines and backgrounds

Unit Learning Outcomes

ULO	Detail	GLO
ULO1	Apply a critical understanding of theoretical and practical knowledge of the structure, functions and operations of International Monetary Systems and globalized financial markets to a variety of cases/contexts (Topics 1-5), with some overall in other topics	GLO1: Discipline knowledge and capabilities GLO4: Critical thinking
ULO2	Critically analyse and evaluate exchange rate determination and risk exposure management of business financing and investment activity. Consider these impacts on multi-national companies and countries, in order to review and implement appropriate risk management strategies. (Topics 6-10), with some overall in other topics	GLO1: Discipline knowledge and capabilities GLO4: Critical thinking
ULO3	Demonstrate individual accountability in collaboratively planning and implementing a research-based team-project	GLO1: Discipline knowledge and capabilities GLO7: Teamwork

Weekly Planner_new

Week	Topic	Special learning activities	Assessment due date
1	Introduction to International Finance-Multinational Corporation and Financial Management	Introductory Tutorial	
2	International financial markets: Euro currency markets	Attempt tutorial questions Topic 1 prior to the tutorial	
3	Foreign investments: FDI and contemporary issues	Attempt tutorial questions Topic 2 prior to the tutorial	**Assignment group formation deadline: 24th March 2024**
4	Managing country risk and debt crises	Attempt tutorial questions Topic 3 prior to the tutorial	
5^	Exchange rate systems and determination	Attempt tutorial questions Topic 4 prior to the tutorial	
6	Foreign exchange markets and trading	Attempt tutorial questions Topic 5 prior to the tutorial	
7*	Foreign exchange derivatives	Attempt tutorial questions Topic 6 prior to the tutorial	
8	Foreign exchange risk management	Attempt tutorial questions Topic 7 prior to the tutorial	
9	MNC's capital budgeting MNC's capital structure and Cost of Capital	Attempt tutorial questions Topic 8 prior to the tutorial	**Written report and individual reflection report (both in MS word format) Submission deadline: Sunday May 12, 2024 by 8pm AEST.**
10	Parity conditions in international finance	Attempt tutorial questions Topic 9 and 10 prior to the tutorial	**MC Test: Opens 10:00 am on Thursday May 16, 2024 and closes 8 pm Friday, May 17, 2024. Examines topics Week 1-10**
11	**Revision and final exam information**		**End of unit assessment during the T1 examination period**

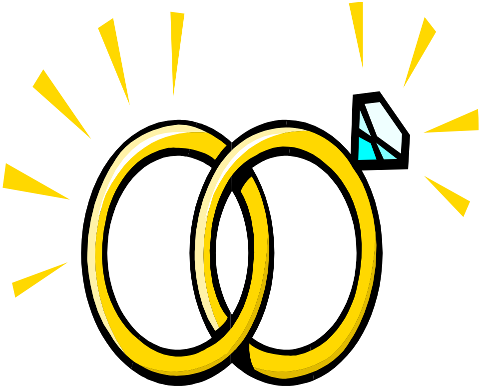
MAF306 Student Expectations

- The Policy on Student Expectations for this unit clearly states that each student:
  - Attends at least 80% of the lecture and/or Listens to the echo recordings for review
  - Prepares for and participates in class, seminar and Cloud Deakin discussions
  - Visits CloudDeakin at least once in two days

- Experience consistently shows that the probability of failure increases when:

  - You do not come to lecture/not listen to echo-lectures
  - You do not prepare for assigned tutorial/discussion questions
  - You do not use CloudDeakin regularly
  - You do not have a study plan
  - You do not attempt to overcome your weaknesses
  - You leave assignments, tutorial work and examination preparation to the last minute.

MAF306 Student Expectations



YOU NEED TO STAY IN CONTACT
and STAY ENGAGED with the unit!!!!!!!

- CloudDeakin statistics show a strong correlation between those that have done extremely well on the tests and those that were well engaged with the unit.

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IF YOU WANT TO KNOW SOMETHING, NO IS NEVER AN ANSWER, SO PLEASE

ASK (AGGRESSIVELY SEEK KNOWLEDGE)

TOPIC 1

Introduction to International Finance and Financial System

Learning Objectives

- Understand what is special about international finance
- International finance and domestic finance
- Internationalization vs globalization
- Broad areas of international finance
- Understand the nature and benefits of globalization
- Outline the series of events that shaped the economic, financial and institutional environment following World War II
- Compare the costs and benefits of internationalisation
- International monetary systems – an overview

- Reading:
  - Shapiro Ch 1 and 3
  - Seminar materials on CloudDeakin site

Introduction

- International finance deals with
  - International financial economics, i.e. causes and effects of financial flows among nations
  - International financial management, i.e. how individual economic units deal with the complex financial environment of international business

International Finance

Includes interactions between:

- International financial markets.
- International banks and banking system.
- International monetary systems.
- Foreign exchange markets.
- International corporate finance and financial flows.
- International foreign investment policies.
- International portfolio diversification and management.

International finance vs domestic finance

- Corporate governance
  - Regulations and institutions are uniquely different for MNCs/MNEs
- Foreign exchange risk
  - MNCs/MNEs face exchange rate risk, currency crises, import/export or foreign competition
- Integration of global markets
  - International joint ventures, market integration
- Political risks
  - MNCs/MNEs face expropriation risk by local government
- Market imperfections
  - Legal restrictions
  - Excessive transaction and transportation costs
  - Information asymmetry
  - Discriminatory taxation
- Expanded opportunity set
  - Expanded Production-Possibility Frontier (PPF) for MNCs/MNEs, i.e., more of everything!

Internationalization vs globalization

- Internationalization
  - 'the process of expanding banking or business activity abroad and replacing domestic banking or business content by international content'
  - markets that transcend national boundaries, typically dealing in cross currencies

- Globalization
  - 'process in which banking and provision of goods and services become worldwide in terms of geographical coverage and universal in terms of their provision'
  - presumes the harmonization of rules and removal of barriers so that all institutions can compete globally.

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International Finance

Three broad areas of study:

- International macroeconomics
- International financial markets
- International financial management

International Macroeconomics

- International monetary system
- Balance of payment
- Parity relationships
- Theories of exchange rate determination
- Open economy macroeconomic policy
  - exchange controls and other restrictions on capital flows
  - export financing programs
- International investment
  - direct investment flows
  - portfolio investment flows
  - development of financing institutions
- Issues and problems
  - international monetary reform
  - promoting financial integration
  - exchange rate instability

International Financial Markets

- International banking
- Eurocurrency market
- Foreign exchange market
  - organization and characteristics of spot, forward, futures, and option markets
  - central bank intervention and market expectations
- International securities markets
  - Foreign bond market
  - Eurobond market
  - Internationalization of stock markets
- Financial swaps
  - currency swaps
  - interest rate swaps
  - commodity swaps

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International Financial Management

- Exchange rate risk management
  - types of forex exposures
  - measuring and managing exposures
  - accounting and tax considerations
- Political risk management
  - types of political risks
  - political risk insurance
  - strategies and techniques for political risk management
- Managing direct foreign investment
  - capital budgeting
  - cost of capital
  - financial structure of affiliates
- Managing portfolio investment
  - portfolio theory and role of diversification
  - analysis of portfolio strategies and techniques

Early Forces Changing Global Markets

- Emergence of the Eurodollar market in 1957 in London
- Formation of the European Economic Community (EEC) in 1957
- Trend towards establishing multinational companies (MNCs) from the mid-1950s
- Rapid industrialisation of the Asian-Pacific Rim Economies
  - Rapid economic progress in Japan since the 1960s
  - The Asian ‘tigers’ in the 1980s - Taiwan, Malaysia, Singapore, Indonesia, Thailand
- Collapse of the Bretton Woods System in 1973 – fixed to floating fx system
- Oil price shocks of 1974 and 1979 (prices increased 400%) with subsequent high inflation and large BOP (Balance of Payment) deficits for most nations
- World debt crisis in 1982 – Latin America – lost decade – high foreign debt

Forces Changing Global Markets 1980s-1990s

- Massive deregulation of financial systems - 1980s.
- Collapse of communism – 1990
- Privatisations of state-owned industries – U.K
- Trade liberalisation
- Economic integration
- Revolution in information technology – 24/7 online.
- Wave of mergers, acquisitions and leveraged buyouts.
- Emergence of free market policies in Third World nations.
- Significant number of nations submitting to the rigors and standards of the global marketplace.
- The rise of China as a global competitor-
    - the most dramatic change in the international economy over the last decade.
    - #1 destination for foreign direct investment (FDI: the acquisition abroad of companies, property, or physical assets).

The Rise of the Multinational Corporation

An MNC is a company with production and distribution facilities in more than one country. Its parent company located in the home country with at least five or six foreign subsidiaries, typically with a high degree of strategic interaction among the units.

- Most begin as *purely domestic businesses*
- They become MNCs in *phases* over time.
  - Begin with *simple, low-risk, low-return strategies.*
  - Progress to *sophisticated, high-risk, high-return strategies.*

Exporting (Early Phase) → Foreign sales office, service facilities, distribution systems (Intermediate Phase) → Overseas Production (Advanced Phase)

The Rise of the Multinational Corporation

MNC is engaged in producing and selling goods or services in more than one country. Its existence is based on the international mobility of certain factors of production:

- Capital raised in London on the Eurodollar market may be used by a **Swiss-based** pharmaceutical firm to finance the **acquisition of German equipment** by a **subsidiary in Brazil**.
- Barbie doll is made in 10 countries!!:
  - Designed in California.
  - Parts and clothing from Japan, China, Hong Kong, Malaysia, Indonesia, Korea and Taiwan – and assembled in Mexico – and sold in 144 countries.

Tell/share anything interesting in relation to this

The Rise of the Multinational Corporation

Doctrine of <u>comparative advantage</u>. This states that each nation should specialize in the production and export of those goods that it can produce with the highest relative efficiency and import those goods that other nations can produce relatively more efficiently.

- Goods and services can move internationally but factors of production – capital, labour and land are relatively immobile – is irrelevant to MNC – factors of production (except land) tend to move more rapidly in search of higher returns.

The Rise of the Multinational Corporation

- Reasons to Go Global:
  - Raw Material Seekers
  - Market Seekers
  - Cost Minimiser Seekers
  - Knowledge Seekers

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Why go Global?

RAW MATERIAL SEEKERS

- Exploit markets in other countries
- Early-day- British – East India Company
- Modern-day counterparts

  BHP Billiton

  Exxon-Mobil

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MARKET SEEKERS

- Produce and sell in foreign markets
- Heavy foreign direct investors
- Representative firms:
  - IBM
  - Nestle
  - MacDonald's
  - Coca-Cola

COST MINIMIZERS

- Seek lower-cost production abroad
  - Outsourcing or owned foreign affiliates
- Motive: to remain cost competitive
- Representative firms:
  - Texas Instruments
  - Motorola

KNOWLEDGE SEEKERS

- Gain access to technology or managerial expertise.
  - For example, German, Dutch, and Japanese firms have purchased U.S.-located electronics firms for their technology.
- Beecham, an English firm, deliberately set out to learn from its US operations how to be more competitive, first in the area of consumer products and later in pharmaceuticals. This helped the company to successfully compete with American and other firms in its European markets.
  - Part of GlaxoSmithKline now

Growth & Development of Global Markets

- Supply side factors
    - Financial activity *leading* real-sector transactions
    - Deregulation of financial markets and services – 1980's
        - USA, UK, Australia, Japan, Europe
        - Floating of currencies
        - Removal of FX and capital controls (Plaza Accord)
        - Reciprocal foreign bank entry
    - International financial institutions
        - Goldman Sachs, Morgan Stanley, JP Morgan, Deutsche Bank, HSBC, Barclay's

Growth & Development of Global Markets

- Supply side factors (cont'd)
  - Financial instruments innovations
    - Euro-loans, euro-notes, index futures, global bonds, securitisation, Islamic financial instruments
  - Modern communications technology
    - reduced cost of international transactions
    - electronic Letters of Credit, online transactions, etc
  - Linkages among major financial markets
    - FX, stock markets and capital markets

Growth & Development of Global Markets

- Demand side factors
  - Growth in international trade, leading to financial activity *following* real-sector transactions
  - Emergence of distinct saver and borrower nations leading to the need to transfer funds from surplus BOP countries to deficit BOP countries (Germany, China, India, Oil-rich Arab countries, Japan → US, Canada, Aust, UK)
    - www.usdebtclock.org
  - Increased cost of financial activity due to domestic regulation
  - Tax considerations – minimal or no tax in some countries and offshore banking centres
    - 'Tax-heavens' in the Caribbean's

Growth & Development of Global Markets

- Demand side factors (cont'd)
  - International portfolio diversification
    - countries
    - currencies
    - take advantage of technology
    - minimise asset price volatility and reduce risk
  - Cost efficient transfer of funds from surplus units (savers) to deficit units (borrowers) wherever domiciled
  - Continued government budget deficits or excessive private debt have forced countries, like the USA and Australia with insufficient domestic savings, to tap world sources

Costs & Benefits of Internationalisation

Costs

- Jobs at risk, wealth gap widens and more stress?
- Loss of autonomy for nations and economies
    - Political considerations
    - Accountability of government in an open economy
- Reduced ability of central banks to control domestic financial conditions
- Inappropriate valuations and excess variability of asset prices (share prices and currency values)
- Potential fragility and instability of financial activity
    - confidence in the financial system
    - ‘systemic risk’ – electronic funds transfer (EFT) and daylight overdrafts
    - political risk
    - foreign exchange risk – currency fluctuations
- Excessive use of scarce resources
    - Labour and capital costs of innovation

Costs & Benefits of Internationalisation

Benefits

- Gains from international trade
    - ‘exchange gains’
    - ‘production gains’
    - comparative vs. absolute advantage
- Foreign capital supporting economic growth
- Liquidity advantages
    - adds “depth” to financial markets
- Efficient markets
    - provides services at minimum cost
    - diversification of risk
    - transfer of funds from surplus units to deficit units wherever domiciled

Internationalisation and International monetary system (IMS)

- IMS plays(ed) a significant role in globalization of markets, international financial economics and financial management
- IMS facilitates the transfer of money from the savers to the borrowers
- IMS - The set of policies, institutions, practices, regulations and mechanisms that determine the rate at which one currency is exchanged for another"….Shapiro (2014) p.89
  - The policies:

    - Monetary policy, Capital flows, tariffs etc.
  - The institutions:

    - IMF, WTO, BIS, ECB
  - The mechanisms:

    - ERs …Floating, Fixed, Zone Arrangements etc

International Monetary Systems

- International Gold Standard (1890-1914), see details in the notes below
- Bretton-Woods System (1946-71), Exchange Gold Standard/Fixed exchange rate, see details in the notes below
- Managed/floating exchange rate regime (1971- present)
- Concurrently a number of ER systems e.g., EU and euro
- Return to the Gold Standard?

International monetary system (IMS)

BREXIT – “should we stay or should we go”?

“If I go there will be trouble, and if I stay it will be double,” sings The Clash’s Mick Jones

Brexit occurred on Jan 31st 2020

https://www.youtube.com/watch?v=xBaS-KxWjTA

IMS Problems

- Conflicts - between internal and external balance
  - internal balance = domestic goals of full employment, price stability, economic growth
  - external balance = goals of currency stability, Balance of payment position
- Monetary co-operation failure
  - Loss of national sovereignty (giving up Monetary Policy independence, national currency etc)
  - Non-homogeneity of nations - different cultures, different economic indicators - i/r, inflation rates, employment rates, etc

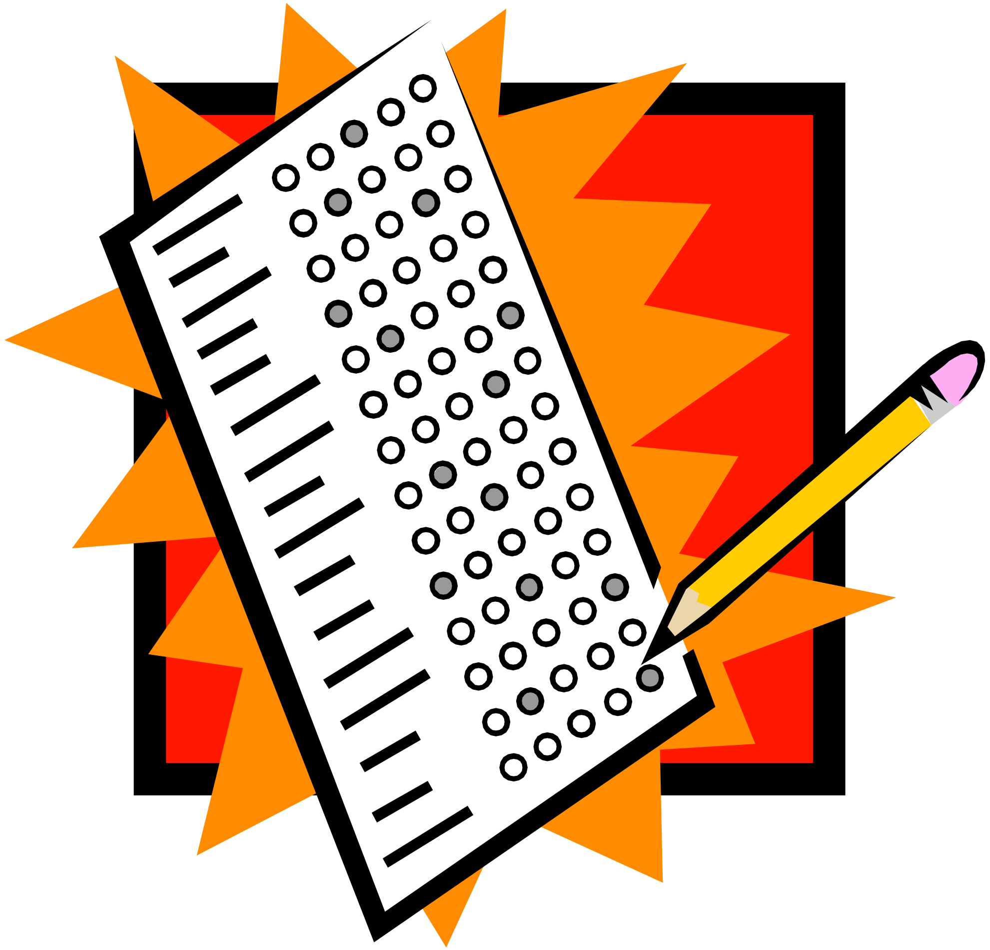
Riddle for the day

I’m a measure of market’s fervour and zeal,
Reflecting investors’ mode, both real and surreal.
When I climb high, optimism’s in flight,
But beware the fall, when fear takes its bite.

What am I?

a) Stock/share
b) Treasury bond/bond
c) Mutual fund
d) Investor/market sentiment
e) Loan/debt

Multiple Choice Questions (run through Mentimeter)



Next week topic

International financial markets: Euro currency markets

MAF306
International Finance and Investment

Topic 2
International Financial Markets and Euro currency markets

Dr Sohel Azad

Important announcements

- Form the group asap by 24 March, otherwise you will be allocated to a random group.
- Assignment details will be uploaded in the unit's clouddeakin site under "Assessment Resources". So, if you fail to miss forming the group, you will miss to start the work on your assignment.
- No hurdle in the exam, it will be run through clouddeakin and not to be proctored.

Topic 2 Learning outcomes

- To understand the role of international banking in financial markets
- To obtain an overview of the global financial markets and the inter-relationships of their components
- To understand the differences between intermediated and non-intermediated international financial markets
- To have a glimpse over eurocurrency markets
- To compare between international bonds, domestic bonds, foreign bonds and global bonds
- To understand the importance and role of the interbank market in determining LIBOR and LIBID
- To know the basics of Islamic financial markets
- To have a glimpse over international stock markets
- An overview of other rapidly expanding products and markets

Reading

- Shapiro Ch 12; pp 348-391, pages may vary
- Other study materials on Cloud Deakin

International Financial Markets (IFM)

- Foreign exchange market
- Euro markets
  - Euro currency market
  - Euro bond market
  - Euro note
  - Euro commercial paper
- International stock markets
- Islamic financial markets
- Other rapidly expanding markets

Intermediated and non-intermediated markets

Intermediated vs non-intermediated international finance

- International ***direct*** finance (non-intermediated)
  - Surplus Unit (SU) → Deficit Unit (DU)
  - No intermediary
  - May involve brokers/underwriters that match SU to DU for a fee

  e.g.,. Euro bonds or foreign bonds, Euro Commercial paper, Stocks
- International ***indirect*** finance (intermediated)

  SU → FI (1) → … → FI(n) → DU

  e.g.,. Eurodeposits, CDs, euroloans, syndicated loans

Intermediated Markets

Intermediated through a commercial bank or a financial institution

	Money Markets (< 12 months)	Capital Markets (> 12 months)
Internal Financial Markets	Domestic savings accounts, lines of credit with domestic borrower	Long term accounts with domestic clients
External Financial Markets	Eurocurrency deposits and loans	Long term accounts with foreign clients

Non-intermediated Markets

Direct to the public, through a broker or investment bank

	Money Markets (< 12 months)	**Capital Markets (> 12 months)**
Internal Financial Markets	Short-term commercial paper (CP) issued in domestic markets	Stocks or bonds issued in domestic markets
External Financial Markets	Euro commercial paper (ECP) issued in international markets	Stocks issued in foreign markets. Foreign bonds issued in foreign markets. Eurobonds

Role of international financial intermediation

Risk intermediation by portfolio diversification and credit analysis

- Credit/default risk
- Liquidity risk
- Interest rate risk

Maturity intermediation - borrow short, lend long

Size intermediation – small, medium or large size loan

Interest rate intermediation - floating to fixed

Currency intermediation - use and need of various currencies

Country intermediation - avoid costs of regulation, taxes etc

Food for thoughts: What do you think about the role of FinTech in intermediation? Is it disrupting or facilitating (international) banks' primary functions?

Features of International Financial Markets

- Involves different currencies
- Largely unregulated
- No barriers to entry/exit
- No specific physical location (electronic)
- Operate on narrow margins
- Net supplier of capital to foreign borrowers
- Financial services to ***both*** domestic and foreign residents
- Higher risk - political risk, currency risk, different legal jurisdictions

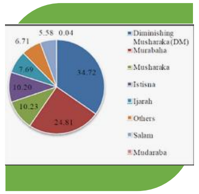
International financial markets

IT'S DIVERSE AND EXPANDING

FOREIGN EXCHANGE MARKETS

EURO CURRENCY MARKETS

INTERNATIONAL DEBT MARKETS



NON-CONVENTIONAL (ISLAMIC) FINANCIAL MARKETS

CRYPTOCURRENCY

GOLD MARKETS

Foreign Exchange Market

- Allows currencies to be exchanged in order to facilitate international trade or financial transactions.
- Largely an interbank market, deals in spot and forward transactions
- The exchange rate system/market has evolved over time
  - Gold Standard (1876-1913), each currency was convertible to gold
  - Period of instability, World War I, the Great Depression and The 1944 *Bretton Woods Agreement* - called for fixed currency exchange rates, pegged to US$
  - By 1971, the U.S. dollar appeared to be overvalued. The *Smithsonian Agreement* devalued the U.S. dollar (from $35 to $38 an ounce of gold) and widened the boundaries for exchange rate fluctuations from ±1% to ±2%.
  - Governments still had difficulties maintaining exchange rates within the stated boundaries. In 1973, the official boundaries eliminated, and the *floating exchange rate system* came into effect

The Euromarkets

- The term <u>Euromarkets</u> encompasses the following:

  1) the Eurocurrency market
  2) the Eurobond market
  3) the Euro-note market
  4) the Euro-commercial paper market

  Prefix 'Euro' does not refer to the European single currency!

Euro-currency markets

- **Euro** – location of the bank or market differs from the country of the currency used
- **Eurocurrencies**
  - A currency deposited in a bank outside its country of origin
    - e.g., **U.S. dollars held in a bank outside of the United States.**
      - eurodollars, euroyen, euro-euros!
- **Eurobanks** – financial intermediaries that accept bank deposits and/or make loans in currencies other than the legal tender of the country where located

Growth of Eurocurrency markets

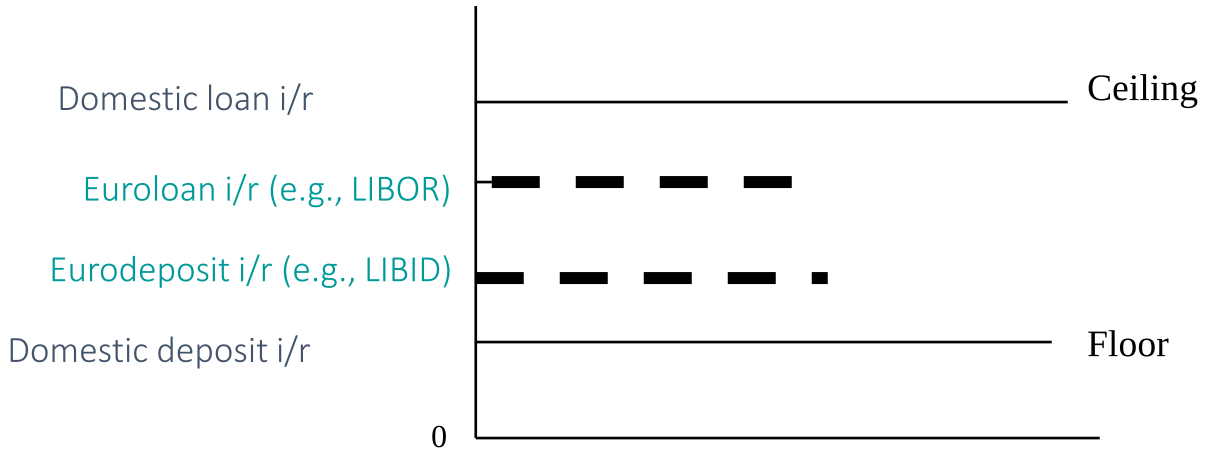
- Political considerations – (Post WW II)
  - Soviet Union placed US$ in French and London banks to avoid seizure of funds by USA
- Failure of Bretton Woods Systems
  - Countries struggled to maintain the fixed FX reflecting their econ conditions
- International Monetary System failure
  - failed to provide adequate liquidity to fund Balance of payment deficits, hence used private sector funds instead (contributed to Debt crisis of 1982)
- Globalisation and economic growth
  - Growth of MNCs & international trade, rapid economic growth from 1960s onward
- Few regulations/relaxation of regulations
  - Less regulation on capital movement, interest rate regulations - low or no taxes on interest income, deposit insurance requirements, or credit allocation regulations; less stringent disclosure requirements
- Others etc: IT revolution, Fintech, financial engineering and derivatives

Eurocurrency markets - features

- Wholesale nature of business
  - Economies of Scale (EOS), and increased efficiency
- Low risk
  - **Relatively short maturities**: Maturities of less than 5 years
  - **Low interest rate risk**:
    - Interest rates tied to a variable rate base such as the London Interbank Offer Rate (LIBOR)
    - Market provides a natural hedge, borrow in one currency and lend in another (currency swap)
  - **Low default risk**: Traded between large commercial banks, investment banks and multinational corporations
- Highly competitive
  - low borrowing costs (LIBOR), as compared to domestic market
  - high deposit returns (LIBID), as compared to domestic market

The Interbank Eurocurrency market

Interest rate spreads in domestic and Euromarkets



Gap between loan and deposit is bank's profit.

LIBOR/TIBOR/EURIBOR/IIBR/SONIA/SOFR/BBSW

Uses:

Benchmark for settlement of interest rate contracts like interest rate swaps, interest rate futures, saving accounts, loans and mortgages and what's not?

- Intercontinental Exchange (ICE), a parent company of the New York Stock Exchange (NYSE) and other famous exchanges and markets from around the world, took over the administration of LIBOR from the British Bankers Association (BBA) in 2014 following the LIBOR scandal.
- The entire LIBOR system has been ceased in 2023 and is replaced by other rates such as, the Sterling Overnight Index Average (GBP SONIA), Secured Overnight Financing Rate (USD SOFR) or Bank Bill Swap Rate (BBSW in Australia) or just interest rate swap rates for different tenors.

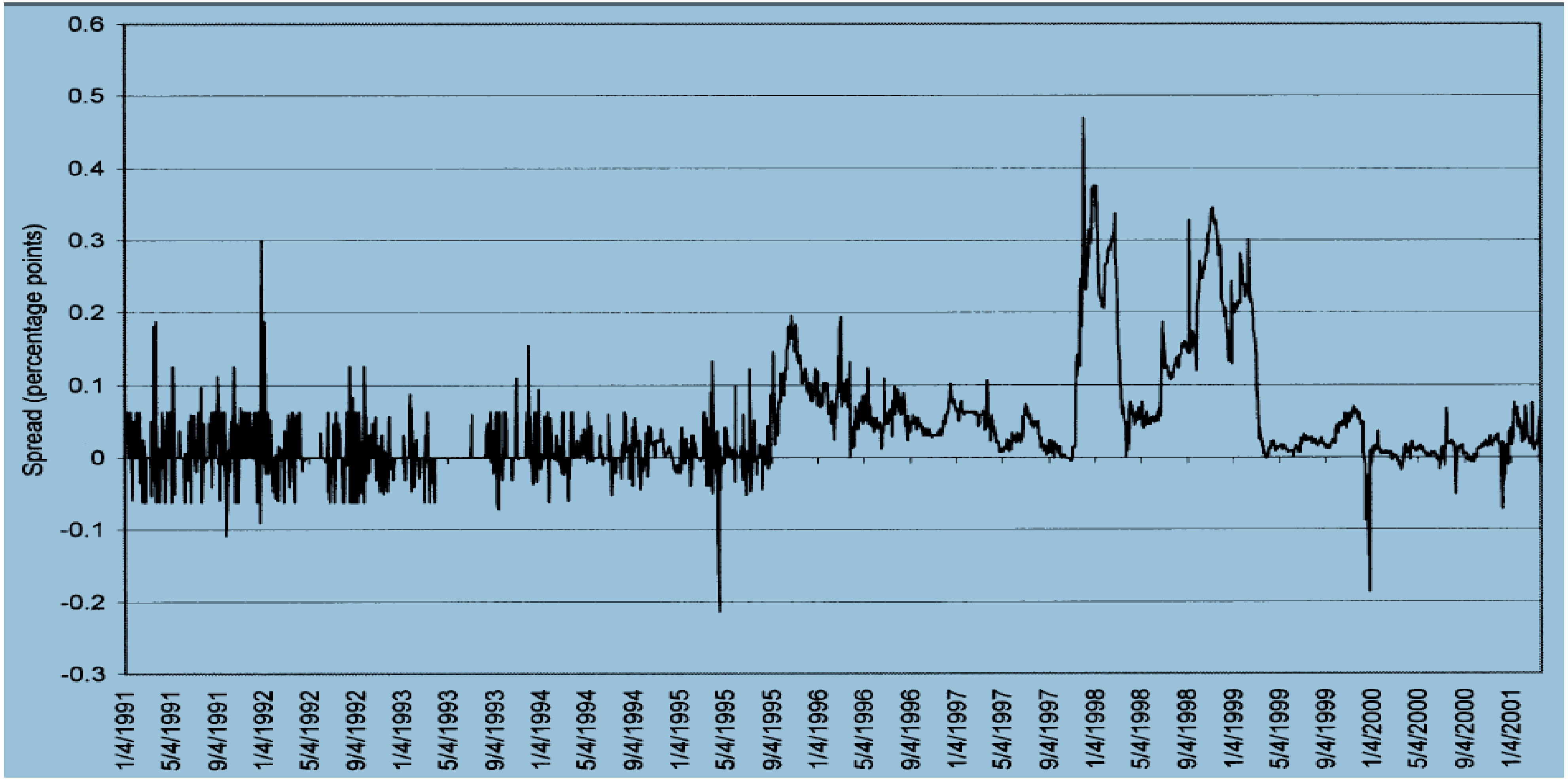
LIBOR vs SOFR

LIBOR	SOFR
Bank to bank lending rate (includes credit risk)	Risk-free rates (does not include credit risk)
Unsecured	Secured with US Treasuries
Based on bank submissions incorporating a limited number of actual transactions and expert judgment	Transaction based
USD$500 million of daily trading of actual transactions in the 3-month wholesale funding market	Over USD$1 trillion of daily trading of actual transactions in the overnight Repo market
Forward-looking term structure	Overnight, backward-looking No term structure currently as of 2021
Currency options include USD, GBP, EUR, JPY and CHF	Currency option is only USD

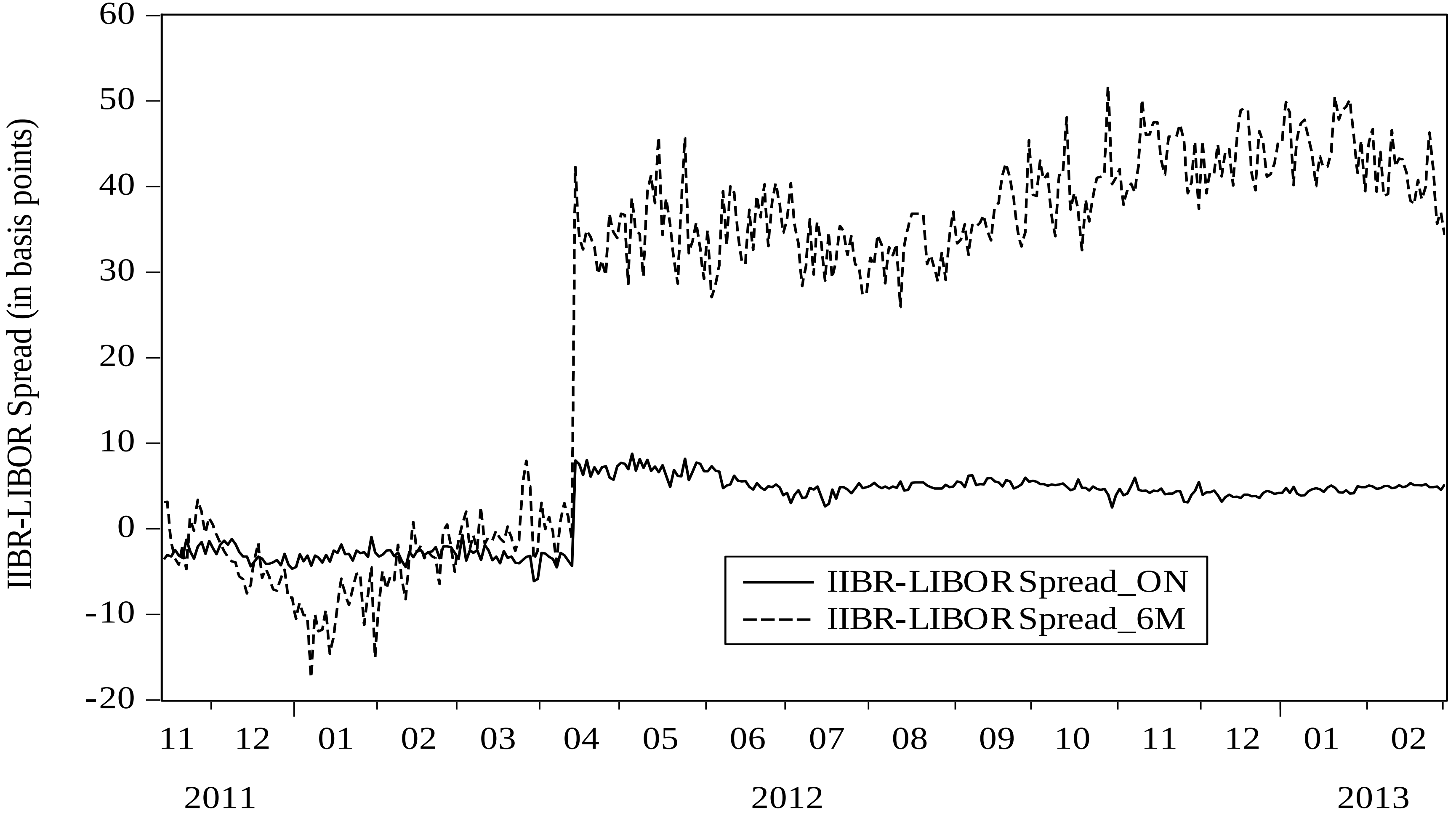
Eurodollar pricing anomaly and arbitrage

- TIBOR-LIBOR spread/***Japan premium***:
  - Presence of Japan premium: 1995-99; 2001~
  - Reasons: banks' rating differences, banks stock price volatility and news
  - Reference: Covrig, Low and Melvin, 2004 (JFQA)
- IIBR - LIBOR spread/***Piety premium***:
  - Islamic interbank benchmark rate (dollar) is higher than London interbank offered rate (dollar), ***piety premium***
  - Presence of stable spread since mid-April 2012: arbitrage?
  - Reasons: timing difference, bank types, ratings or news
  - Reference: Azad et al, 2018 (Emerging Markets Review)

Japan premium: TIBOR-LIBOR spread



Piety Premium: IIBR-LIBOR Spread



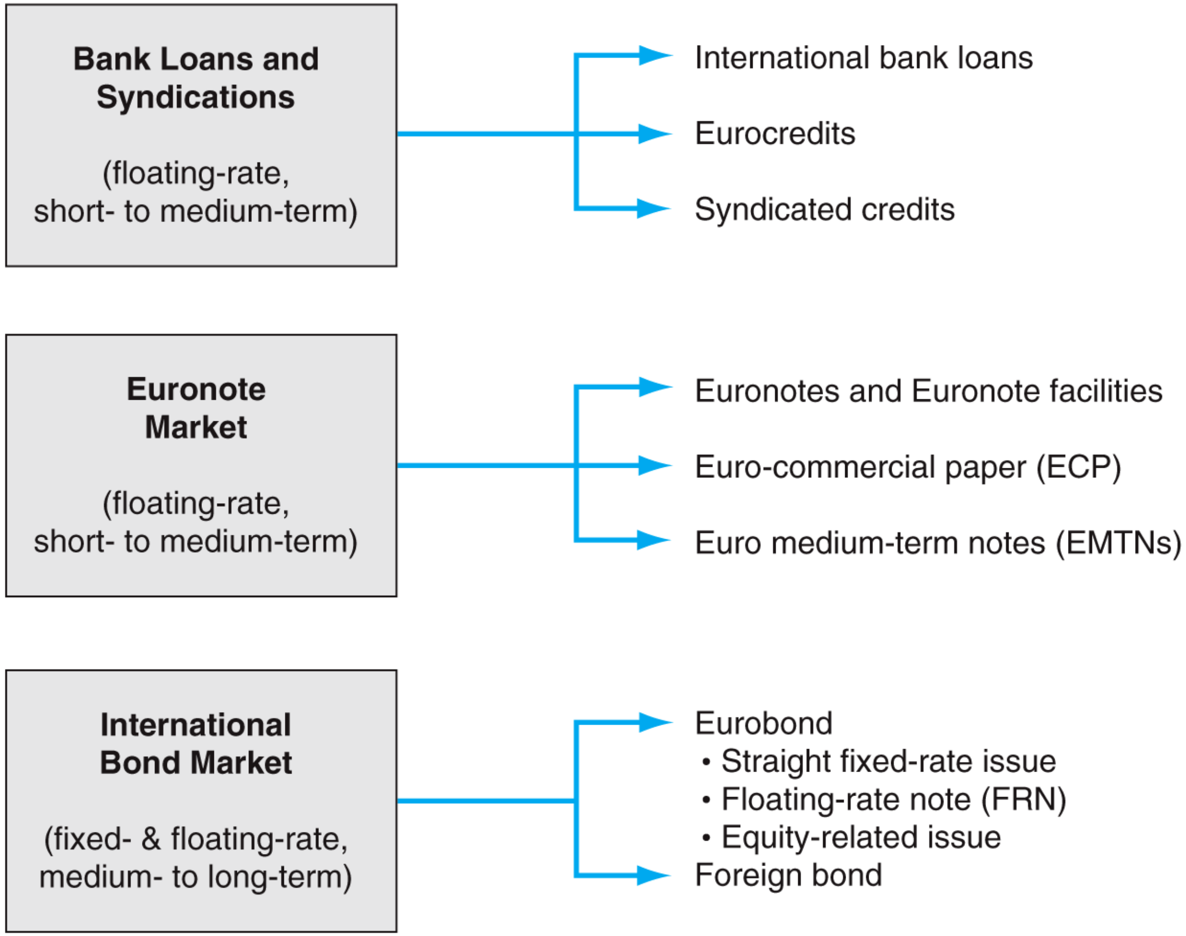
Arbitrage profit

Arbitrager can take advantage of the pricing anomaly in the two interbank markets on a particular day:

- Say, London interbank offered rate (LIBOR in US$) in London for 6 months maturity is 1%
- Islamic interbank benchmark rate (IIBR in US$) in Makkah for 6 months maturity is 1.20%
- Hypothetical cost of arbitrage is 0.05%

- With a hypothetical amount of US$1M an arbitrager can make a profit of: (0.012-0.01-0.0005) x $1m = $1500

International Debt Markets and Instruments



International Debt Markets: Short-term

- **Bank loans and syndications:**
  - International bank loans sourced in the Eurocurrency markets with narrow spread, less than 1%.
  - Syndication of loans - lead bank and many participant banks spreads risk among a number of banks
- **Euro credits :**
  - Short-Term bank loans to MNCs, sovereign governments, international institutions and banks
  - Denominated in Eurocurrencies
  - Variable interest rates, tied to LIBOR
- **Note Issuance Facility (NIF) –** Low-cost substitute for (syndicated) loan, similar to US commercial paper; Allows borrowers to issue own notes; Placed/distributed by investment banks – underwritten
- **Euro notes** - unsecured short-term debt, issued by corporations and governments
- **Euro-commercial paper (CP)** - A kind of euro notes not underwritten by guarantor bank – no guaranteed purchaser

International Debt Markets: Long-term

The Foreign Bond Markets

- Foreign bonds are issued in a domestic market by a foreign borrower
  - foreign refers to the origin of the borrower
  - issued in the currency of the country where sold
  - by foreign companies or governments, e.g., Yankee bonds, samurai bonds, Kangaroo bonds, Matador bonds, Panda bonds

- Global bonds trade in the Eurobond market and in one or more internal bond markets simultaneously
  - issued by large and AAA-rated borrowers
  - denominated in one or more actively-traded currencies
  - first issued by the World Bank in 1980, and first Corporate Bond by Japanese electrical Co in 1992

International Debt Markets: Long-term

- Eurobonds “placed” outside the borders of the country issuing the currency (i.e., sold in a currency different to the domestic currency to that of country where marketed)
  - Form of direct finance
    - “Placed” on market rather than formal issue, avoids regulation relating to new issues
    - Up to 75% in USD
    - Sold by syndicates in many countries simultaneously
    - Fixed or floating i/rate
    - Interest income is not subject to withholding tax

International Debt Markets: Short-term vs Long-term

Eurocurrency Loans

- Cost of borrowing
    - variable rates
    - Lower floatation costs (0.5%)
- Maturities – average of 5 years
- Size - millions
- Loans have greater flexibility egs. Drawdowns / repayments at any time
- Loans obtained faster 2-3 weeks
- Borrowing can be done in multiple different currencies
- Arranged as syndicated loans – lead bank and up to 20 participating banks

Euro bond

- Cost of borrowing
    - Fixed rate or floating rated
    - Higher floatation costs (2.25%)
- Maturities tend to be longer
- Size – billions
- Eurobond must be drawn down in one sum and repaid on fixed dates
- Eurobonds slower to issue
- Issued in one currency, majority in US$
- Bonds much greater volume

Non-conventional (e.g., Islamic) Financial Markets

- **Market is expanding rapidly**.
  - Reasons: stability, less volatility, better asset quality, risk sharing, profit loss sharing instead of interest, risk diversification.
- Was more resilient during the GFC, why? (https://www.imf.org/external/pubs/ft/survey/so/2010/RES100410A.htm)
- Video: Four Things You Need to Know about Islamic Finance
- **Main financial products**: Banking, Sukuk (i.e., bonds), superannuation and stocks
- **Countries/Markets:**
  - Gulf Cooperation Council (GCC) - the epicentre
  - South Asia; East and South East Asia including Malaysia, Indonesia, China, Hong Kong, Japan,
  - Australia (MCCA, Crescent Wealth, Amanah, Insaaf, Ijara, Icfal, Hejaz, NAB's halal business loan, Islamic Bank Australia)
  - Other western world: to name a few - Denmark, France, Germany, Italy, Ireland, Luxemburg Switzerland, UK, USA.

Traditional Bonds vs Sukuk

- Sukuks are Islamic bonds or Islamic investment certificates used to raise money for projects that need large-scale investments. Sukuk bonds are based on Islamic principles and laws in relation to the prohibition of interest, alcohol, gambling, speculation and pork related investments. Because of these reasons, sukuk bonds are also categorized as **socially responsible investments (SRI)** and green bonds.
- The first corporate sukuk was issued by a non-Muslim company in 1990 in Malaysia
- Unlike bond, sukuk are like investment certificates consisting of ownership claims in a pool of assets. Sukuk give claims to both cash flows and asset ownership.
- Thus, Sukuk are similar to the conventional concept of securitization, a process in which ownership of the underlying assets is transferred to a large number of investors through certificates indicating proportionate value of the relevant assets.

International equity markets and benchmarks

There are many international equity markets around the world. However, based on the size & liquidity, global reach, capital formation, price discovery, economic indicators and innovation and tech, following are the major international equity markets and their respective benchmarks:

- New York Stock Exchange (NYSE) and NASDAQ in the United States:
  Benchmarks: S&P 500 Index, Dow Jones Industrial Average (DJIA), NASDAQ Composite Index
- London Stock Exchange (LSE) in the United Kingdom:
  Benchmark: FTSE 100 Index
- Tokyo Stock Exchange (TSE) in Japan:
  Benchmark: Nikkei 225 Index
- Hong Kong Stock Exchange (HKEX) in Hong Kong:
  Benchmark: Hang Seng Index
- Shanghai Stock Exchange (SSE) and Shenzhen Stock Exchange (SZSE) in China:
  Benchmark: SSE Composite Index (Shanghai), SZSE Component Index (Shenzhen)
- Euronext in Europe:
  Benchmark: EURO STOXX 50 Index

Rapidly expanding markets and products

- ***Peer-to-Peer (P2P) lending*** – no middlemen such as, banks or financial institutions
- ***Crowdfunding*** – large pool of investors, bypass IPOs and listing requirements
- ***DeFi (Decentralized Finance)***: DeFi platforms are growing rapidly, offering a wide range of financial products and services such as lending, borrowing, and trading, all built on blockchain technology.
- ***Cryptocurrency***
  - Digital currency operating outside the control of the central bank
  - Originated in early 1980s, became more popular after GFC
  - Hundreds of different cryptocurrencies – 1500+, but dominated by just a few
  - Bitcoin is the largest (market cap) followed by Ethereum, Ethereum 2, Tether and so on
- ***Digital Assets and Tokenization:*** *Apart from cryptocurrencies, there is a growing trend towards tokenization of assets such as real estate, art, and even intellectual property rights. This enables fractional ownership and increases liquidity for traditionally illiquid assets.*

Coinbase: https://www.coinbase.com/price

Bitcoin price history

(in AU$, Market cap 700 Billion, March 2023)

Source: [Coinmarket]

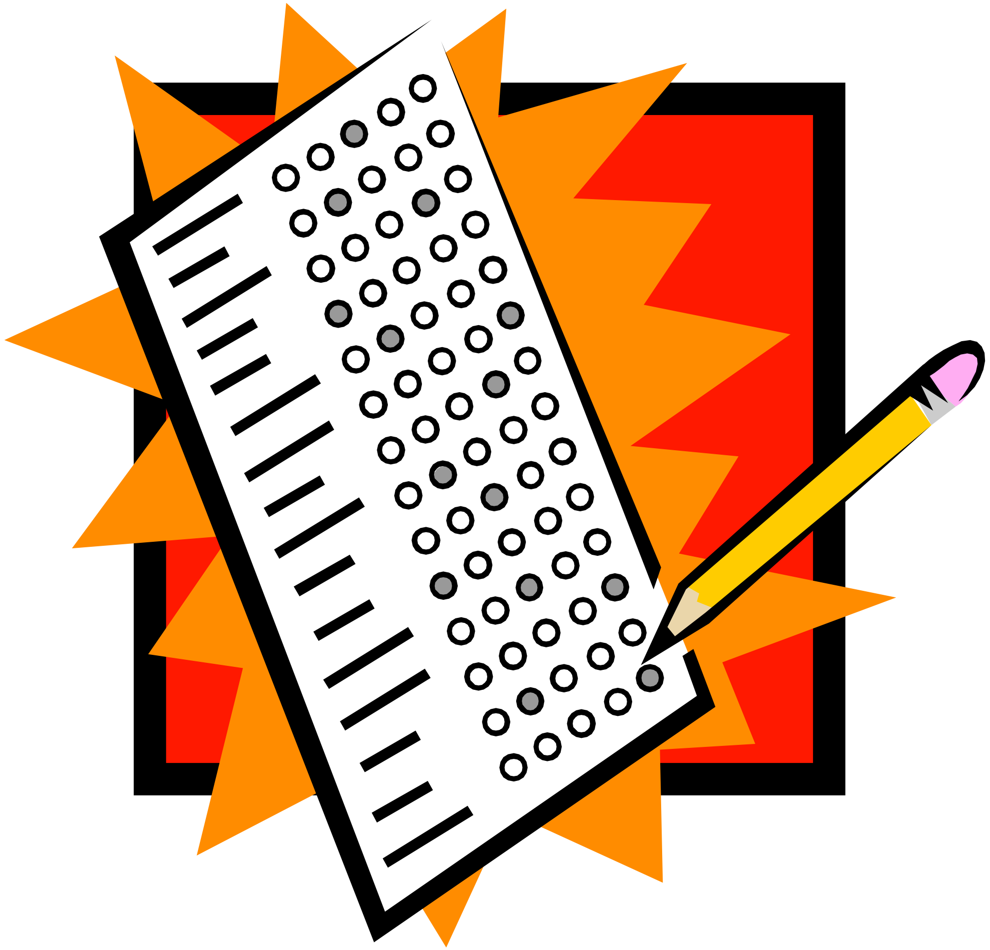
Are Cryptocurrencies Viable For Widespread Use?

Pro’s of Cryptocurrencies	Con’s of Cryptocurrencies
Cryptocurrency has low transaction costs compared to other digital payment methods like PayPal.	Cryptocurrency is only accepted by certain vendors.
Cryptocurrency makes trading anywhere in the world easy. It’s a decentralized currency. This opens up financial options for people in countries that don’t have access to financial services.	Not having a central bank control cryptocurrency adds to its volatility as no central force can step in to correct the markets (although this can differ by coin).
Transactions are quick, permanent, and hard to fake, this eliminates a lot of the fraud issues banks deal with.	If something goes wrong with a transaction or if a coin is lost there is no way to recover it. If someone does steal coins there is no way to rectify the issue.
Cryptocurrency isn’t inflationary. With coins like bitcoin there is a set amount that will ever be created.	Any given cryptocurrency lacks the flexibility of centralized currency due to its non-inflationary nature.
Single currency use e.g., Traders from different countries/locations can use the same currency without additional transaction costs.	Prone to hacking, phishing attacks, and malware, irreversible nature of transactions means that funds lost or stolen are often unrecoverable

Gold – the precious yellow metal

- The critics of fiat money advocate for gold or crypto
- Gold has been used as a store of value, a safe haven and a means of exchange for millennia.
- The 17th Century British Mercantilist Sir William Petty described gold, silver and jewels as wealth “at all times and all places”
- In crisis periods, investors tend to put their money in gold.
- Investors have traditionally used gold as a hedge against inflation or a falling dollar.

MCQ to be run through Mentimeter



MAF306: Topic 3
Foreign Investments:
FDI, Investment Vehicles & Managing Portfolios

Dr. Sohel Azad

Learning resources

- Shapiro Ch 16, Ch 1
- Eiteman et al Ch 17
- Other: CloudDeakin resources, Newspaper articles, news (Bloomberg)

Learning Objectives

- FDI: An overview
- Justifying international trade/FDI: The theories
  - Why, where and how?
- Sustaining and transferring competitive advantages
- FDI theories: OLI paradigm and FDI entry
- How to invest abroad: alternative methods for FDI
- Investment vehicles/opportunities
- Risk and diversification
- Choosing assets: what to consider

Overview of FDI

Types of International Investment

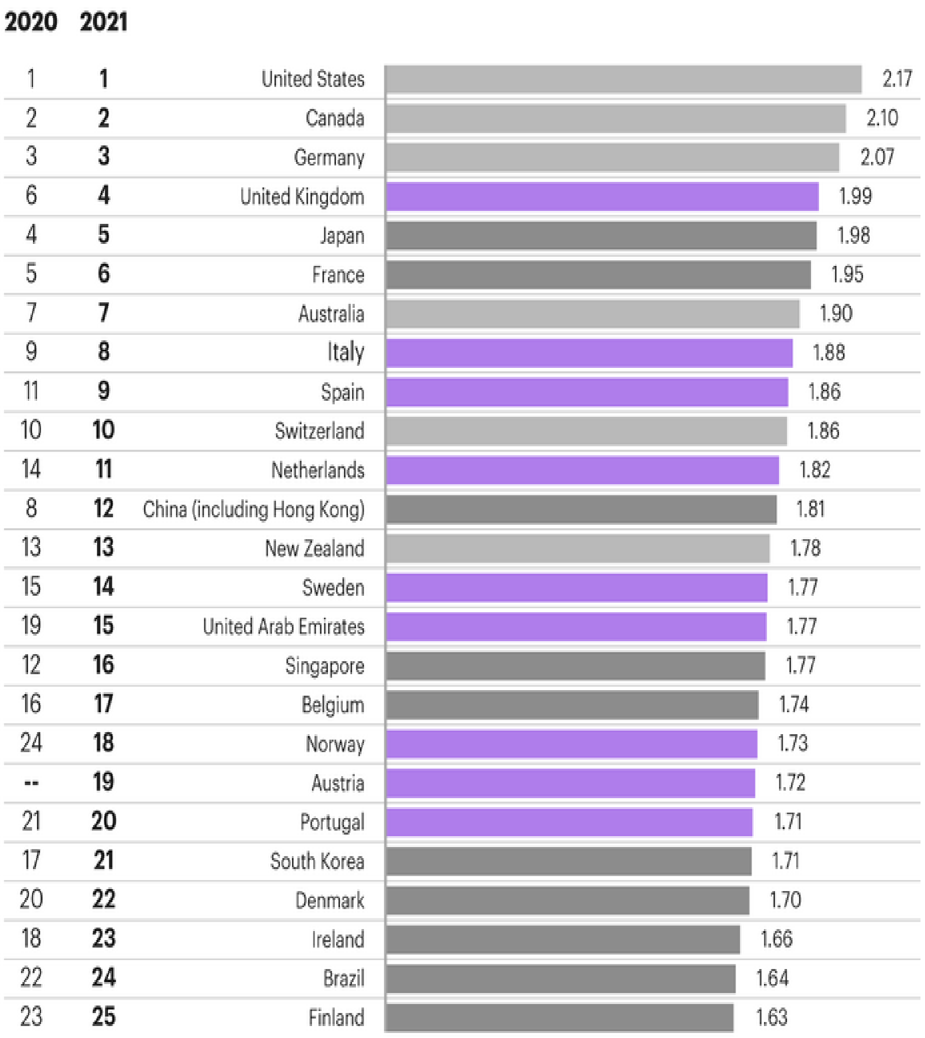
- **Foreign Institutional Investment (FII)**
  - Purchase of stocks or bonds issued by foreign Company’s and Government
  - $<$ 10% ownership
  - Long Term returns, diversified risk
- **Foreign direct investment (FDI)**

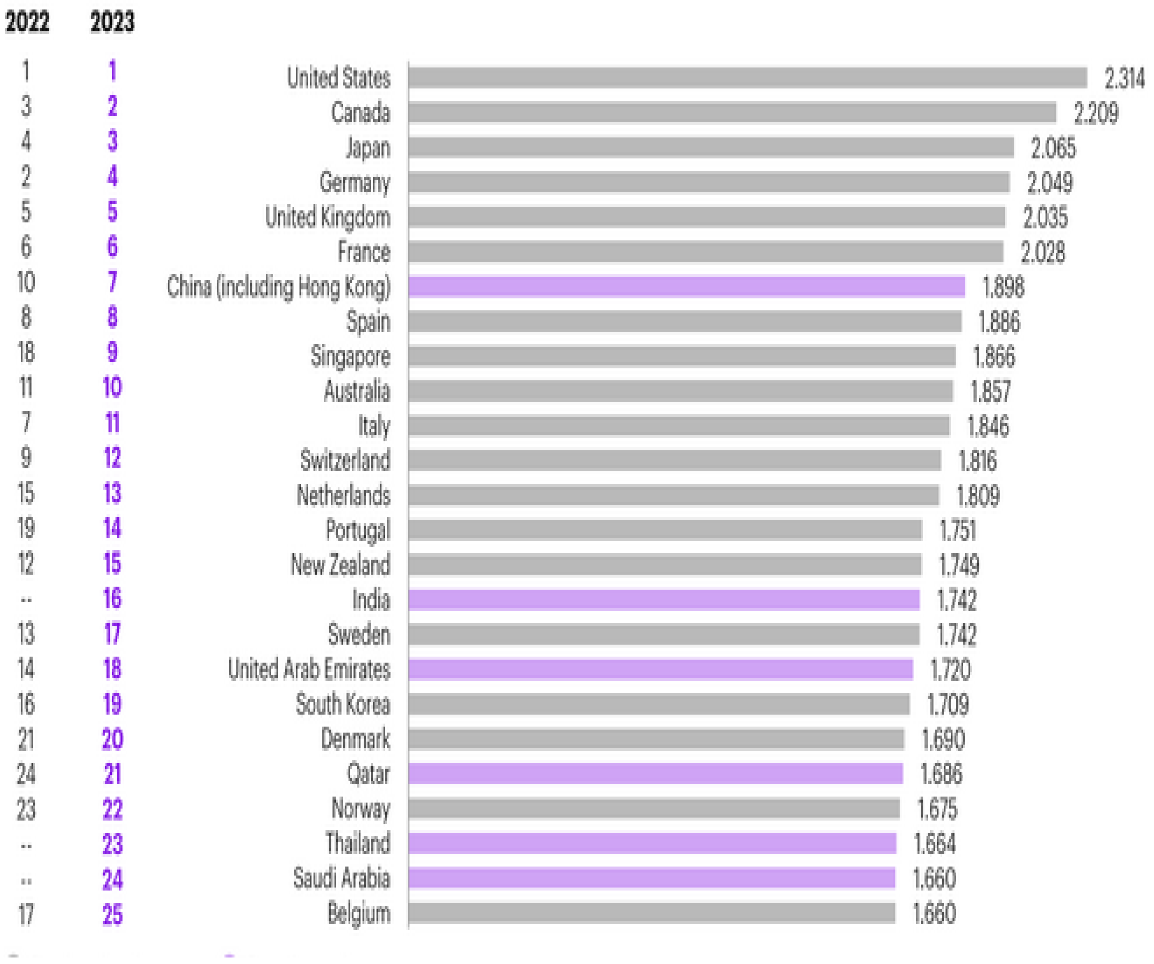
  Foreign residents acquire ownership (10%-25% min^{m}) of assets to control production and distribution in a Company
  - Mergers
  - Acquisitions
  - Joint ventures
  - Greenfield

787 Dreamliner structure suppliers

Selected component and system suppliers.

FDI destinations: recent survey from executives (source: **A.T. Kearney 2020 and 2023 FDI Confidence Index**)



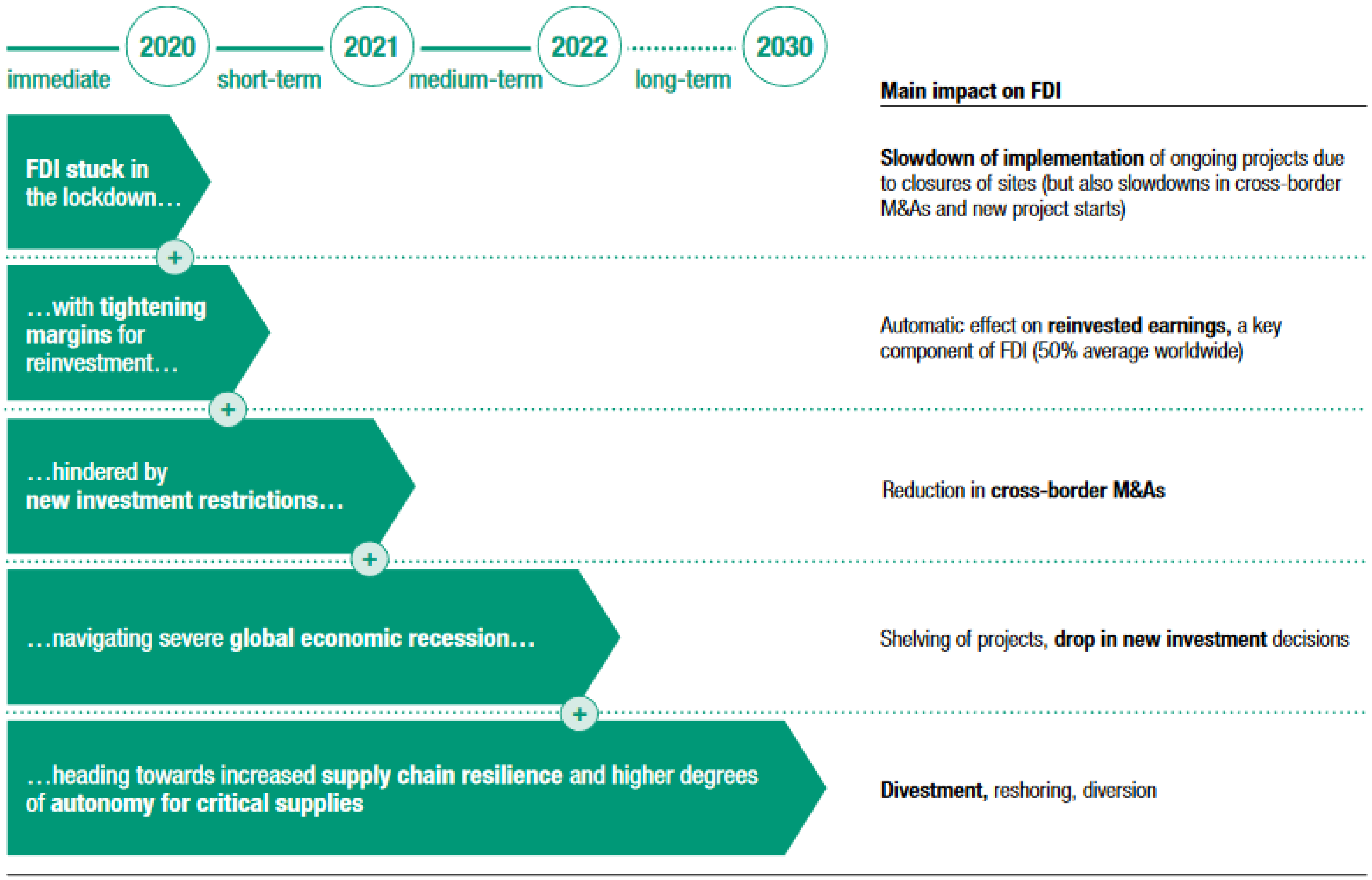


Impact of Covid-19 on FDI

Immediate impacts:

FDI stuck in the lockdown. The physical closure of places of business, manufacturing plants and construction sites to contain the spread of the virus causes immediate delays in the implementation of investment projects. Some investment expenditures continue (e.g. the fixed running costs of projects), but other outlays are blocked entirely.

Announcements of greenfield projects are also delayed. Similarly, many mergers and acquisitions (M&As) are temporarily suspended.



Source: UNCTAD.

When and why is FDI justified?

When and why is FDI justified?

- offers more opportunities than a purely domestic portfolio
- attractive investments overseas
  - Raw material, lower cost, new market…
- impact on efficient portfolio with diversification benefits
  - Extend the efficient frontier
  - Argument: risk diversification
- combined entity should create synergy – more value than the sum of its parts (i.e., 2+2=5)

When and why is FDI justified: Theories

- The *theory of competitive advantage* provides a basis for explaining and justifying international trade
- The above theory comes from the classical international trade theory "*theory of comparative advantage*", which was first developed by Adam Smith and David Ricardo in 1817. The theory was taken from the book The Wealth of Nations (1776), where Adam Smith brought the concept of ***absolute advantage***
- The theory has some underlying assumptions, such as -
    - Free trade
    - No uncertainty
    - No market frictions
        - Perfect competition
        - No government interference
        - Costless information

The Theory of Comparative Advantage

- The features of this theory are as follows:
  - Exporters in Country A sell goods or services, which are unrelated to importers in Country B
  - Firms in Country A specialize in making products that can be produced relatively efficiently, given Country A's endowment of factors of production (land, labor, capital, and technology)
  - Firms in Country B do the same with different products (may be with different factors of production)

The Theory of Comparative Advantage

- Because the factors of production cannot be transported (?), the benefits of specialization are realized through international trade
- The *terms of trade*, the ratio at which quantities of goods are exchanged, shows the benefits of excess production
- Neither Country A nor Country B is worse off than before trade, and typically both are better off (albeit perhaps unequally)

The Theory of Comparative Advantage

- Assume, Thailand is more efficient than Brazil at producing both sports shoes and stereo equipment
- With one unit of production (a mix of land, labor, capital, and technology), efficient Thailand can produce either 12 shipping containers of shoes or 6 shipping containers of stereo equipment
- Brazil, being less efficient in both, can produce 10 containers of shoes or 2 containers of stereo equipment with one unit of input

The Theory of Comparative Advantage

- A production unit in Thailand has an *absolute advantage* over a production unit in Brazil in both shoes and stereo equipment
- Thailand has a larger *relative advantage* over Brazil in producing stereo equipment (6 to 2 = 3) than shoes (12 to 10 = 1.2)
- As long as these ratios are unequal (*3* & *1.2*), *comparative advantage* exists
- The following exhibit illustrates total world (in this example) production and consumption if there was no trade and if each country completely specialized in one product

The Theory of Comparative Advantage

- **Production if no trade (using 1000 units of inputs)**

	Shoe Production	Stereo Production
Thailand produces & consumes	300 x 12 = 3,600 ctrs	700 x 6 = 4,200 ctrs
Brazil produces & consumes	300 x 10 = 3,000	700 x 2 = 1,400
Total world production & consumption	6,600 ctrs	5,600 ctrs

- **Production after complete specialization**

	Shoe Production	Stereo Production
Thailand produces only stereo equipment		1,000 x 6 = 6,000 ctrs
Brazil produces only shoes	1,000 x 10 = 10,000	
Total world production & consumption	10,000 ctrs	6,000 ctrs

❖Clearly the world in total is better off because there are now 10,000 containers of shoes (instead of 6,000) and 6,000 containers of stereo equipment (instead of 5,600)

❖International trade is needed to distribute the production across international boundaries!

Today’s implications of the Theory

- International trade implications of comparative advantage theory in 19th century and today are different:
  - Countries do not appear to specialize only in those products that could be most efficiently produced by that country’s particular factors of production
  - At least two of the factors of production (capital and technology) now flow easily between countries (rather than only indirectly through traded goods and services)
  - Modern factors of production are more numerous than this simple model
    - Today’s comparative advantage is based more on services, and their cross-border facilitation by telecommunications and the Internet
  - Comparative advantage shifts over time

Examples of global outsourcing of comparative advantage in intellectual skills

Country	Activity
China	Chemical, mechanical and petroleum engineering services; business and product development centres for companies like GE.
Costa Rica	Call centres for Spanish-speaking consumers in many industrial markets; Accenture has IT support and back-office operations call centres.
Eastern Europe	American IT service providers operate call centres for customer and business support in Hungary, Poland and the Czech Republic and Romanian and Bulgarian centres for German-speaking IT customers in Europe.
India	Software engineering and support; call centres for all types of computer and telecom services; medical analysis and consultative services; Indian companies such as Tata, Infosys and Wipro are already global leaders in IT design, implementation and support.
Mexico	Automotive engineering and electronic sector services.
Philippines	Financial and accounting services; architecture services; telemarketing and graphic arts.
Russia	Software and engineering services; R&D centres for Boeing, Intel, Motorola and Nortel.

Sustaining, transferring and protecting competitiveness

Sustaining & Transferring Competitive Advantage

- Some of the competitive advantages enjoyed by MNEs/MNCs are:
  - Economies of scale and scope
  - Managerial and marketing expertise
  - Advanced technology
  - Financial strength
  - Differentiated products
  - Competitiveness of their home market

Sustaining & Transferring Competitive Advantage

- Economies of scale and scope:
  - Can be developed in production, marketing, finance, research and development, transportation, and purchasing
  - Large size is a major contributing factor (due to international and/or domestic operations)
- Managerial and marketing expertise:
  - Includes skill in managing large industrial organizations (human capital and technology)
  - and knowledge of modern analytical techniques and their application in functional areas of business
- Advanced technology:
  - Includes both scientific and engineering skills

Sustaining & Transferring Competitive Advantage

- Financial strength:
  - Demonstrated financial strength by achieving and maintaining a global cost and availability of capital
  - This is a critical competitive cost variable that enables them to fund FDI and other foreign activities
- Differentiated products:
  - Differentiated products originate from research-based innovations or heavy marketing expenditures to gain brand identification
- Competitiveness of the home market:
  - A strongly competitive home market can sharpen a firm's competitive advantage relative to firms located in less competitive ones

Protecting Competitive Advantages: HOW??

- limit the scope of technology transfer to include only non-essential parts of the production process
- limit the transferability of technology by contract
- limit dependence on any single partner
- use only assets near the end of their product life cycle
- use only assets with limited growth options
- trade one technology for another
- remove the threat by acquiring the foreign partner….

FDI theories: Choice and modes of FDI entry

The OLI theory and choice of FDI entry

- The OLI Paradigm (Buckley & Casson, 1976; Dunning 1977) is an attempt to create an overall framework to explain why MNEs choose FDI rather than serve foreign markets through alternative modes such as licensing, joint ventures, strategic alliances, management contracts and exporting
  - Ownership – Ownership advantage is attributed to the Industrial Organization Theory (1959) - a firm must first have some competitive advantage in its home market - “O” or *owner-specific* – which can be transferred abroad: brand names, patents, copyrights, proprietary technology or technological processes, and marketing and management expertise; cheaper access to raw materials, added advantages of R&D
  - Location – is attributed to Location Theory (1972) - essential inputs are immobile, so firms look for location-specific natural resources, man-made resources, low taxes, low wage costs, high labor productivity, or state-supported monopolies
  - Internalization – is attributed to Internalization Theory (1983) - the firm will maintain its competitive position by controlling the entire value-chain in its industry – “I” or *internalization*; This leads to FDI rather than licensing or outsourcing; takes the benefits of market imperfections like taxes, capital controls but assumes FX risks

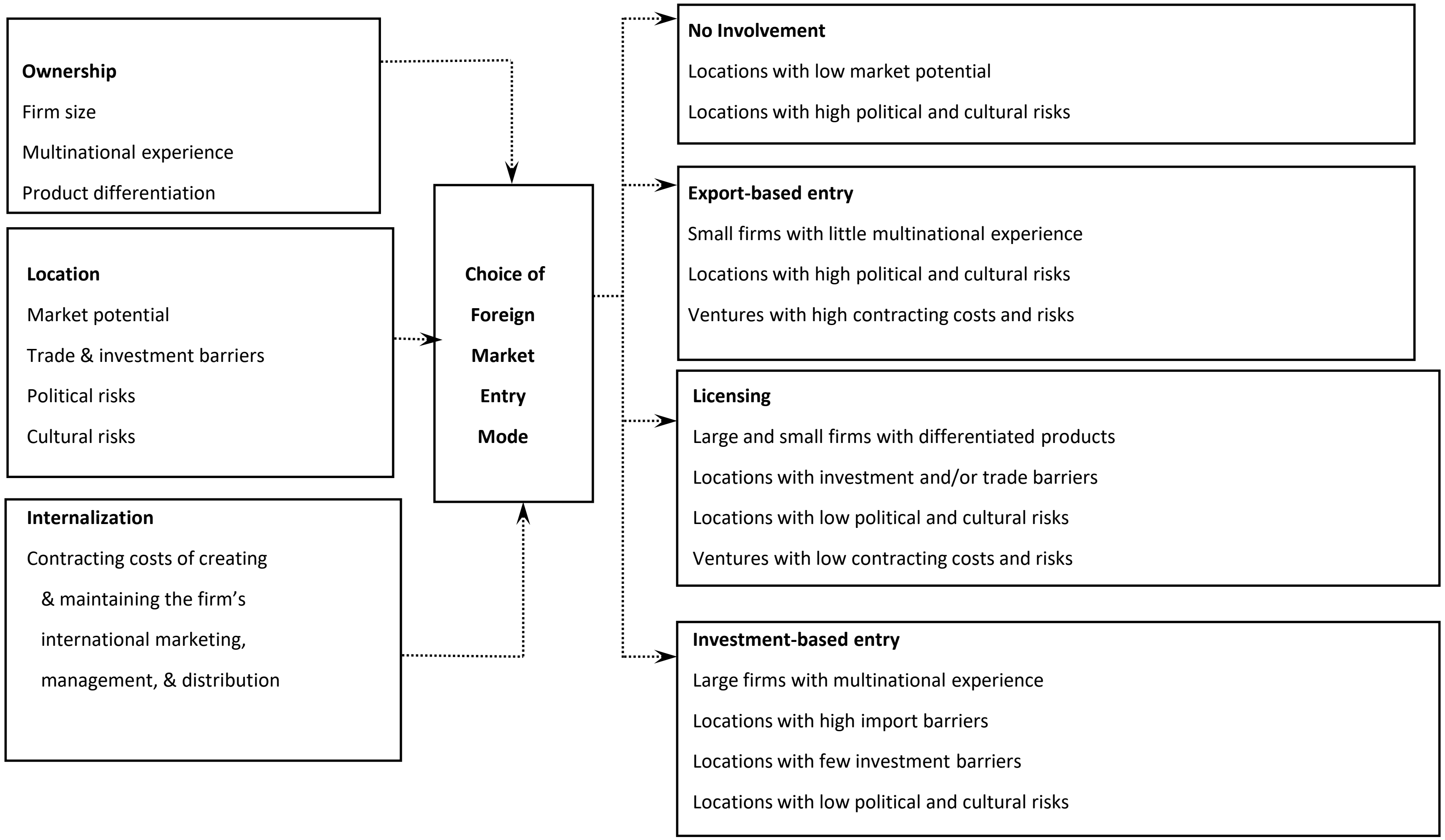
The OLI theory and choice of FDI entry

- The Eclectic Approach (1988)
  - the why, where and how of the MNC
  - synthesis of industrial organization, locations and internalisation theories
  - now known as the *OLI Framework*

- **Why?** Ownership-specific (O) advantages – can be transferred abroad
- **Where?** Location-specific (L) advantages – firm attracted to features of the foreign market
- **How ?** Market internalization (I) advantages – firm owns and controls the entire production process

OLI/eclectic approach - leads to FDI rather than licensing or outsourcing

The OLI theory & choice of FDI



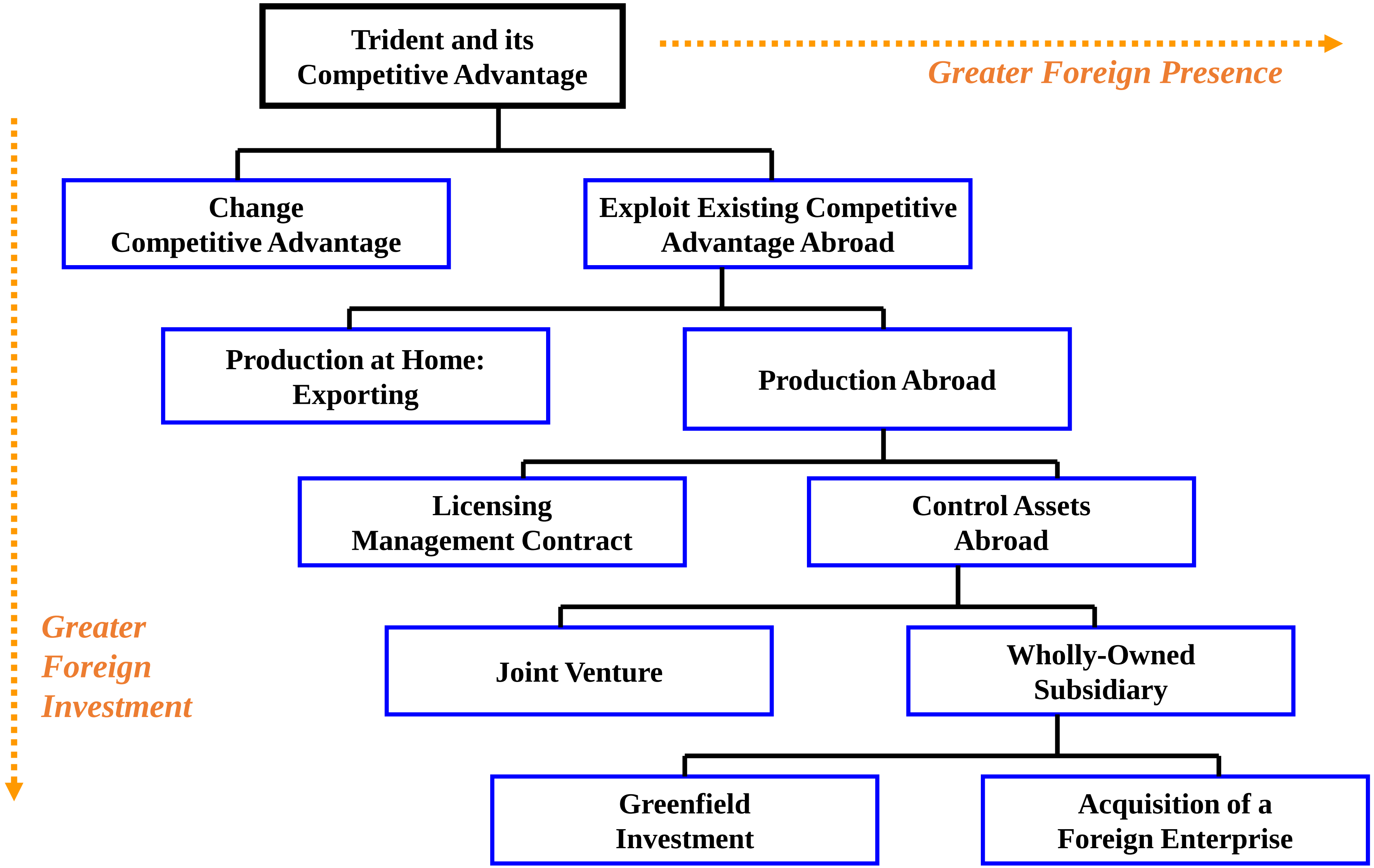
Behavioral theories: where to invest

- Two related behavioral theories behind FDI that are most popular are
  - ***Behavioral approach to FDI***
  - ***International network theory***
- ***Behavioral Approach*** – Observation that firms tended to invest first in countries that were not too far from their country in psychic terms
  - This included cultural, legal, and institutional environments similar to their own
- ***International network theory*** – As MNEs grow they eventually become a network, or nodes that operate either in a centralized hierarchy or a decentralized one
  - Each subsidiary competes for funds from the parent
  - It is also a member of an international network based on its industry
  - The firm becomes a transnational firm, one that is owned by a coalition of investors located in different countries

How to Invest Abroad: Modes of FDI

- Exporting vs. production abroad
- Licensing/management contracts vs. control of assets abroad
- Joint ventures vs. wholly owned subsidiary
- Greenfield investment vs. acquisition
  - A *greenfield investment* is establishing a facility “starting from the ground up” (usually require extended periods of physical construction and organizational development)

Modes of FDI

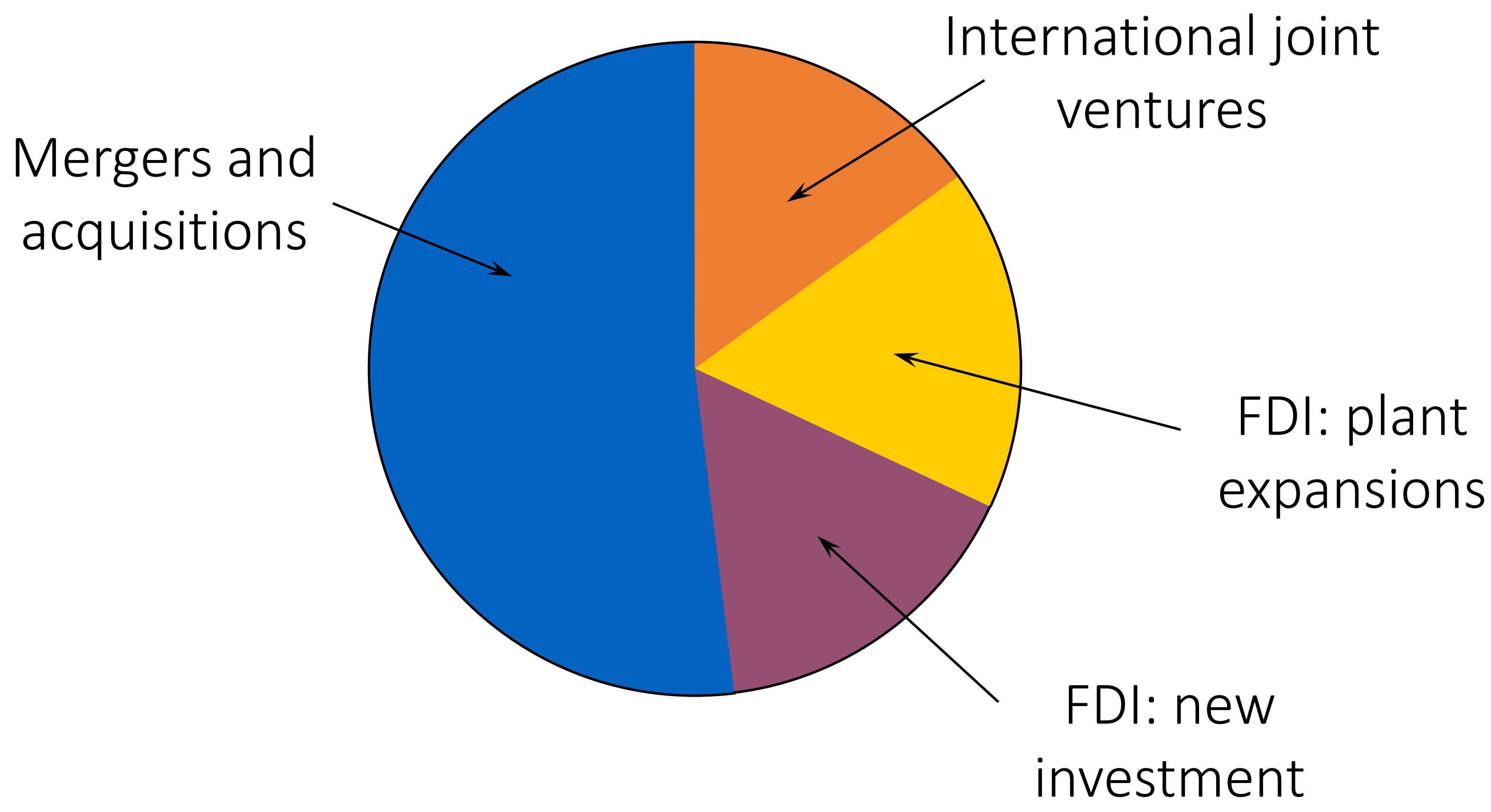

Trident and its Competitive Advantage
Greater Foreign Presence
Change Competitive Advantage
Exploit Existing Competitive Advantage Abroad
Production at Home: Exporting
Production Abroad
Licensing Management Contract
Control Assets Abroad
Greater Foreign Investment
Joint Venture
Wholly-Owned Subsidiary
Greenfield Investment
Acquisition of a Foreign Enterprise

Modes of FDI

- Export entry
  - agents or distributors (foreign or domestic)
  - foreign sales branches or subsidiaries

- Contract-based entry
  - licensing – eg. Nike
  - franchising – recipe for the product eg. McDonald’s
- Investment-based entry*
  - foreign direct investments → “Greenfield” Investment
  - mergers and acquisitions (M&A)
  - strategic alliances or joint ventures – with host Company or host Government
  - real estate purchase

Investment-based entry*



Source: Ernst & Young

Pros and Cons of FDI

- Pros/advantages:
  - Liquidity, flow of investment and funds
  - Employment generation
  - Increase in exports
  - Access to resources
  - Diversification of risk
  - Exchange rate stability
  - Tax advantages
  - More/new customers

Cons:

- Expropriation/Political risk
- Host country interference
- Loss of proprietary information
- Capital outflow/flight
- Unhealthy competition
- Effects on environment
- Hinders domestic investment
- Exchange rate risk
- Loss of control over business

Investment vehicles and managing portfolios

Investment vehicles*

- Investment companies
- Mutual funds
- Exchange traded funds (ETFs)
- Hedge funds
- Real estate investment trust (REITs)
- Direct or cross listing
- Depository receipts
- Cryptocurrency
- Commodity futures and gold
- Green/environmental financing/investments
- Religious stocks/bonds/asset class
- International fixed income (e.g., bond portfolios)

*See the additional slides uploaded in the cloud site

Managing Portfolios

- Allocating funds across different securities or assets, markets, Why?
  - To achieve a specific target-return while taking on the least possible risk.
- Bread & Butter for portfolio managers at Fidelity, Vanguard, Blackrock etc.
- Even MNCs indulge in portfolio management
- International Portfolio Management - is it different?
  - Different countries, different currencies, ever-changing exchange rates, volatile markets etc.

Risk, Return and diversification

- What about investing in more than one country
  - Diversification
  - Correlation of returns
- Why not invest only in developed countries?
  - What about return?
- Why not invest only in developing countries?
  - What about liquidity, safety (credit risk)
  - Dilution of risk

Risk and return around the world

- Data (2000-2018) indicate that emerging markets generally have higher risks and returns than developed markets.
    - India and Brazil, for instance, offer higher returns than the USA or Switzerland but much higher standard deviation of returns.
- Where to invest?
    - Options are diverse, have to research
    - In the *Merchant of Venice*, Shakespeare has the merchant Antonio say:

My ventures are not in one bottom trusted,

Nor to one place: nor is my whole estate

Upon the fortune of this present year;

Therefore, my merchandise makes me not sad.

Source: Shapiro

Risk of international investing

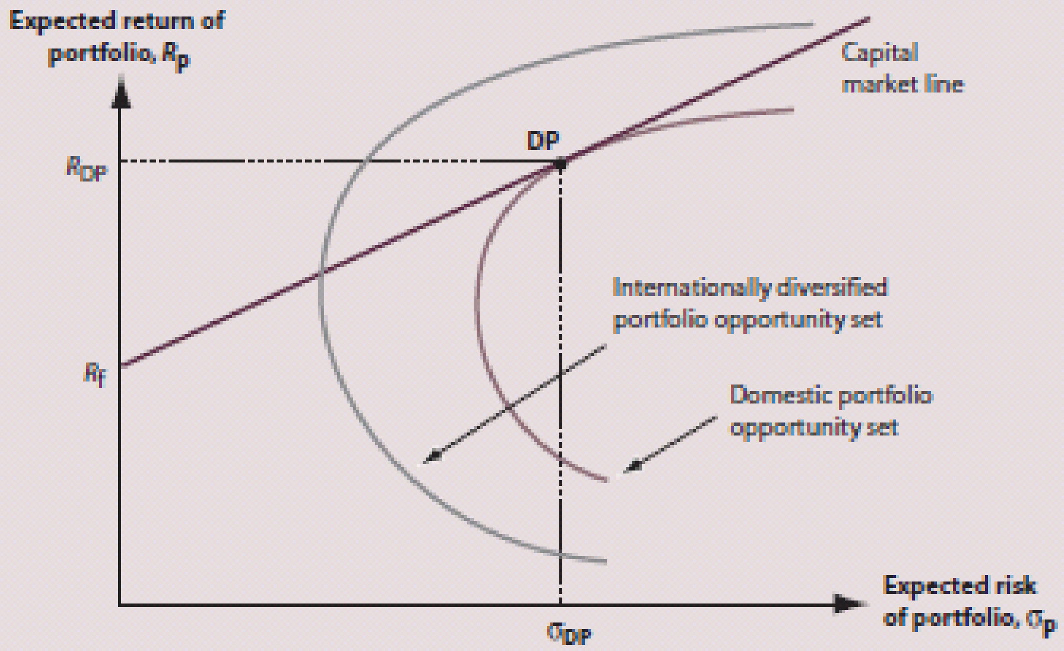
- *Changes in currency exchange rate:* dividends received on or proceeds from the sale of international investments may be reduced or increased when the dividends or proceeds are converted into dollars.
- *Currency controls* may restrict or delay an investor's ability to move currency out of a country.
- *Dramatic changes in market value.* Assets/firms value/shares could change dramatically
- *Political, economic, and social events* may influence foreign markets more than the U.S. market.
- *Lack of liquidity:* foreign markets may have lower trading volumes, abbreviated hours of operation, and restrictions on the amount and type of stocks investors may purchase.
- *Less information:* Disclosure policies can be more lax in some foreign countries relative to the U.S.
- *Legal recourse:* even if investors have legal recourse in the U.S., judgments may not be enforceable in foreign countries. Thus, investors may have to rely on the legal system of the foreign company's home country.
- *Different market operations:* Foreign markets may have different periods for clearance and settlement and may not report stock trades as quickly as U.S. markets.

Return/benefits of international diversification

- International investments offer more opportunities than purely domestic investments.
  - More than half of the world's market capitalization is outside of the U.S. (After NYC, the largest equity markets are in Tokyo, Shanghai, Hong Kong and Paris)
  - 75% of the Fortune 500 (largest MNCs) are located outside the U.S.
  - Certain products are only produced outside the U.S.
- International investments offer opportunities to diversify holdings
- *Basic rule of portfolio diversification* – the broader the diversification, the more stable the returns.
- The expanded universe of international securities implies the possibility of achieving a better risk-return tradeoff than investing solely in U.S. securities.
- While nearly 75% of investment risk can be eliminated in a fully diversified U.S. portfolio, all companies in a country are subject to the same cyclical economic fluctuations (*systematic* risk), which cannot be diversified away.
- By diversifying across countries whose economic cycles are not perfectly aligned (e.g., an oil price shock that hurts energy importers helps oil-exporting countries), investors may further reduce risk by diversifying away some of the national systematic risk.

International vs domestic diversification

FIGURE 16.4 The internationally diversified portfolio opportunity set



The addition of internationally diversified portfolios to the total opportunity set available to the investor shifts the total portfolio opportunity set to the left, providing lower expected risk portfolios for each level of expected portfolio return.

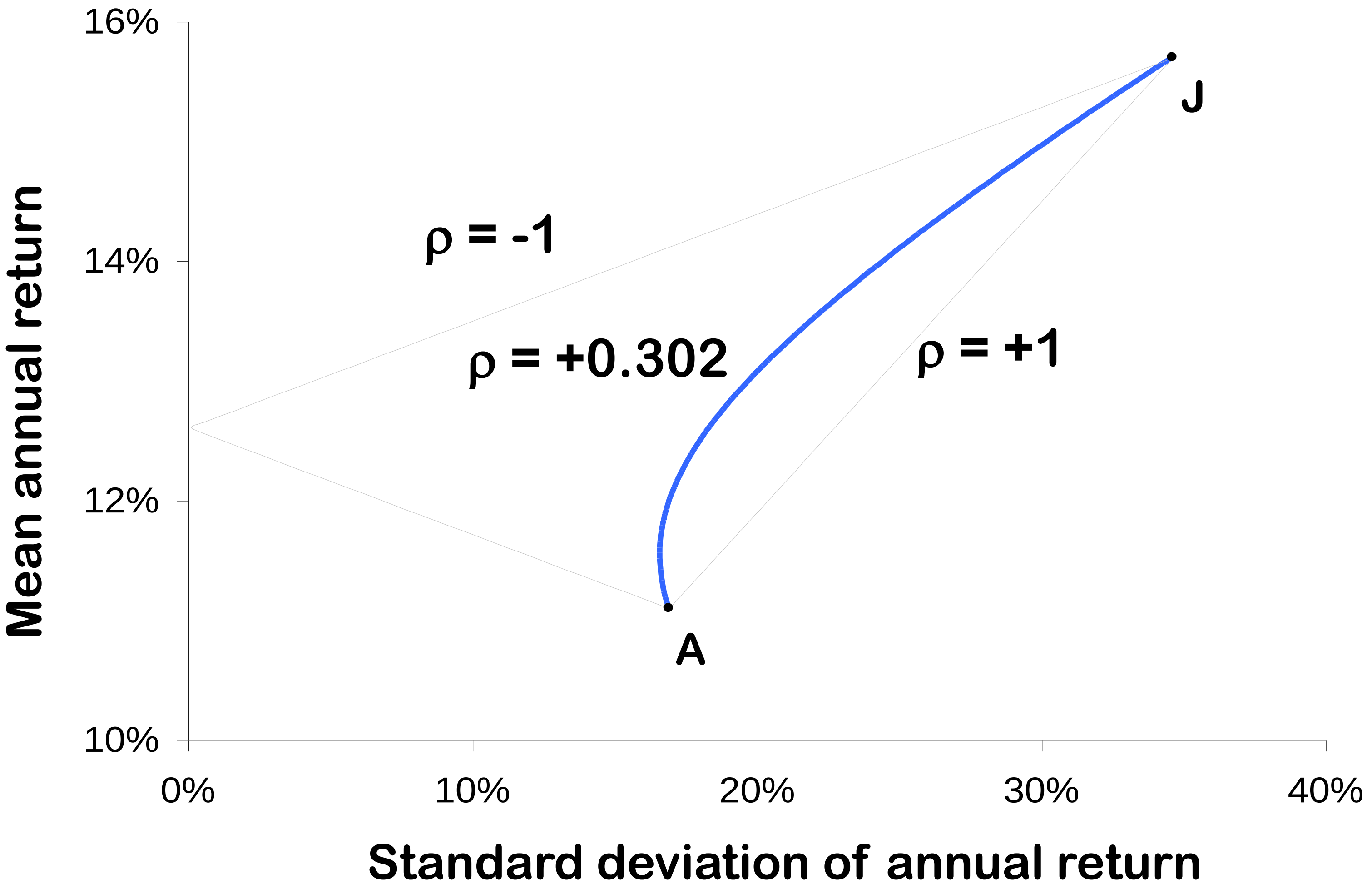
Potential for…

- higher returns
- lower portfolio risk

The limits of (international) diversification

- As number of assets becomes very large, portfolio standard deviation does not approach zero
- Fama (1976) finds that portfolio risk falls very slowly after about 12-16 shares are included
- Most risk-reduction benefits from diversification can be achieved in a 15-20 asset portfolio
- Not all risk can be diversified. Common/systemic risk cannot be diversified away, hence, investor gets the reward for that. Asset specific risk can be diversified
- The extent to which risk is reduced by portfolio diversification depends on the correlation of assets in the portfolio. Higher correlation between assets limit the benefit of diversification
- As the number of assets increases, portfolio variance becomes more dependent on the covariances (or correlations) and less dependent on individual assets' variances.

Asset correlation and diversification



If r = +1.0, no risk reduction is possible

If r = –1.0, complete risk reduction is possible

If r = 0, risk is less than some relationship

Construction of a portfolio

- The efficient frontier (Mean-variance efficiency)
  - Given risk-aversion, each investor will try to secure a portfolio on the efficient frontier. The efficient frontier is determined on the basis of dominance.
- Dominance implies that a portfolio is efficient if:
  - No other portfolio has a higher return for the same risk, or
  - No other portfolio has a lower risk for the same return.

Calculating Risk and return from diversification

Estimating the benefits of an international portfolio

- Let α be the fraction invested in U.S. stocks and $1 - \alpha$ be the fraction invested in foreign stocks.
- Let R_{usa} be the expected return on U.S. stocks and R_{world} be the expected return on international stocks ex. the U.S.
- Let σ_{usa} the standard deviation on U.S. stocks and σ_{world} be the standard deviation on international stocks ex. U.S.
- Let $\rho_{usa,\, world}$ be the correlation between returns on the U.S. and international portfolios

Return on an international portfolio (r_p):

$$R_p = \alpha R_{usa} + (1-\alpha) R_{world}$$

Risk on an international portfolio (σ_p):

$$\sigma_p = \left[\alpha^2 \sigma_{usa}^2 + (1-\alpha)^2 \sigma_{world}^2 + 2\alpha(1-\alpha)\sigma_{usa}\sigma_{world}\rho_{usa,world}\right]^{\frac{1}{2}}$$

Example will be done in the seminar

Return from foreign portfolio investing in stocks/bonds*

- **Total dollar return can be divided into 3 elements:**
  - Dividends/interest income
  - Capital gains/losses
  - Currency gains/losses
- **One-period total dollar return R$ on stock is computed as:**

$R_\$$ = Foreign Currency Return x Currency Gain/Loss - 1

$$R_\$ = \left[1 + \frac{P_{t=1} - P_{t=0} + DIV}{P_{t=0}}\right](1 + g) - 1$$

$P_{t=1}$ *and* $P_{t=0}$ foreign currency stock price (now, $P_{t=1}$) vs (previous, $P_{t=0}$)

DIV = dividend income in foreign currency

g = percent change in dollar value of foreign currency

*Return from foreign portfolio investing in bonds can be calculated using the same formula with different notations

Example: América Móvil of Mexico

- América Móvil is one of the largest telecom companies in the world with over $50 billion in annual sales. It is traded on the BMV *(Bolsa Mexicana de Valores)* exchange in Mexico City and on the NYSE in the name of AMX.
- The company's operations were hurt by the COVID-19 pandemic in the first quarter of 2020: The operator registers its income in Mexican pesos, but generates a lot of its international debt in US dollars and Euros. Given that the peso has slumped since the start of the year – and that attempts to contain the outbreak of Covid-19 have escalated – America Movil's financing expenses in Q1 were inflated, significantly impacting its bottom line.
- The stock price went from MXN15.33 to MXN14.05 from January 2, 2020 to March 31, 2020 (Q1 2020). The company makes a dividend payment of MXN0.19 in June and November.
- The MXN/USD went from 18.9 to 23.4 from January 2, 2020 to March 31, 2020. What was the dollar return on América Móvil?

$$R_\$ = \left[1 + \frac{P_1 - P_0 + DIV}{P_0}\right](1+\mathrm{g}) - 1 = \left[1 + \frac{14.05 - 15.33 + 0}{15.33}\right]\left(1 + \frac{18.9 - 23.4}{23.4}\right) - 1$$

$$R_\$ = \underbrace{(1 - 8.35\%)}_{\text{Loss of 8.35\% on AMX}}\underbrace{(1 - 19.23\%)}_{\text{Depreciation of the peso by 19.23\%}} - 1 = -25.97\%$$

The shifting sands of portfolio analysis

- Expected returns & variance (risk premia) vary over time
  - Expected returns are low (high) when economic conditions are strong (weak)
- Financial market volatilities vary over time
  - Volatility also varies over time
  - Conditional volatility models such as GARCH can capture time-varying volatility
- Cross-country correlations also vary over time
  - Observed correlations are higher during market downturns or turmoil period than the tranquil period (called contagion, i.e., domino effect)

Home bias puzzle

- **Portfolio theory can argue either way**
  - Tilt toward more domestic assets
    - Domestic stock portfolios can hedge domestic inflation risk [Adler and Dumas (1983), JF]
    - Financial institutions with domestic liabilities have an incentive to hold domestic assets [Griffin (1997), JPM]
  - Tilt toward fewer domestic assets
    - Labor income is highly correlated with other domestic assets, so investors' portfolios of tradable assets (that have less correlation with labor income) should be tilted away from domestic assets [Bexter and Jerman (1997), American Economic Review]

Home bias puzzle

- **Violations of the perfect market assumptions**
  - Market frictions
    - Government controls
    - Taxes
    - Transactions costs
  - Investor irrationality
  - Unequal access to market prices
  - Unequal access to information

Investor irrationality

- Heuristics (decision rules)
  - While heuristics can simplify decision-making, they also lead to cognitive biases
- Frame dependence
  - The form in which a problem is presented can influence decisions
    - ❑ Most individuals are ***risk-averse*** when facing profits/gains
    - ❑ Many individuals are ***risk-seeking*** when facing losses.
- Overconfidence
  - ❑ lead to higher trading volume [Barber and Odean (2001), QJE]
  - ❑ men trade 50% more than women
  - ❑ excessive trading – men lose 2.65% and women lose 1.72% of returns per year
- Regret avoidance
  - ❑ tendency to hold losers & sell winners [Shefrin and Statman (1985), JF]
    - ❑ *…sell Winners Too Early and Ride Losers Too Long*

Unequal access to information

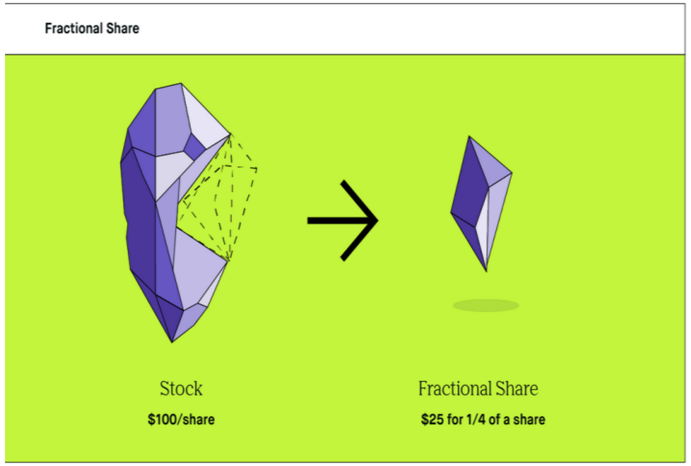
- It is difficult to get and interpret information from distant markets
- Once invested, it is difficult to monitor the actions of distant managers

Empirical evidence

- Individuals prefer investments that are culturally similar and geographically nearby [Coval and Moskowitz (1999), JF]

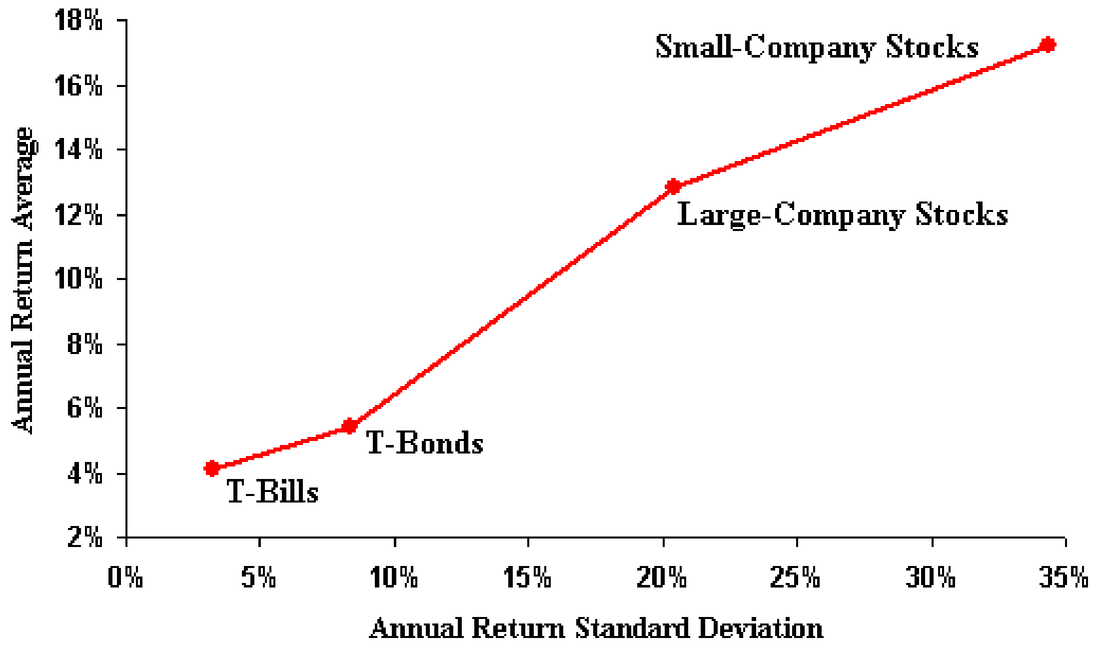
Choosing assets: What to consider

- Valuation: undervalued or overvalued?
- Market timing/trend: Bullish, bearish or crisis markets?
- Asset diversity: small, medium or large, or even Penny or Fractional stock?
- Growth stocks?
- Sectoral or regional diversity or home bias
- Investment duration: short-term or long-term
- Analyse, analyse and analyse



Large vs. small stocks

- Eun et al (2008, JFQA)
  - large-cap stocks receive dominant allocation. However, returns to large-cap stocks or stock market indices co-move, limiting diversification benefits
  - In contrast, small-cap stocks and stocks of locally oriented companies do not. So gains are higher for these stocks



MAF306: Topic 4

Country Risk Management
Debt Crises

Dr. Sohel Azad

Important announcements

- Groups have been formed, no group swaps are allowed at this stage except permitted by the UC for valid reasons
- A zoom session will be run on Wednesday (27th March) 5pm through seminar zoom link to briefly explain on how to proceed on the assignment. You are more than welcome to join for any specific questions that you might have or listen to the recorded session.
- Country names will be sent to you in a separate e-mail.

Learning resources

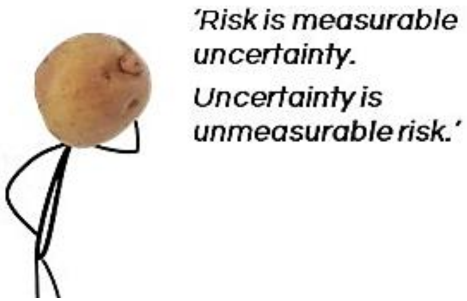
1. Lecture and seminar materials
2. Shapiro Ch 6/Ch14
3. Support materials in cloud Deakin

Learning Objectives

- Country risk
  - Overview of risks faced by MNCs
  - Definition, factors, indicators of country risk & economic health
  - Understand the assessment/measurement of country risk
  - Outline strategies for managing country risk
- Debt crises
  - An overview of recent debt crises
  - Resolving debt crisis: Review of debt management strategies

Overview of risk

- Risk: what do you think (uncertainty?)
- The probability/chance that an actual outcome of a future event may be different to the expected outcome (sales, profit, equity premium, etc….)
- Examples of Risk Disasters: Long Term Capital Management, **Barings Brothers** or Barings Bank, Allied Irish Bank, Celsius (crypto) etc.



Mostly attributed to Rogue Traders (see a recent Australian story)

Overview of risk

Risks that need to be measured and managed by MNC

- *Interest rate risk*.

  The exposure to changes in interest rates. A movement in interest rates will affect the value of financial assets and liabilities and impact upon associated cash flows.

- *Foreign exchange risk.*

  The risk that the value of assets and liabilities denominated in foreign currencies will be affected by changes in exchange rates and that associated cash flows will also change.

- *Liquidity risk.*

  The risk that an organisation will not have sufficient cash and/or liquid assets to meet its day-to-day operating

- *Credit risk.*

  The risk that principal, interest or credit due to the organisation will not be repaid. Non-receipt of expected future cash flows will also impact upon future liquidity.

Overview of risk

– *Capital risk*.

The ability of the organisation to raise capital as required to meet its future growth prospects. Also includes the return on equity expectations of shareholders and investors.

– *Investment risk*.

Directly related to interest rate risk. The ability of treasury to maximise return on financial assets, particularly within the securities portfolio.

– *Cybersecurity risk.*

MNCs handle large amounts of sensitive data, such as confidential information about their operations, customers, and employees. They must invest in cybersecurity measures to protect their data and systems.

– *Country risk*.

The risk that economic, political or social conditions will change in another jurisdiction such that the value of assets or liabilities and associated cash flows will be adversely affected.

International Country Risk

Country risk:

defined as the performance variance arising due to any unexpected changes in the political, economic and financial environment of a host country.

Thus, risks are diverse and could emanate from changes in:

- Political environment
- Economic environment
- Financial environment

Country Risk Analysis

- Country risk analysis (CRA) is the assessment of the risks and rewards associated with making investments and doing business in a country.
- Nonbank MNCs analyze country risk to determine the safety of their investments from political interference and economic disruptions.
- Banking MNCs analyze to determine which countries to lend to, the currencies in which to denominate loans etc.
- Concerned with sovereign risk: the inability of governments to service their debts.

International Country Risk

Political / Sovereign risk

Host government will unexpectedly change the ***rules of the game*** under which businesses operate

Political risk includes…

- External political risk
  - Exposure to loss as a result of war
  - Terrorism
- Internal or Social political risk
  - civil war
  - racial, religious riots
  - ethnic polarizations/revolution

International Country Risk

Examples of sovereign/political risks

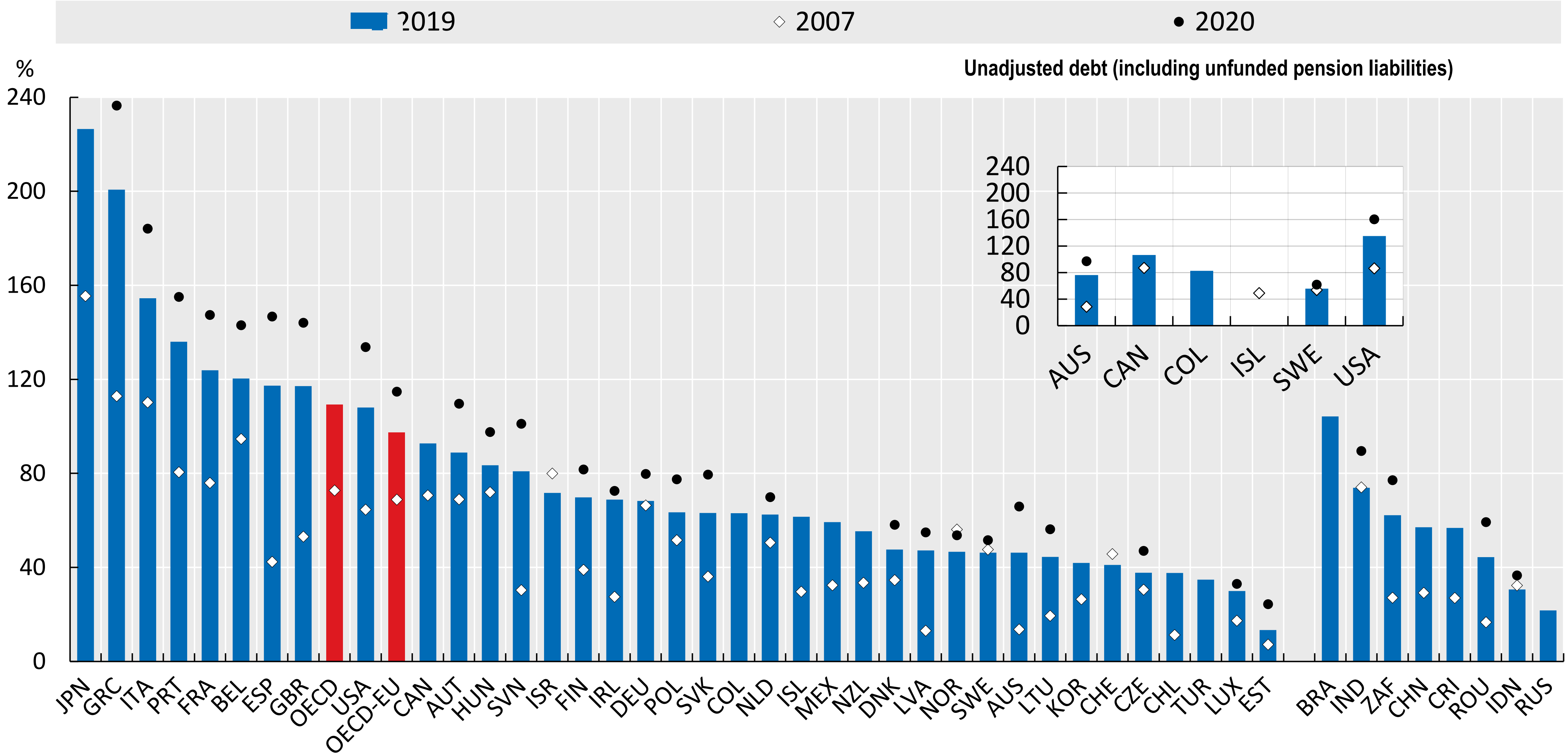
- Govt as direct debtor or guarantor may repudiate foreign debt eg. Mexico 1982, Greece 2010, US debt crisis 2011 (credit rating down)
- Expropriation risk
- Disruptions in operations – Govt changes to tariffs, changes to tax laws, labour laws and contracts
- Protectionism - local content requirements
- Blocked funds – Forex and capital controls on capital outflow
- Loss of intellectual property rights
- Asset freezes
- Nationalization of industries – Venezuela
- Unfavourable fixed ER – where ER is fixed at too low/high a value e.g., China
- Corruption – ‘rent-seeking’

International Country Risk

Financial risk and Economic Risk

- Refers more generally to unexpected events in a country's financial, economic, or business life/opportunities/environment,
- Financial risk is more related to the ability to service debt, whereas economic risk is more related to general macroeconomic conditions
  - Deterioration in exports prices vs import prices
    - deteriorating 'terms of trade'
  - Increasing production costs
  - Increasing energy prices (oil)
  - Capital flight
  - Fiscal irresponsibility: budget deficit (see next slide)
  - Macroeconomic instability: inflation, interest rate, GDP growth, unemployment etc.

Fiscal irresponsibility: Govt debt as % of GDP



Source: OECD

Measurement of Country Risk

Various Methods to Measure Country risk

Business Environment Risk Intelligence (BERI)	ICRG: International Country Risk Guide (http://www.prsgroup.com/ICRG.aspx):	Political Risk Services (PRS)	Control Risks' Country Risk Forecast
Eurasia Group Stability Index	Economist Intelligence Unit	Euromoney	Institutional Investor
Standard & Poor's Rating Group	Moody's Investors Services	Transparency International	World Bank

Business Environment Risk Intelligence (BERI)

Low Risk (70–100)	POR Combined Score	High Risk (40–54)	POR Combined Score
1 Switzerland	82	26 Czech Republic	49
2 Singapore	79	27 Italy	49
3 Netherlands	75	28 Thailand	49
4 Japan	74	29 Russia	48
5 Norway	74	30 South Africa	48
6 Taiwan (R.O.C.)	73	31 Kazakstan	47
7 Germany	71	32 India	46
8 Austria	70	33 Egypt	45
		34 Hungary	45
Moderate Risk (55–69)		35 Argentina	44
9 Belgium	66	36 Brazil	44
10 Finland	65	37 Iran	44
11 Sweden	65	38 Philippines	44
12 United States	64	39 Poland	44
13 Canada	63	40 Vietnam	43
14 Ireland	63	41 Greece	42
15 Denamrk	62	42 Peru	42
16 France	62	43 Venezuela	42
17 China (P.R.C.)	61	44 Indonesia	41
18 United Kingdom	60	45 Mexico	41
19 Malaysia	59	46 Turkey	41
20 Australia	58	47 Ukraine	41
21 Korea (South)	58	48 Colombia	40
22 Spain	58	49	
23 Chile	55	**Prohibitive Risk (0–39)**	
24 Portugal	55	50 Pakistan	38
25 Saudi Arabia	55	51 Romania	37

Source: Business Environment Risk Intelligence. Reprinted with permission.

Assigning a score to the country risk:
Timokhina et al (2019) with cross-reference of Krasnov et al 2002 and Hollensen, 2008

<table>
<tr><th>Indicators</th><th colspan="2">Method of assigning a score to each indicator</th></tr>
<tr><td rowspan="4">1. Political stability
2. Economic growth
3. Labour cost/productivity
4. Short-term credit
5. Long-term loans/venture capital
6. Attitude towards the foreign investor and profits
7. Nationalization
8. Monetary inflation
9. Balance of payments
10. Enforceability of contracts
11. Bureaucratic delays
12. Communications (phone, fax, internet)
13. Local management and partner
14. Professional services and contractors</td><td colspan="2">Increase of the score, depending on the countries' position in the international ranking</td></tr>
<tr><td>In the 1st third of the ranking
In the 2nd third of the ranking
In the 3rd third of the ranking</td><td>+ 2
+ 1
+ 0</td></tr>
<tr><td colspan="2">Increase of the score, depending on the countries' position in relation to each other in the international ranking</td></tr>
<tr><td>The 1st place among three countries
The 2nd place among three countries
The 3rd place among three countries</td><td>+ 2
+ 1
+ 0</td></tr>
<tr><td rowspan="2">15. Currency convertibility</td><td colspan="2">Increase of the score, depending on the currency convertibility degree in the country</td></tr>
<tr><td>Fully convertible currency
Partially convertible currency
Inconvertible currency</td><td>+ 4
+ 2
+ 0</td></tr>
</table>

Measurement of Country risk

- ICRG - International Country Risk Guide:

(The International Country Risk Guide (ICRG) | PRS Group)

1. Political risk component
2. Economic risk component
3. Financial risk component

Political risk components

1. Government stability (12)
2. Internal conflicts (12)
3. External conflicts (12)
4. Corruption (6)
5. Investment profile (12)
6. Military in politics (6)
7. Religious tension (6)
8. Law and order (6)
9. Ethnic tension (6)
10. Democratic accountability (6)
11. Bureaucracy quality (4)
12. Socio eco conditions (12)

Sum of these components add up to 100 points and can be stratified as follows:

Below 50 - Very high political risk

50 – 59.9 - High risk

60 - 69.9 - Moderate risk

70.79.9 - Low risk

80 – 100 - Very low risk

Economic risk components

1. GDP per capita – (5 points)
2. Real GDP growth – (10 points)
3. Annual inflation rate – (10 points)
4. Budget balance as a percentage of GDP – (10 points)
5. Current account as a percentage of GDP – (15 points)

Other:

- interest rates
- unemployment rate
- money supply expansion (printing money)

Sum of these components add up to 50 points and can be stratified as follows:

Below 25 - Very high risk

25 – 29.9 - High risk

30 - 34.9 - Moderate risk

35 - 39.9 - Low risk

40 – 50 - Very low risk

Financial risk components

1. Foreign debt as a percentage of GDP (10 points)
2. Foreign debt as a percentage of exports of goods and services (10 points)
3. Current account as a percentage of exports of goods and services (5 points)
4. Net international liquidity as months of import cover (15 points)
5. Exchange rate stability (10 points)

Sum of these components add up to 50 points and can be stratified as follows:

Below 25 - Very high risk

25 – 29.9 - High risk

30 - 34.9 - Moderate risk

35 - 39.9 - Low risk

40 – 50 - Very low risk

Other measures of country risk

Debt-Service Ratio

- Debt service to export earnings - signals likelihood of inability to pay
- Debt service amount - contractually **fixed** interest and principal payments
  - Export earnings fluctuate/volatile
  - Short-Term fluctuations in export price $\rightarrow$ liquidity problem
  - Long-Term prices permanently depressed or loss of market for export $\rightarrow$ insolvency problem

Import Cover

- Ratio of official reserves to imports
  - Measures the % of annual imports which can be met by reserve holdings

Structural Measure

- Emphasizes specific component structure of BOP
  - Interest and income paid to foreign residents
  - Level of exports relative to imports (type of imports)
  - Capital inflow to finance borrowing

Common characteristics of high country risk

- High govt. deficit to GDP
- Expansionary monetary policy combined with a fixed FX rate
- High govt. spending
- Price controls, interest rate ceilings, trade restrictions and rigid labor laws
- High tax rates affecting incentives to work, save, and invest
- High level of state-owned firms
- Public-sector spending and regulations
- Pervasive corruption that impedes development, discourages foreign investment
- The absence of basic institutions

Common characteristics of low country risk

- A structure of incentives that rewards risk-taking in productive ventures
- A legal structure that stimulates the development of free markets
- Minimal regulations and economic distortions
- Incentives to save and invest
- An open economy
- Stable macroeconomic policies

Managing/reducing Country Risk

1. Negotiate the environment with the host country prior to investment
   - A) The investment environment
   - B) The financial environment

2. Structure foreign operations to minimize country risk while maximizing return

3. Obtain political risk insurance
4. Considering FX risk management tools/derivatives

Managing/reducing Country Risk…cont.

1A. Negotiate the investment environment

- Taxes – taxable bases, tariffs, tax breaks
- Labour issues – labour laws
- Concessions – access to restricted markets, restriction of import-competing firms
- Obligations and restrictions – required infrastructure programs eg. Roads, hospitals etc
- Provisions for planned investment divestiture – build-operate-transfer project to govt after pre-specified time
- Performance assurances and remedies against expropriations, delays
- International arbitration of disputes

Managing/reducing Country Risk…cont.

1B. Negotiate the financial environment

- Rules governing:
  - Cash flow remittance,
  - transfer prices,
  - royalties,
  - loan repayments,
  - dividends etc
- Access to capital markets
- Subsidized financing
- Corporate governance environment
  - Host country restrictions on ownership
  - Performance assurances and remedies
  - International arbitration of disputes

Managing/reducing Country Risk…cont.

2. Structure foreign operations to minimize risk while maximizing return

- Limit the scope of technology transfer to foreign affiliates to include only non-essential parts of the production process
- Limit dependence on any single partner
- Enlist local partners
- Use more stringent investment criteria
- Plan for disaster recovery

Managing/reducing Country Risk…cont.

3. Obtain political risk insurance

Insurable risks include -

- Expropriation risk, due to
  - war
  - revolution
  - insurrection
  - civil disturbance
  - terrorism
- Repatriation restrictions
- Currency inconvertibility

Political risk insurers

- Government export credit agencies
  - U.S. Overseas Private Investment Corporation (OPIC)
  - U.K. Export Credits Guarantee Department
- International
  - U.S. Overseas Private Investment Corporation
  - World Bank
  - Multilateral Investment Guarantee Agency
- Private
  - Lloyd's of London
  - American International Group (AIG)

Managing/reducing Country Risk…cont.

4. Considering FX risk management tools

FX derivatives include but not limited to

- Contractual hedging
  - Futures, forwards, options, swaps
- Natural hedging
  - Risk sharing, matching/exposure netting, Lead-lag,
- Partial hedging

International Debt Crises

International Debt Crises

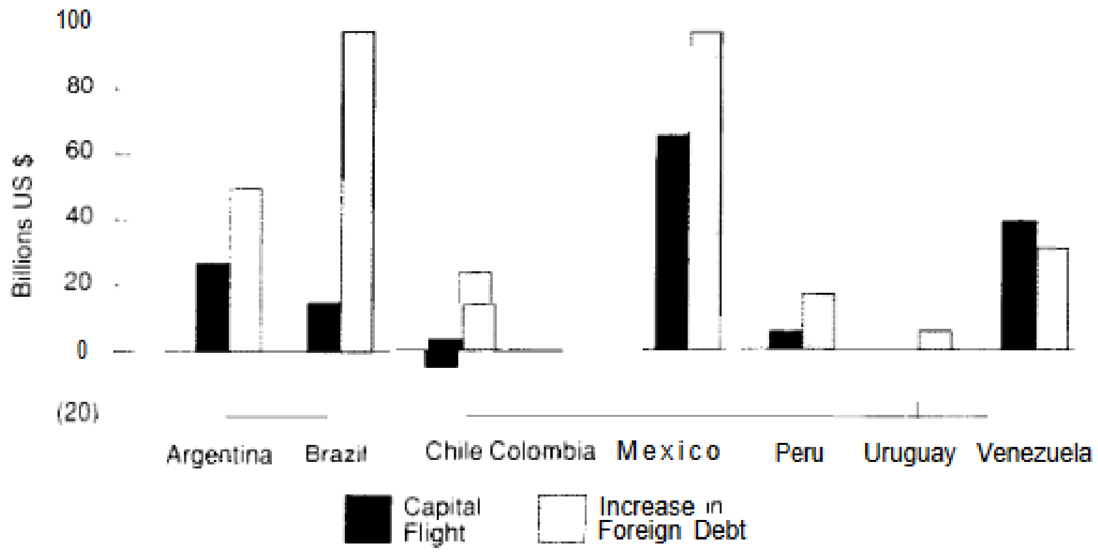
Major debt and financial crises

- Latin American Debt Crisis and Capital Flight 1983-87
- Mexican peso crisis of 1995
- Asian currency/financial crisis of 1997
- Russian ruble crisis in 1998
- Argentinian peso crisis of 1998
- Turkish lira crisis 2001 and 2005; 2018-2022
- PIIGS: Euro crisis 2010 ongoing

Contributing factors in those crises:

- A fixed or pegged exchange rate system leading to overvalued local currency
- A large amount of foreign currency debt
- Loose monetary policy and housing bubble & burst
- Excessive govt. spending; Deficit budget
- Trade deficit/imbalance
- Rogue speculators

Latin American Debt Crisis and Capital flight: 1973-1987



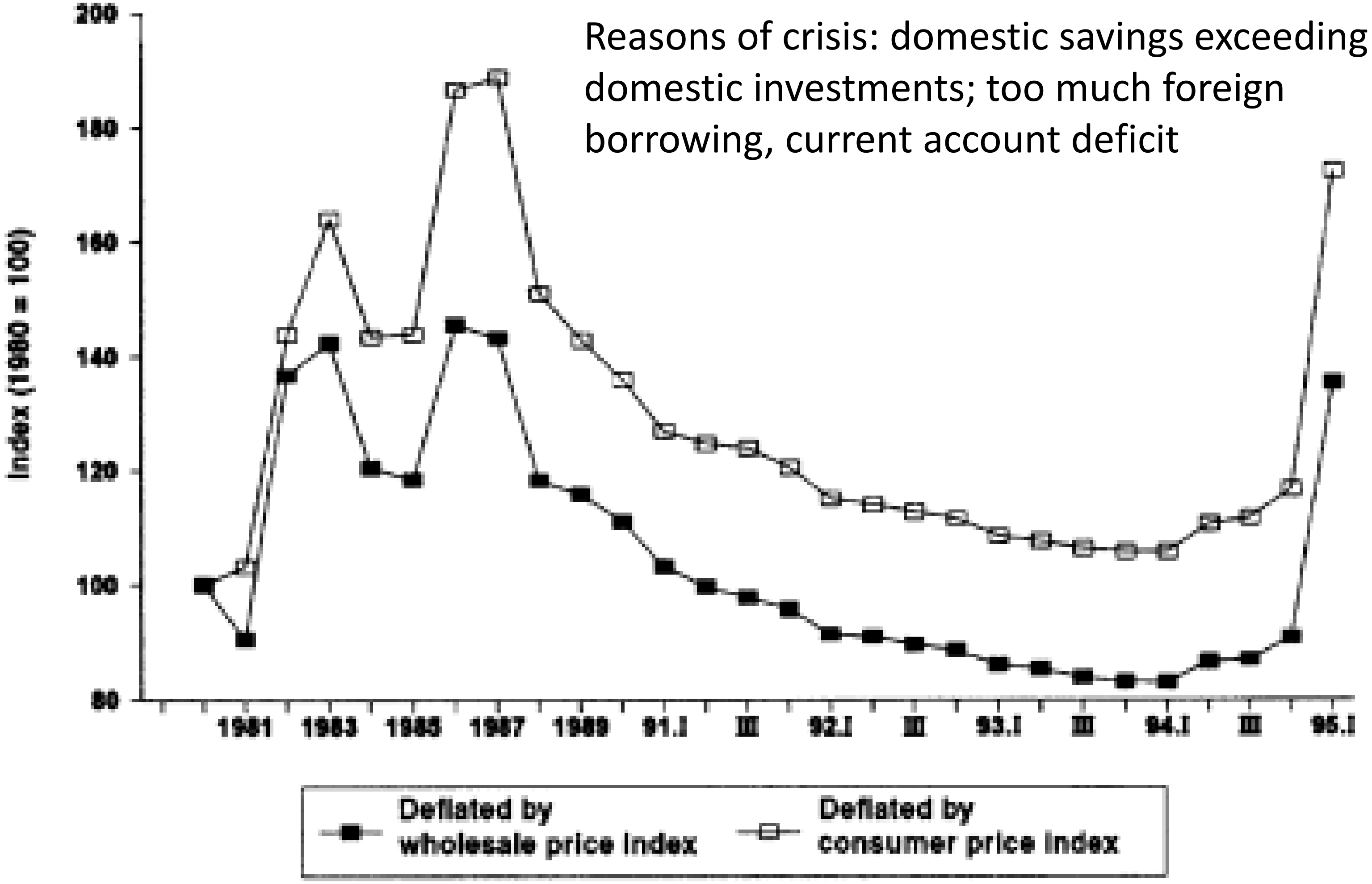
Common reasons/factors

- Inappropriate economic policies with unfavorable investment climate
- Discriminatory treatment of local and foreign investors
- Local investors deprived of government's explicit guarantees of loans by international banks
- Elite local investors taking loans from overseas
- Loan pushing: International banks encouraging oil exporting countries to supply funds to Latin America and Third World
- Removal of capital control

Source: Pastor, Capital Flight and the Latin American Debt Crisis, 1987
https://files.epi.org/page/-/old/studies/capital_flight-1989.pdf

Mexican peso crisis

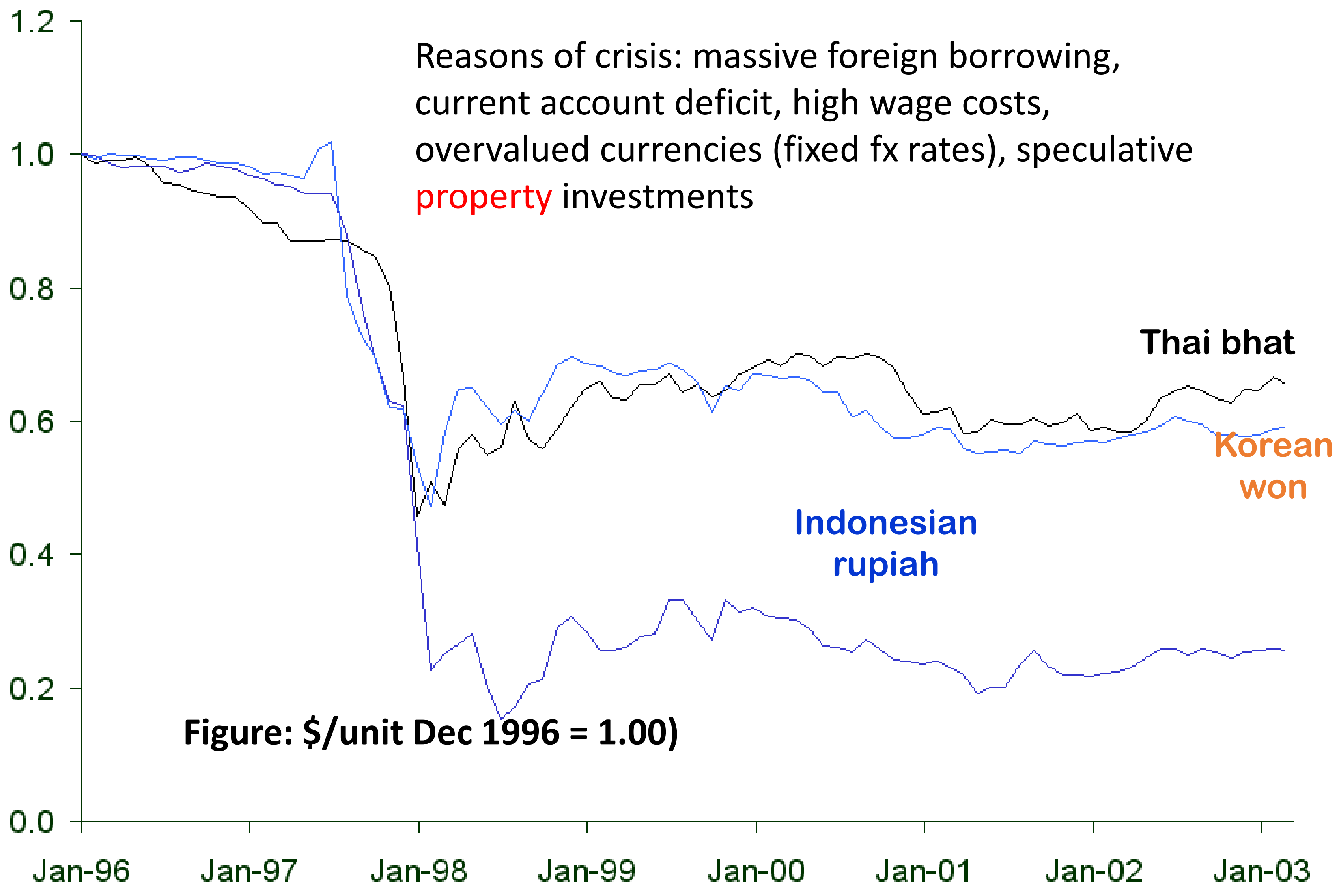
Reasons of crisis: domestic savings exceeding domestic investments; too much foreign borrowing, current account deficit



Source: Banco de Mexico.

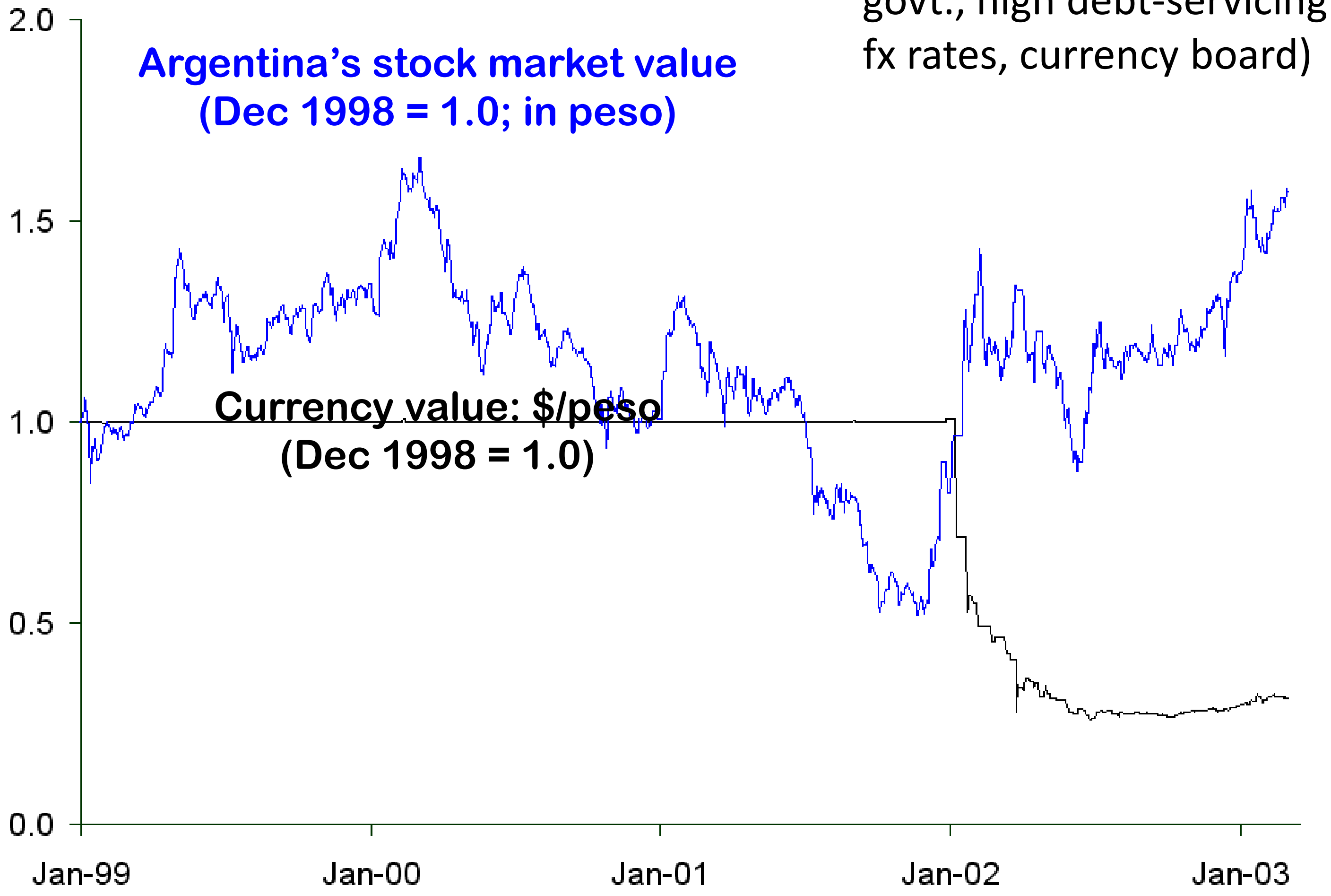
Figure: Real exchange rate

Asian currency crisis



Argentina's currency crisis

Reasons of crisis: deficit budget financed by foreign govt., high debt-servicing costs, overvalued peso (fixed fx rates, currency board)



Recent Major Crises

Global Financial Crisis

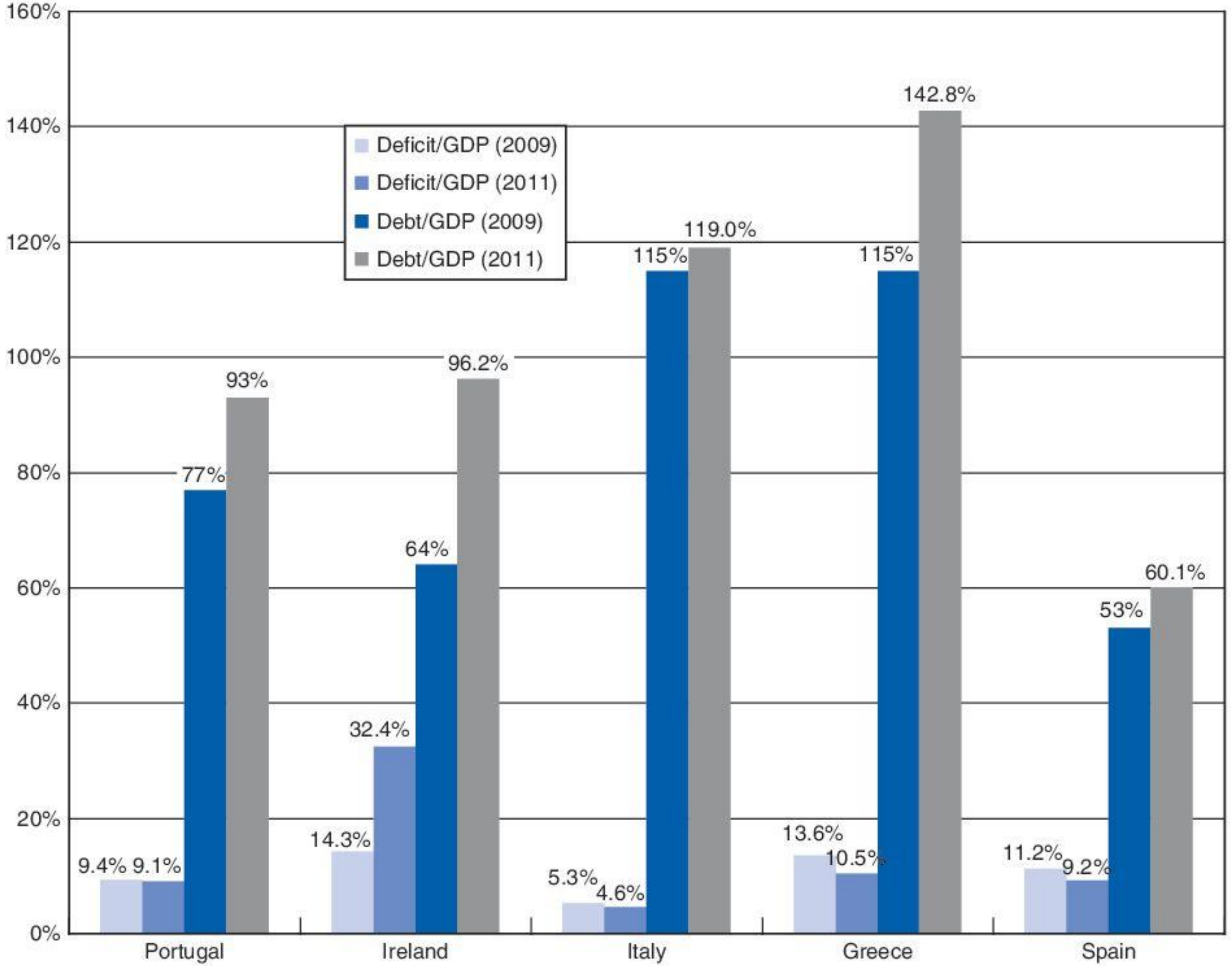
1. Reasons of crisis:
   - Series of events: Loose monetary policy (early 2000s), particularly by the U.S. Federal Reserve, and global imbalances, cheap credit together with the easy availability of funds caused bubble in real estate prices in the U.S. and contagion to a number of countries & financial markets.
   - The housing market began to falter in late 2005, with increasing signs of collapse throughout 2006.
   - Many other factors such as subprime mortgages, weak regulatory structures, high leverage and moral hazard (rescuing a firm in the case of bankruptcy - Lehman brothers) exacerbated the effects of the crisis.
   - Banks limited access to funding to even the lowest indebted firms – liquidity crisis
2. Potential remedies
   - Change in the governance structure of central banks, restricted monetary policy?
   - Measures to reduce global imbalances,
   - Changes in banking regulation, etc
   - Allow the firms to fail than guaranteeing a bail out?

[Inside Job (2010 Full Documentary Movie) - YouTube]

Recent Major Crises

- US debt crisis 2011:
  - US *sovereign credit rating* downgrade by S&P (from AAA to AA+, Moody's first gave AAA in 1917)
  - Reasons/Sources: Structural problem; Transmission from Europe?
- PIIGS debt crises
  - Portugal, Italy, Ireland, Greece, and Spain were hit hard by GFC
  - High level of public and private debt, bursting of housing bubble
- Sri Lankan debt crisis: a crisis in the era of geopolitical competition
  - China holds around 20% of the sovereign default loan
  - https://www.bbc.com/news/business-61505842

Swine Flu: PIIGS in Trouble
Greece debt-GDP turned out to be 180% (2015)



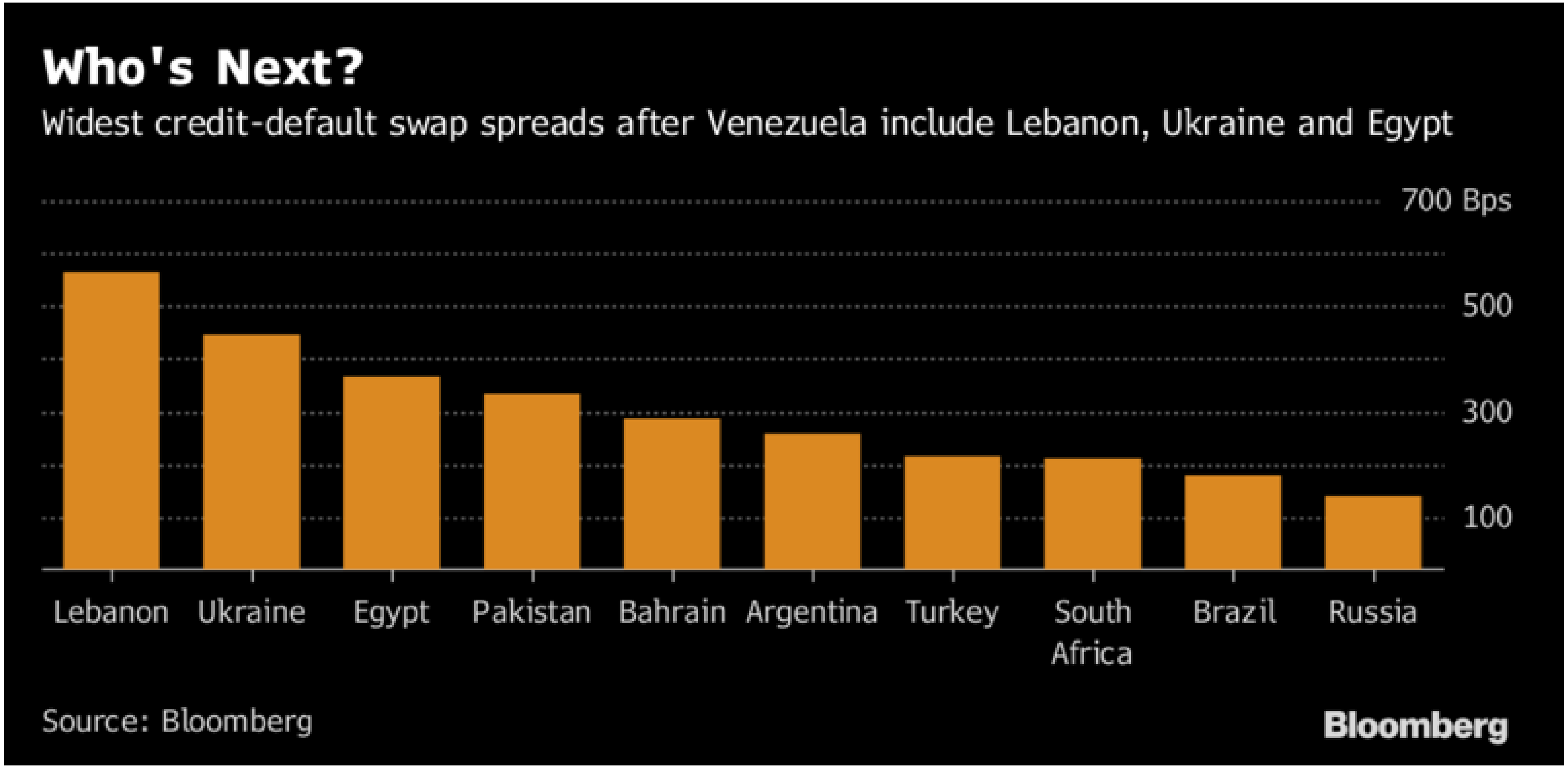
Serial debt defaulters

Ecuador (1832, 1868, 1911, 1931, 1982, 1999, 1914, 2008, 2020)

Argentina (1830, 1890, 1982, 2001, 2020)

Paraguay (1827, 1874, 1892, 1920, 1932, 1986, 2003)

Recent debt defaulters and CDS spreads



Highest CDS spreads after the Venezuela default (Nov 17, 2017).
In 2020, Lebanon and Argentina Defaulted

Resolving crisis: New Mexican Model?

EXHIBIT 6.6 MEXICO'S ECONOMIC POLICIES: THEN AND NOW

Old Mexican Model	New Mexican Model
Large budget deficits	Fiscal restraint
Rapid expansion of money supply	Monetary discipline
Nationalization	Privatizations
Restricted foreign direct investment	Attract foreign direct investment
High tax rates	Tax reform
Import substitution	Trade liberalization
Controlled currency	End currency controls
Price and interest rate controls	Prices and interest rates set by market
Government-dominated economy	Reduced size and scope of government

[8] Alvaro Vargas Llosa, "The Return of Latin America's Left," *New York Times*, March 22, 2005, http://www.independent.org/newsroom/article.asp?id=1483.

Resolving crisis: Use Debt Management Strategies –
to stabilize the economy and promote economic growth

Short-term strategies

- ↓ budget deficits
- cut Gov't expenditure & ↑ taxes - Austerity
- ↓ inflation
- tighten monetary policy & ↑ i/r*
- ↓ Demand for imports (impose trade restrictions – quotas, tariff barriers)*

Long-term strategies

- ↑ level of export
- ↑import-competing production
- Link repayment schedules (i/r & principal) to GDP or export earnings
- Devalue currency*

*** EU countries can't do!**

Debt Management Strategies…cont

Official Approach - IMF & Central Banks

- **PARIS CLUB**
  - Restructuring of official loans - maturity stretched and grace periods
  - Pressure applied to banks to lend NEW funds
  - Debtor countries must submit to IMF adjustment program

Private Response - Commercial Banks

- **LONDON CLUB**
  - Collective bank renegotiations and new loans to be provided by private sector lenders
  - Provided subject to the nation undertaking IMF Conditionality

Debt Management Strategies…cont

Debt-for-debt swaps

Exchange short-term debt (already in arrears) for long-term debt with a percentage of write-off (haircut) – extension/restructure

The new long-term debt may be backed by industrialized nation and/or supranational agency – e.g., EU

Direct cash buyback

The debtor country redeems the debt from the creditor at a deep discount.

- Need foreign reserve (hard currency) to implement
- Dampen future access to international capital market
- Barriers are syndicated loans where more than one creditor is involved in providing the finance.

Debt Management Strategies…cont

Debt-for-equity swaps

Banks sell their loans at a discount to debtor governments in debtor country currency, then use the local currency to purchase equity in the stock market of debtor country.

Advantages:

- ↓ debt burden for debtors – can retire hard currency loans for their own
- ↓ debt exposure for creditors

Disadvantages:

- Heavy discounting of the debt
- Loss of autonomy in debtor nations as infrastructure becomes increasingly owned by foreign investors.

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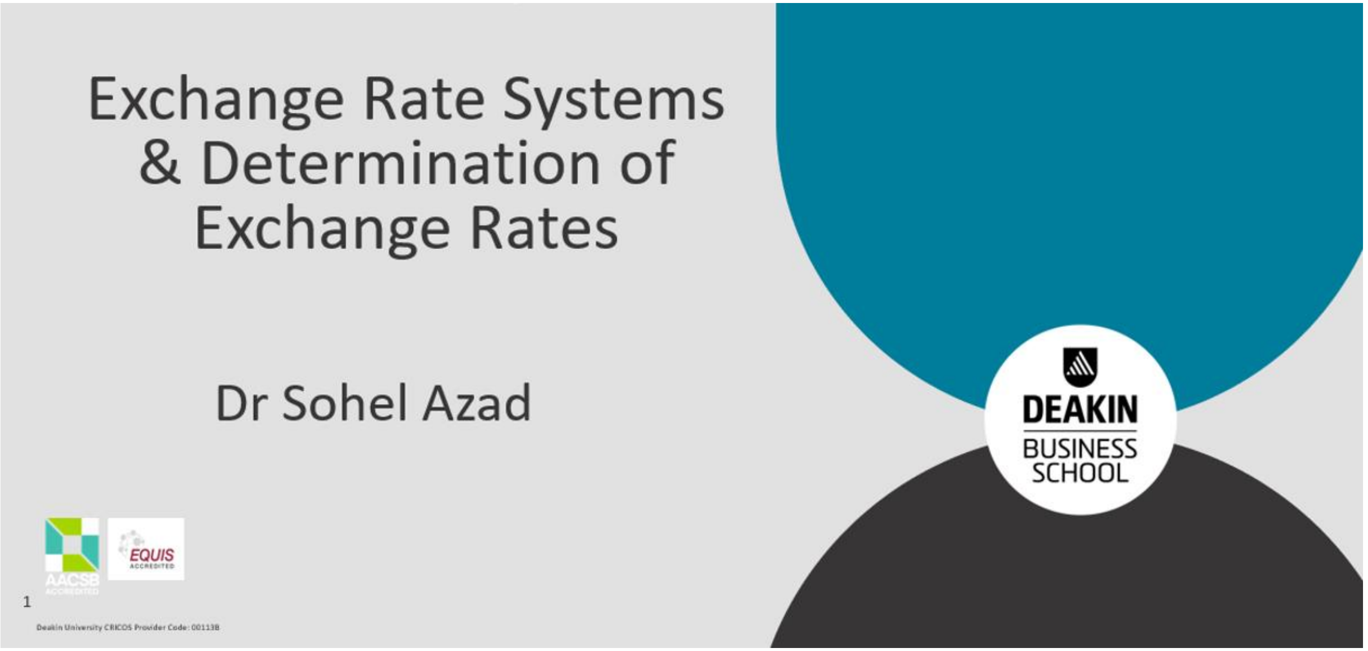


Learning Objectives

- Introduction: What are exchange rates?
- Compare between several alternative exchange rate systems and regimes
- Understand currency boards and dollarization – pros and cons
- Determination of exchange rates:
  - Equilibrium exchange rate
  - Factors affecting FX rates
  - Describe the motives and different forms and consequences of central bank intervention in the FX market
  - Evaluate the effectiveness of foreign currency exchange rate market intervention

Reading:

- Shapiro Ch 2, Ch 3
- Related materials in CloudDeakin site

Introduction

- What is an exchange rate?
  - Price of one currency in terms of another currency.
  - An exchange rate is a PRICE! Currency is a commodity!
- Examples of prices:
  - 1 Loaf of Bread = $3.95
  - 1 Gallon of milk = $3.79
  - 1 Pound of Beef = $14.99
  - 1 British pound = $1.65
  - 1 Swiss franc = $3.68
- Determined by *supply and demand, and many more factors!*
- Why is forex a problem? Volatility (risk)? How to manage this (i.e., forex or currency) risk? Will discuss in FX risk management

Classifications of Alternative Exchange Rate Regimes

Pegged (fixed) Exchange Rate

↓

Floating (flexible) Exchange Rate

- Exchange Arrangements (EU, ECCU, CFA Franc zone)
- Currency boards (Hong Kong, Argentina)
- Pegged rate within a horizontal band (Denmark, Iceland)
- Crawling pegs (Costa Rica, Nicaragua)
- Managed floats (Norway, Pakistan, Singapore and some previous USSR countries)
- Independently floating (Aust, US, UK, NZ)

https://www.imf.org/external/np/mfd/er/2004/eng/0604.htm

Exchange Rate Regimes

Fixed Rate System (e.g., Bretton Woods System, July 1944)

- rate determination
  - government committed to maintaining target rates
  - future value of currency is known with certainty
  - if rates threatened, central banks buy/sell currency to maintain stated value.
  - monetary policies coordinated – each member must accept the group's joint inflation as its own
    - Mundell's 'Impossible Trinity' (fx stability, financial integration and monetary independence)
- some government controls
  - on global portfolio investments
  - ceilings on granting of credit to foreign firms
  - import restrictions

Exchange Rate Regimes

Single currency peg

- Where the currency is formally fixed (by central bank) to a major currency
- e.g., Hong Kong dollar is pegged to the US$

Basket currency peg

- e.g., China's move of the Yuan, from a currency board to a crawling peg in 2005, followed immediately by the Malaysian Ringgit

❖ A pegged currency is set by supply and demand just as a floating currency is.

- How?
  - It is fixed by central banks trading in the FX market.
  - Central banks trade in large quantities thus holding supply and demand relatively constant.

Exchange Rate Regimes

Exchange arrangements with no separate legal tender

- Euro area where its member nations share a single currency (same legal tender), 19/27 countries in EU. There are other countries which use euro as their official currency (e.g., Andorra, Kosovo, Monaco)
- Members belong to a monetary and currency union, e.g., cooperative arrangements, but no cooperative fiscal policy
- Others
  - East Timor, El Salvador, Micronesia, Panama (US$)
  - African nations – CFA Franc zone
  - ECCU – East Caribbean Currency Union use the EC$ linked to the US$

Exchange Rate Regimes

Currency Board system:

- No central bank, No discretionary monetary policy – market forces determine money supply
- Compels government to follow responsible fiscal policy
- exchange currency at a fixed ER, combined with restrictions on the issue of money
- issues notes & coins, convertible on demand, at a fixed rate, into a foreign reserve currency
- domestic currency is backed by foreign currency
- reserves = high-quality, interest-bearing securities
- Argentina, China, and Malaysia have used Currency Boards
- Currently using – Brunei, Bulgaria, Hong Kong and Singapore

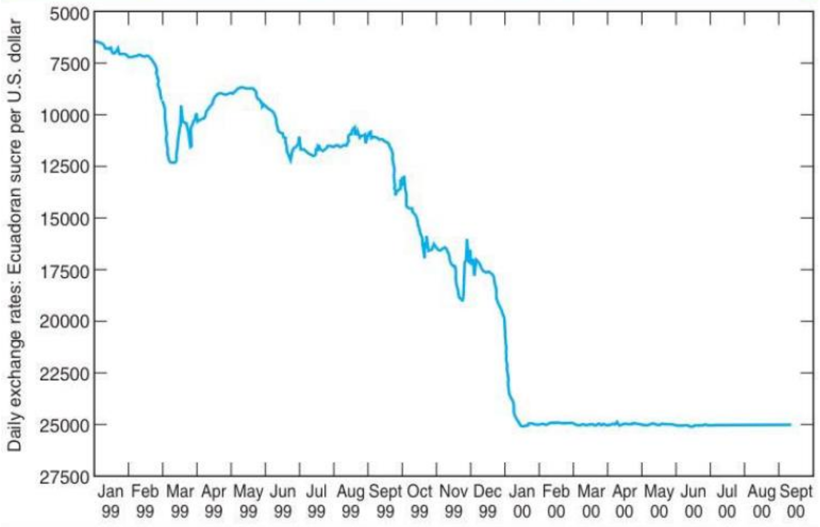
Exchange Rate Regimes

Dollarization

It is the complete replacement of the local currency with the currency of a foreign country e.g., Panama, El Ecuador and Liberia use the US dollar as their official currency.

- Downside of dollarization – loss of central bank's autonomy – however, better than monetary chaos !!
- No substitute for sound economic policies.
- May lead to ***stabilization*** of the foreign exchange market.

EXHIBIT 2.9 DOLLARIZATION STABILIZES THE SUCRE



Source: Pacific Exchange Rate Service, pacific.commerce.ubc.ca/xr/plot.html. © 2000 by Prof. Werner Antweiler, University of British Columbia, Vancouver BC, Canada. Time period shown in diagram: 1 Jan 1999–9 Sep 2000.

Exchange Rate Regimes

Target-Zone Arrangement

- Rate determination
  - market forces constrained to upper and lower range of rates
  - members to the arrangement adjust their national economic policies to maintain target
- Limited flexibility
  - The ECU before becoming the Euro in 1999
  - Where the currency is pegged at a fixed rate but the ER is allowed to fluctuate within a very small band of +/- 1% around the central rate
  - In EU case, margin got as wide as 15%!

Exchange Rate Regimes

Crawling Pegs

- Currency is adjusted periodically in small amounts at a fixed, pre-announced rate or in response to select indicators
- Greater flexibility with adjustments to indicators – e.g., adjusted in light of economic growth, inflation, unemployment, balance of payments, interest rates.

Exchange Rate Regimes

Freely Floating (Clean Float)

- forces influenced by:
  - market forces of supply and demand determine rates such as:
    - price levels
    - interest rates
    - economic growth
- without direct interference by government authorities:
  - rates fluctuate over time randomly
  - future value of currency is unknown

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Exchange Rate Regimes

Managed Float (Dirty Float)

- market forces set rates unless excess volatility occurs
- then central bank determines rate by intervening in FX market
- Central Bank interventions
  - Smoothing out daily fluctuations
    - Crawling Peg is a variant of this approach
  - 'Leaning against the wind'
  - Unofficial pegging

Exchange Rate Regimes

Current System is a hybrid system:

- major currencies use floating on a managed basis,
- some currencies freely floating, and
- others move in and out of various fixed-rate systems.

Exchange Rate Regimes

Arguments FOR Flexible Exchange Rate

- Smoother adjustment to external shocks, such as trade balance
- Less international reserves required.
- Central banks not required to defend inappropriate exchange rate.
- Autonomy of Fiscal Policy (FP) and Monetary Policy (MP).
- Market efficiency – ER reflects market fundamentals and expectations
- Efficient & inexpensive foreign exchange risk elimination available.

Exchange Rate Regimes

Arguments AGAINST Flexible Exchange Rate

- Increased volatility leads to increased prices, a reduction in world trade and a reduction in world Standard of Living
- Inherently inflationary - no external discipline on monetary and fiscal policy
- Destabilising speculation leads to **overshooting** of the exchange rate
- Misaligned exchange rate leads to incorrect decisions for resource allocation
- Small economies lack sophisticated financial markets to cover against exchange rate risk leads to a bias against foreign trade

Choosing an ER system depends on policy objectives and trade-offs

- Liquidity
- International competitiveness
- Balance of payments problems – e.g., USA deficit, demanding that China revalue the Yuan
- Economic stability – e.g., inflation objective, eco growth
- Exchange rate stability!!

The Impossible Trinity – The Trilemma

- Attributes of an ideal monetary system….
  - Exchange rate stability
  - Independent Monetary policy
  - Full capital market integration – free flow of capital

- Impossible to achieve all three – IMS (International Monetary System) must give up one of the 3 goals!

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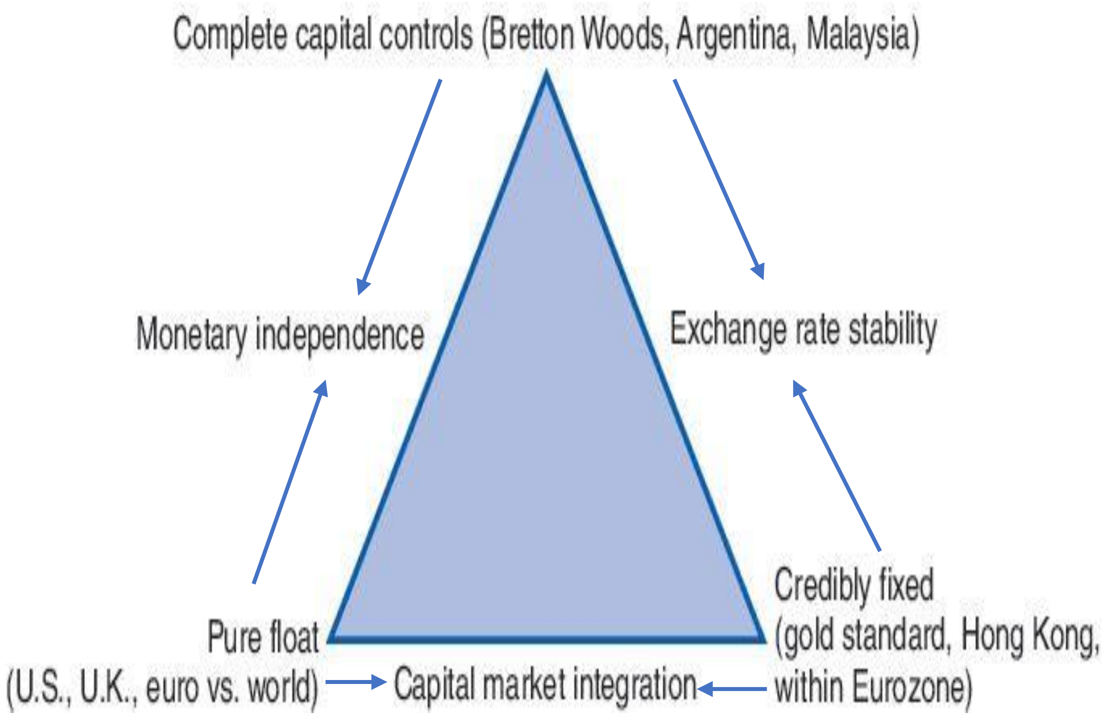
The Trilemma and ER Regime Choice



A country cannot be on all sides of the triangle, it must give up one of the attributes to achieve one of the 'states' described by the corners of the triangle

The Impossible Trinity – The Trilemma

- Choose pure floating exchange rate eg. USA, capital market integration and monetary independence but will lose ER stability – can't have a fixed ER!
- Choose Exchange Rate stability eg. Hong Kong, can get capital market integration – can't have monetary independence
- China chooses both monetary independence and ER stability – must impose capital controls
- Countries sitting inside the triangle – try to juggle all 3 – China and India
- Ignore the principles of the trilemma – currency crash!! – Brazil, Mexico, Russia, Indonesia, Korea and Thailand

Determination of exchange rate

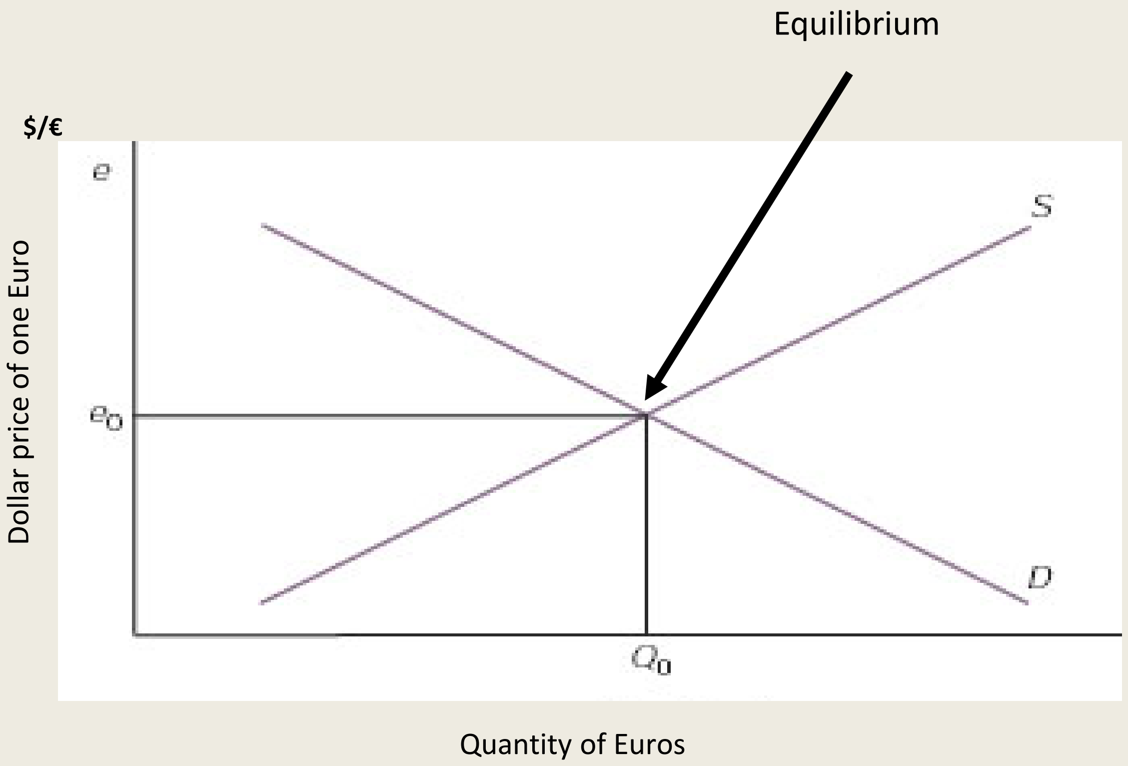
Equilibrium Exchange Rates

Setting the Equilibrium Exchange Rate:

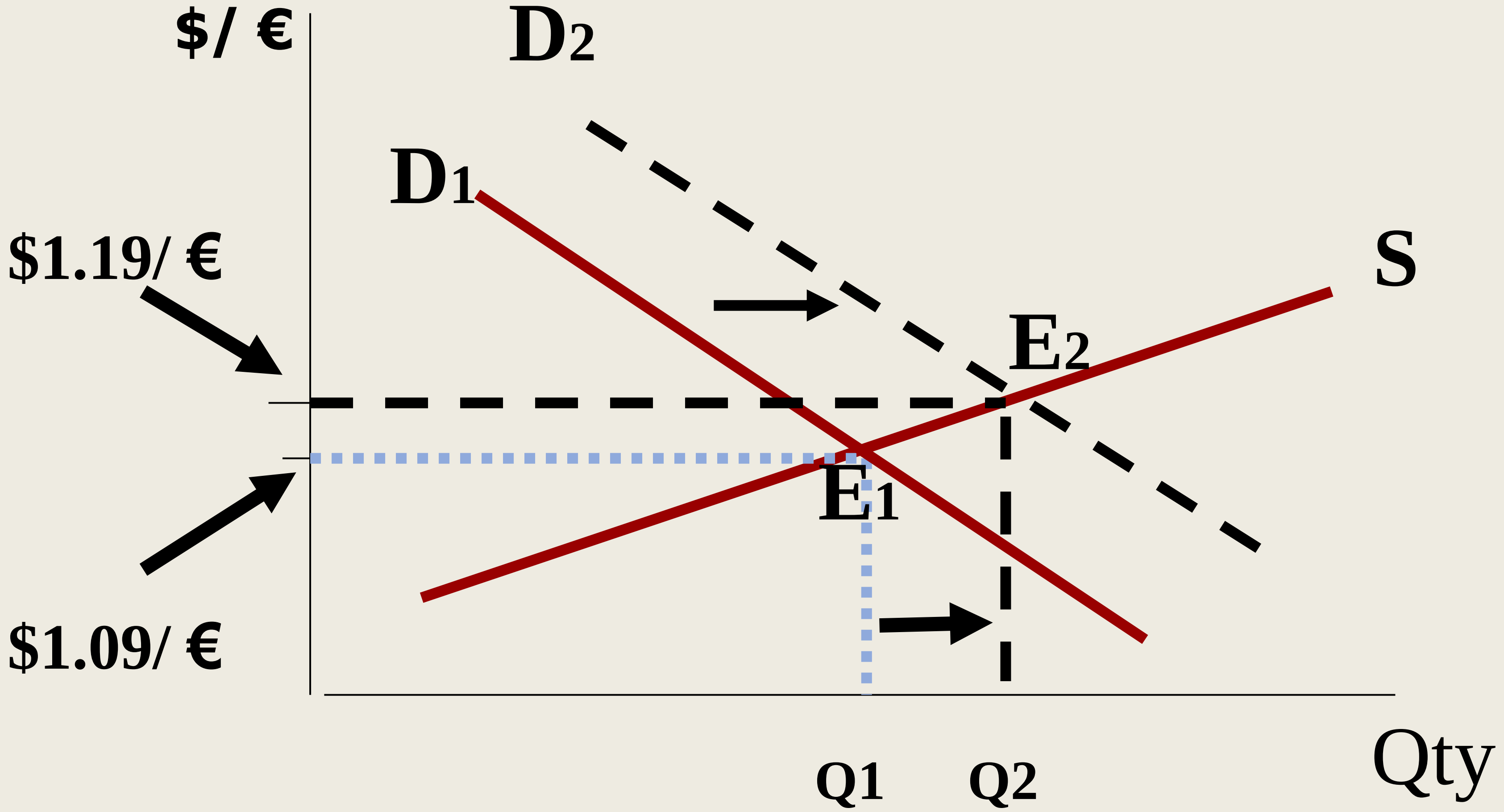
The exchange rate is the price of one unit of foreign currency expressed as a certain price in local currency.

For example \$1.15/€ means the euro in the U.S. is worth \$1.15. The first mentioned currency (US \$) is called the "reference currency" or "term currency" or "base currency", the other currency (€) is the "commodity currency".

The $/€ Equilibrium Rate



Change of equilibrium



Equilibrium Exchange Rates

Calculating Exchange Rate Changes

- Direct quote:

$$\frac{e_1 - e_0}{e_0}$$

- Indirect quote:

$$\frac{e_0 - e_1}{e_1}$$

Where:

e_0 = old currency value (open price)

e_1 = new currency value (close price)

Equilibrium Exchange Rates

EXAMPLE: € change against US$

If the US$ value of the € goes from \$1.10/€ (e_0) to \$1.20/ € (e_1),

then the € has appreciated against the US\$ by (direct quote):

$$(1.20 - 1.10)/1.10 = 9.1\%.$$

conversely, the \$ has depreciated against the € by (indirect quote):

$$(1.10 - 1.20)/1.20 = - 8.3\%.$$

The reason the euro appreciation is unequal to the amount of dollar depreciation is due to the fact that the value of one currency is the inverse of the value of the other one. Hence the percentage change in currency value differs because the base from which the change is measured differs.

Factors Affecting Exchange Rates

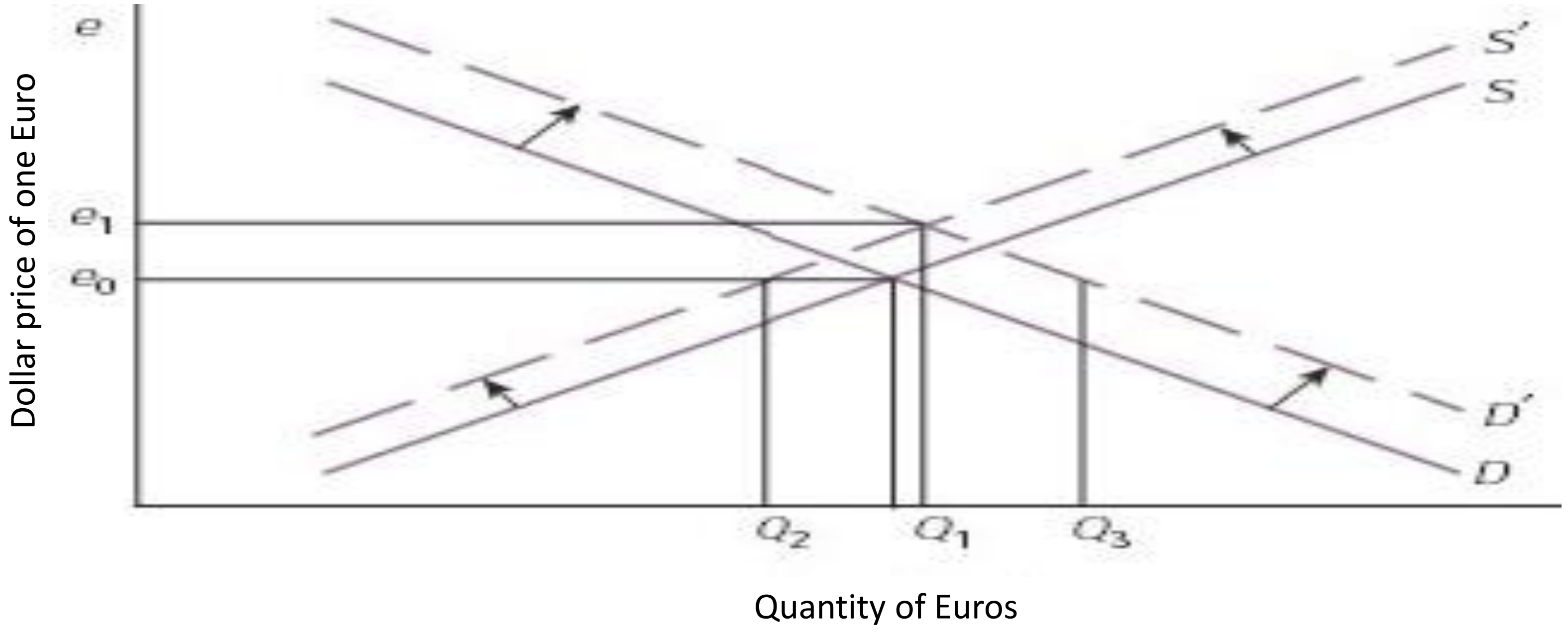
The list is non-exhaustive

- Inflation rate; Interest rate
- Balance of payment
- Economic growth (GDP growth)
- Country risk: political and economic risk
- Speculation
- Analysts forecast, market expectation/reaction
- Government/Central bank intervention
- Monetary policy and fiscal policy
- Commodity prices
- Credit ratings
- Industrial relations
- Religion/culture

Factors Affecting Exchange Rates

Relative Inflation Rates

- High inflation rates tend to cause the domestic currency to depreciate in value relative to currencies of other countries.
- If country **X** has high inflation compared to country **Y**, country **X’**s exchange rate will depreciate in comparison to country **Y** .



Factors Affecting Exchange Rates

Relative Interest Rates

- An increase in real interest rates in one nation relative to other nations, all else being equal, will lead to an appreciation of that nation's currency relative to other nations.
  - Relation between real and nominal interest - Fisher effect
- Higher interest rates encourage capital inflow and increase demand for the currency, leading to an appreciation in value of that currency.

Relative Economic Growth Rates

- A nation with high economic growth (GDP) will attract investment capital seeking to acquire domestic assets. This results in increased demand for the domestic currency and a stronger currency, other things being equal. The opposite also holds.

Factors Affecting Exchange Rates

Political and Economic Risk

- Investors prefer to hold lesser amounts of riskier assets; thus, low-risk currencies – those associated with more politically and economically stable nations – are more highly valued than high-risk currencies.

Speculators

- Many a times the prime reason for the exchange rate movements is the speculation or expectation of market participants. Speculators anticipate the event before the actual data is out and position them accordingly in order to take advantage when actual data is out, the initial positioning and actual profit affects the exchange rate. E.g. Speculators (hedge funds) think that currency of a country is overvalued and will depreciate in near future, and then such speculators will pull out the money from particular country resulting in lower demand for currency which results in depreciation of currency.

Factors Affecting Exchange Rates

Exchange rate expectations

- Psychological factors
- Market optimism, expectations, impressions, trust and faith formation
- Accumulation of information

Government intervention

- market perceptions and expectations of changes to monetary policy (impacts i/r and hence ER) and fiscal policy – *monetary easing, i.e., printing money!!*
  - Zimbabwe hyperinflation, Argentina, Post GFC in USA & UK.
  - USA, Australia in March 2020 – Response to Covid

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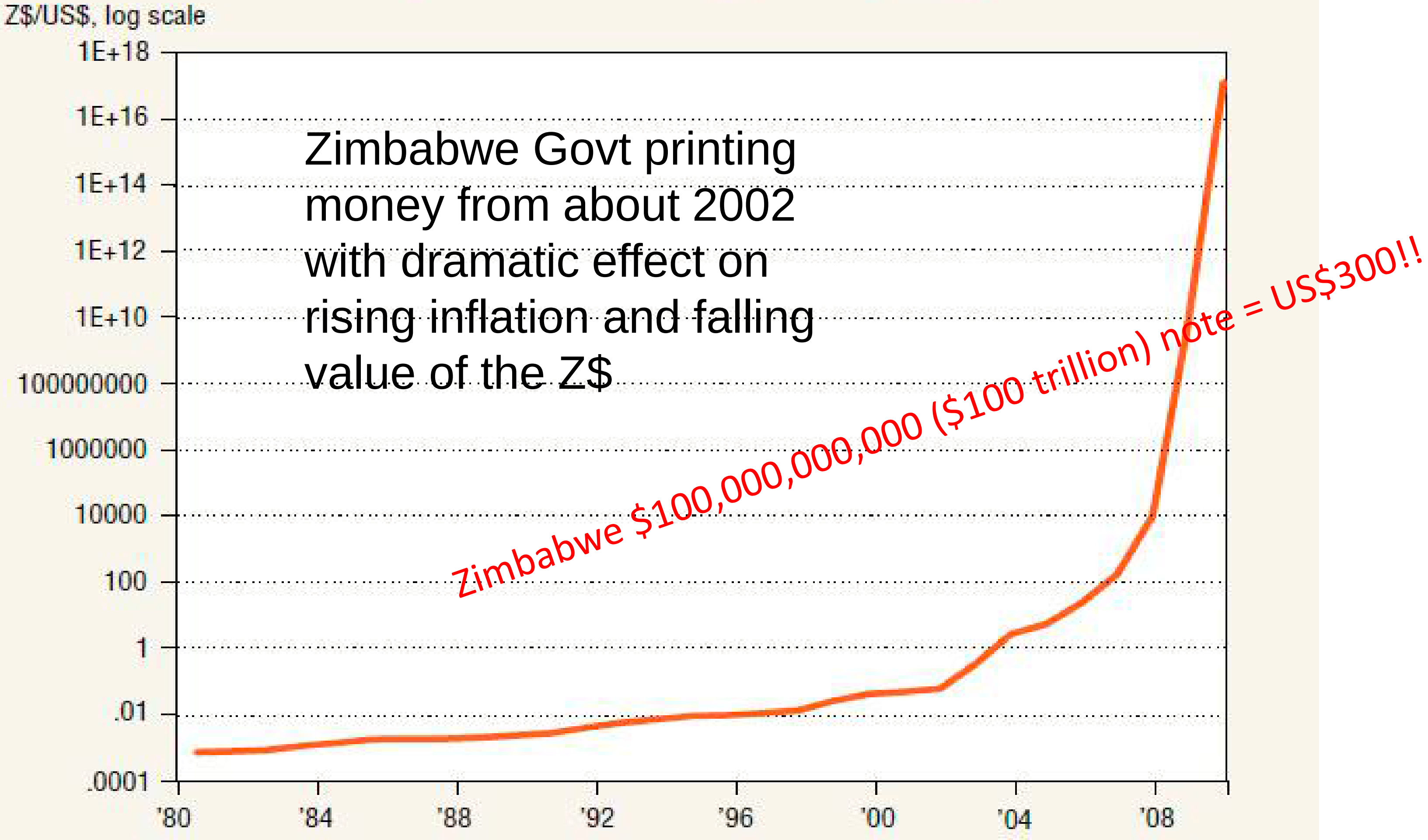
Chart 5

Zimbabwe Dollar Depreciates Sharply During Hyperinflation Era



SOURCE: Alan Heston, Robert Summers and Bettina Aten, Penn World Table Version 7.0, Center for International Comparisons of Production, Income and Prices at the University of Pennsylvania, May 2011.

[Zimbabwe and Hyperinflation: Who Wants to Be a Trillionaire? | Marginal Revolution University (mru.org)]

Factors Affecting Exchange Rates

Other factors:

- Exchange rate **overshooting**: e.g., **reactions (to news, information) that may start with a fundamental change but go too far**
- **Trading activities** of banks, commercial and investment (past and present bank failures, SVB)
- **Trading activities** of HEDGE FUNDS/**Speculators**
  - E.g., George Soros' hedge fund, Soros Fund Management reportedly held large positions in the NZ and A$; Engaged in billions of dollars speculation
- **Large one-off transactions** e.g., in 2000, NAB sold Michigan National Bank to ABN AMRO for AUD 6 billion – this moved the market about two cents
- **Political announcements, events**
  - Treasurer's may 'talk-up or talk-down' the value of the A$

FX Market Intervention

Foreign Exchange Market Intervention

FX Market Intervention - ***deliberate*** attempts by central banks/ monetary authorities to influence their respective exchange rates:

- Central banks purchase & sale foreign exchange to influence their currency's value
- Purpose - to increase the market demand for one currency or increase the market supply of another – cause the home currency value to *rise* or *fall*

Foreign Exchange Market Intervention

Rationale of/why Intervention?

– Smooth exchange rate movements (avoid excessive volatility): speculative, rumors
– Signal to the market a ‘psychological benchmark’
– Respond to temporary disturbances and provide short-term stability.
– Central bank ‘buying time’ for reassessment of economic policy
– Prevent depreciation due to higher domestic inflation
– Build-up foreign exchange reserves when the domestic currency is appreciating (i.e., sell $A/buy foreign currencies)

Foreign Exchange Market Intervention

Unsterilized vs Sterilized intervention

Unsterilized intervention:

Monetary authority (central bank) intervenes in the foreign exchange market to affect the domestic money supply and thus interest rates without taking offsetting actions (buying and selling domestic assets such as T-bond) to neutralize the impact on the money supply.

Monetary base is not affected by fx transactions. That is, the notes and coins in the circulation do not change.

- **If A$ too LOW in value**
  - Buy A$ results in a decrease in money supply and increases interest rates – may lead to deflation
- OR
- **If A$ too HIGH in value**
  - Sell A$ results increase money supply and decreases interest rates – may lead to inflation

Foreign Exchange Market Intervention

Sterilized intervention:

As opposed to unsterilized intervention, Monetary authority (central bank) intervenes in the foreign exchange market to affect the domestic money supply and thus interest rates **by taking offsetting actions** (buying and selling domestic assets such as T-bond) to neutralize the impact on the money supply.

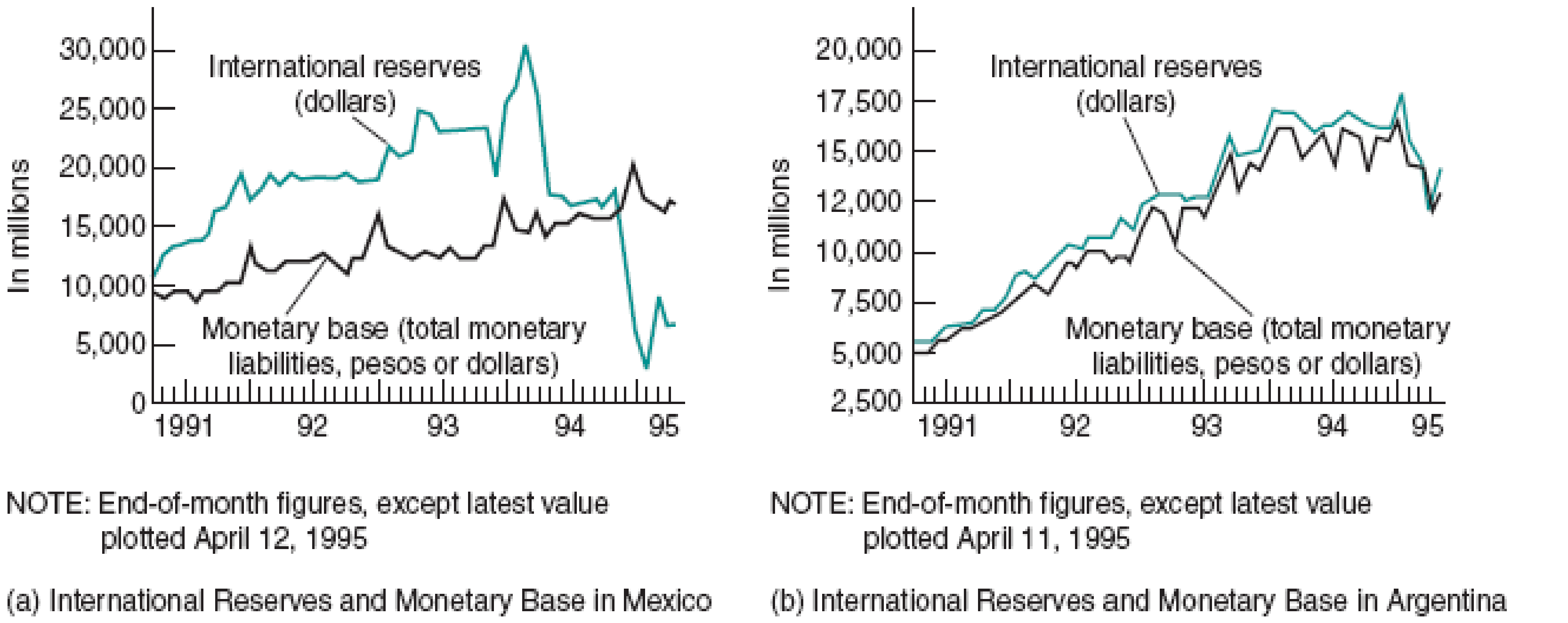
- Can neutralize/sterilize the earlier effects on monetary conditions. How?
    - Purchase of A$ in FX is neutralized by buying government bonds from the public, results in an increase of public's cash thus increasing the reserves in the banking system and the money supply
        - Reverses the impact on monetary conditions and interest rates
        - Interest rates fall back to original level
- Exchange rate is permanently affected only if investors view securities as imperfect substitutes.

Foreign Exchange Market Intervention

- Results of sterilization:
  - FX reserves should increase or decrease.
  - Balance sheet structural change of assets but no change in domestic money supply.
- Evidence
  - Mexico’s sterilization in 1994 – international reserves fell sharply but Money base unchanged.
  - Argentina’s currency board precluded the ability to sterilize, money base forced to closely match its US$ reserves.
  - China – 2003: In a 3-month period issued 250 billion Yuan in short term notes (base money supply changed) to sterilize the Yuan created by its FX market intervention!

Sterilisation vs Non-sterilisation

Mexico has a dramatic drop in international reserves as it sterilizes its currency interventions (sells US$ and buys pesos), while money base is unchanged. Argentina was on a currency board therefore unable to sterilize, money base follows US$ reserves.



Source: Wall Street Journal, May 1, 1995, p. A14. Reprinted by permission. © 1995 Dow Jones & Company, Inc. All rights reserved worldwide.

Exhibit 2.14 : Mexico and Argentina Follow Different Monetary Policies p. 80

Eeffectiveness of FX Market Intervention

Divided opinions on whether intervention is successful or unsuccessful

Some argue Unsuccessful because:

- Efficient market hypothesis – market gets it right by itself (Market forces of Supply and Demand).
- Currency prices are determined by ALL available information.
- Central Bank intervention is destabilizing and costly. Hence, attempts to affect real exchange rate via intervention will fail.
- No simple relationship between changes in exchange rate and international competitiveness, employment or current account deficits.
- Current account deficits DO NOT cause currency depreciation, nor does currency depreciation by itself help reduce a current account deficit! Look at the historical evidence.

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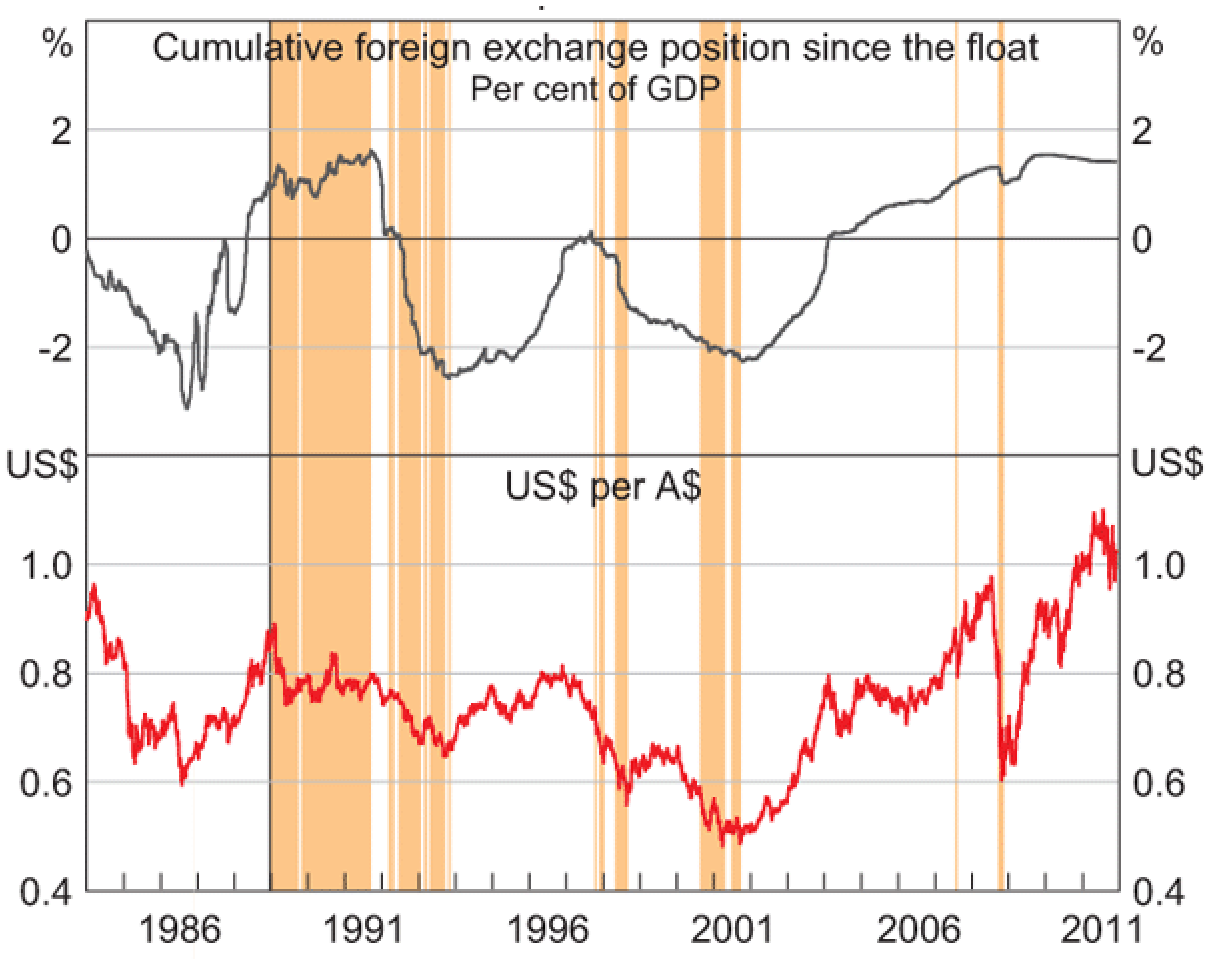
Effectiveness of FX Market Intervention

Some argue intervention Successful

- Efficient Market Hypothesis has not explained all price change adequately
- Psychological factors not captured in the model
- Not possible to explain overshooting and speculation
- 1985 September – Plaza Accord – to depreciate US$ vs G5 nations
- 1987 February – Louvre Accord – to halt depreciation of US$ vs G5 nations

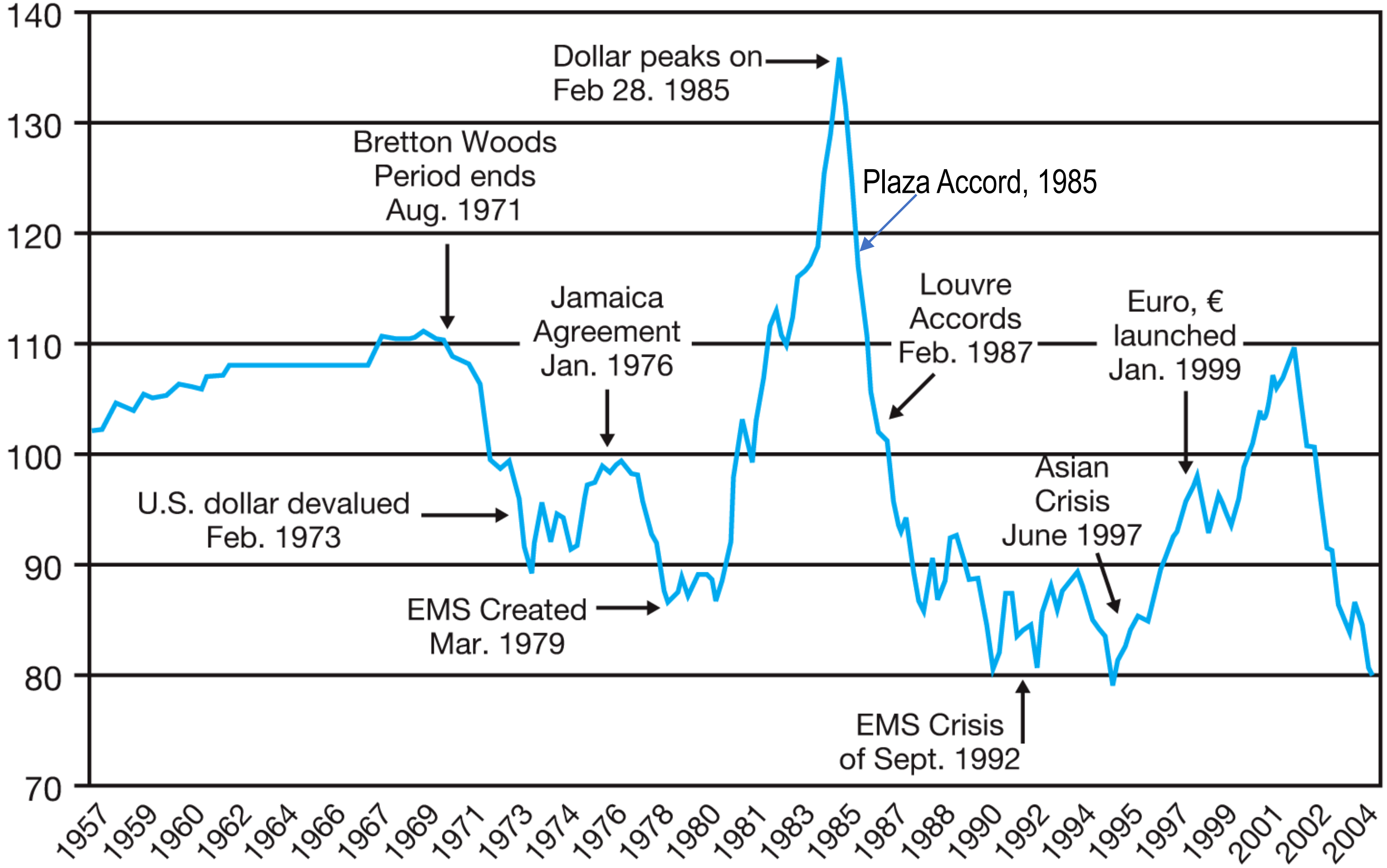
RBA Intervention in FX markets

RBA's foreign exchange market intervention (shaded area) has evolved since the float of the Australian dollar in 1983.



Source: https://www.rba.gov.au/publications/bulletin/2011/dec/7.html (**Graph 1**)

EXHIBIT 2.2 The IMF's Nominal Exchange Rate Index of the U.S. Dollar and Significant Events, 1957–2005



Source: International Monetary Fund, *International Financial Statistics,* http://www.imfstatistics.org.

Multiple Choice Question 1

Of the following, exchange rates depend the most on:

a. relative inflation rates
b. relative interest rates
c. relative trade deficits
d. a and b

Multiple Choice Question 2

Suppose, on Friday, July 13, 2012, the A$ was worth Yen 97. Over the weekend the A$ devalued against the Yen to Yen 87. By how much had the Yen changed <u>against the A$</u>?

a. 11.49%

b. -11.49%

c. 5.21%

d. -10.31%

Answer A
Indirect quote, Yen change in %
= (Beginning – ending rate)/Ending rate

(97 – 87)/87 = 11.49%
Yen appreciated

Multiple Choice Question 3

Suppose, on Friday, July 13, 2012, the A$ was worth Yen 97. Over the weekend the A$ devalued or fell against the Yen to Yen 87. By how much had the A$ has depreciated against the Yen?

a. 11.49%

b. -11.49%

c. 5.21%

d. -10.31%

Answer D
Direct quote, A$ change in %
= (Ending – Beginning rate)/Beginning rate

(87 – 97)/97 = -10.31%
A$ depreciated

Multiple Choice Question 4

Suppose, on Friday, July 13, 2012, the GBP was worth A$1.5000. Over the weekend the GBP revalued to A$1.5300. By how much had the GBP changed <u>against the A$</u>?

a. +2.00%

b. -2.00%

c. +1.96%

d. -1.98%

Answer A

Direct quote, GBP change in %
= (Ending – Beginning rate)/Beginning rate
(1.5300 – 1.5000)/1.5000 = 2%
GBP appreciates by 2%

INTERNATIONAL FINANCE AND INVESTMENT

Student

https://go.blueja.io/j9JkUYJZg0mw5qmx5W72qA

To access the evaluation, scan this QR code with your mobile phone.

Gentle Reminder: only MCT

- Opens 10:00 am **Thursday May 16, 2024**, and closes 8 pm **Friday May 17, 2024**.
- Examines topics Week 1-10
- A total of 22 MCQs; 10 marks, which is 10% of the total assessment in the unit
- Test duration: 1 hour once you log in
- Test will be run through CloudDeakin
- A practice test is uploaded, you can attempt it multiple times
- Review all MCQs uploaded under each topic
- All relevant resources, especially, seminar answers are released for test preparation

MAF306: Week 10

MNC's Capital Budgeting
MNC's Capital Structure & CoC

* The Bachelor of Commerce at Deakin University is internationally EPAS accredited

MNC's Capital budgeting

- Review of basic algebra of capital budgeting
- Recipes of discounting foreign cash flow and relevance to international parity condition (IPC) theory
- An example
- Resources:
  - Shapiro: Chapter 17
  - Eiteman et all/Multinational Business Finance: Chapter 18
  - K.C. Butler / Multinational Finance : Chapter 13 (5^{th} ed), see cloud Deakin resources

The algebra of cross-border capital budgeting

Let's begin with
"Domestic" NPV calculations

$$V_0^d = \Sigma_t \; E[CF_t^d] \,/\, (1+i^d)^t$$

1. Estimate future cash flows $E[CF_t^d]$
   - Include only incremental cash flows
   - Include all opportunity costs
   - Exclude all sunk costs

V_0^d = Net Present Value; CF_t^d = Cash Flow; i^d = interest or discount rate
Superscript d = domestic

The algebra of cross-border capital budgeting

"Domestic" NPV calculations

$$V_0^d = \Sigma_t \; E[CF_t^d] \; / \; (1+i^d)^t$$

1. Estimate future cash flows $E[CF_t^d]$

2. Identify a risk-adjusted discount rate

- Discount nominal CFs at nominal discount rates and real CFs at real discount rates
- Discount equity CFs at equity discount rates and debt CFs at debt discount rates
- Discount CFs to debt and equity at the WACC (Weighted Average Cost of Capital)
- Discount cash flows in a particular currency at a discount rate in that currency

The algebra of cross-border capital budgeting

"Domestic" NPV calculations

$V_0^d = \Sigma_t \; E[CF_t^d \,] \,/\, (1+i^d\,)^t$

1. Estimate future cash flows $E[CF_t^d]$
2. Identify a risk-adjusted discount rate
3. Calculate net present value V_0^d
   - Based on expected future cash flows and the appropriate risk-adjusted discount rate

MNC's capital budgeting

...discounting foreign currency cash flows

Recipe 1: Discount in the foreign currency (project's perspectives)

Discount in the foreign currency at i^f and convert the foreign currency NPV to a domestic currency value $V_0^d|i^f$ at the spot exchange rate $S_0^{d/f}$

Recipe 2: Discount in the domestic currency (Parent's perspectives)

Convert foreign cash flows into the domestic currency at expected future spot rates and then discount at the domestic rate i^d to find $V_0^d|i^d$

$S_0^{d/f}$ = spot exchange rate; i^f = interest or discount rate; Superscript f = foreign

Recipe 1 Discount in the foreign currency

1. Estimate future cash flows $E[CF_t^f]$
2. Identify i^f
3. Calculate net present value
   - Calculate $V_0^f = \Sigma_t \, E[CF_t^f] \, / \, (1+i^f)^t$
   - Convert to the domestic currency using spot rate

$$= S_0^{d/f} \, V_0^f \;\rightarrow\; V_0^d|i^f$$

V_0^f = Net Present Value in Foreign currency

Recipe 2 Discount in the domestic currency

1. Estimate $E[CF_t^d] = E[S_t^{d/f}]\ E[CF_t^f]$
2. Identify i^d
3. Calculate net present value in domestic currency

   $= V_0^d | i^d$

Two recipes are same if IPC holds: Wendy's Neverland Project – print it from CloudDeakin

- £10,000 (Cr40,000) investment for the ship at time t=0; so spot rate is Cr4/£
- £6,000 (Cr24,000) investment for inventory at time t=0 (assume LIFO)
- Expected nominal revenues of Cr30,000, Cr60,000, Cr90,000, and Cr60,000 in years 1 through 4
- Variable operating costs are 20% of sales
- Cr2,000 of fixed operating costs at the end of the first-year increase at the *rate of inflation* (which is 36.14% for Neverland and 8.91% for UK) thereafter
- Forward premium on the pound is 25% (using the expected inflation rate in both countries, i.e., 1.3614/1.0891 = 1.25 = 25%), pound is expected to appreciate at 25% per year.
- The ship is expected to retain its Cr40,000 real value
- Income & capital gains taxes are 50% in each country
- Inventory sold for Cr24,000 in real terms at t=4
- The ship is owned by the foreign affiliate and depreciated straight-line to a zero-salvage value
- All cash flows occur at year-end
- Use the given discount rates (in Cr, £) for NPV of this project (nominal discount rate in Cr = 50% and in £=20%; while real discount rate is 10.18% in both Cr and £)

Investment & disinvestment cash flows
(in Neverland crocs)

	t=0	. . .	t=4
Ship	-40,000		
Inventory	-24,000		
Sale of ship			137,400*
- Tax on sale			-68,700*
Sale of inventory			82,440*
- Tax on sale			-29,220*
Balance sheet			
cash flows	-64,000		121,920

*Workings on the next slide

Workings for the previous slide

***Sale of the ship and tax on the sale of ship**

Sale of ship $= E[CF_4^{Cr}] = CF_0^{Cr}\,(1+p^{Cr})^4$

$= Cr40{,}000(1.3614)^4$

$= Cr137{,}400$

Tax on sale of ship $= (T^{Cr})\,(MV^{Cr}\text{-}BV^{Cr})$

$= (0.5)(Cr137{,}400\text{-}0) = Cr68{,}700$

***Sale of inventory assuming LIFO (last-in-first-out) inventory accounting and tax on the sale of inventory**

Sale of inventory $= E[CF_4^{Cr}] = CF_0^{Cr}\,(1+p^{Cr})^4$

$= Cr24{,}000(1.3614)^4 = Cr82{,}440$

Tax on sale of inventory $= (T^{Cr})\,(MV^{Cr}\text{-}BV^{Cr})$

$= (0.5)(Cr82{,}440\text{-}Cr24{,}000)$

$= Cr29{,}700$

Operating cash flows
(in Neverland crocs)

	t=1	t=2	t=3	t=4
Revenues	30,000	60,000	90,000	60,000
- Variable costs	-6,000	-12,000	-18,000	-12,000
- Fixed cost	-2,000	-2,723	-3,707	-5,046
- Depreciation	-10,000	-10,000	-10,000	-10,000
Taxable income	12,000	35,277	58,293	32,954
- Taxes	- 6,000	-17,639	-29,147	-16,477
Net income	6,000	17,639	29,147	16,477
+ Depreciation	10,000	10,000	10,000	10,000
Operating CFs	16,000	27,639	39,147	26,477

Recipe 1: Discounting in crocs

	t=0	t=1	t=2	t=3	t=4
Bal sheet CFs	-64,000				121,920
Operating CFs		16,000	27,639	39,147	26,477
$E[CF_t^{Cr}]$	-64,000	16,000	27,639	39,147	148,397

$$V_0^{Cr} = -NCO_0 + \frac{NCI_1}{1+d} + \frac{NCI_2}{(1+d)^2} + \cdots + \frac{NCI_n}{(1+d)^n}$$
$$= \sum_{t=0}^{n} \frac{NCI_n}{(1+d)^n}$$

V_0^{Cr} = -Cr137 @ i^{Cr} = 50%

Recipe 2: Discounting in pounds

	t=0	t=1	t=2	t=3	t=4
$E[CF_t^{Cr}]$	-64,000	16,000	27,639	39,147	148,397
$*F_t^{Cr/£}$	4	5	6.25	7.8125	9.7656
$E[CF_t^{£}]$	-16,000	3,200	4,422	5,011	15,196

$V_0^{£} | i^{£}$ = -£34 at $i^{£}$ = 20% (using spot rate $S_0^{Cr/£}$ = Cr4/£)

*Pound is expected to appreciate at 25% (premium on pound)

Role of International parity conditions (IPC)

As mentioned, two recipes are same if IPC holds,

i.e. $(1+i^d)^t = (1+i^f)^t (F_t^{d/f} / S_0^{d/f})$

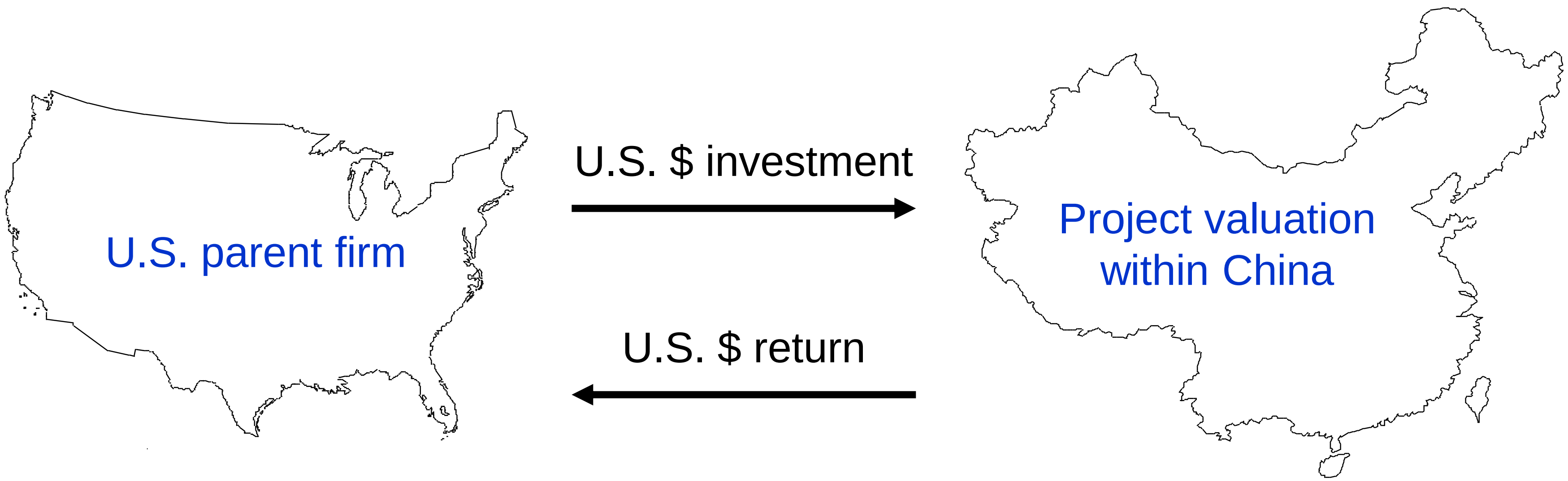
$F_t^{d/f}$ = forward rate $S_0^{d/f}$ = spot rate

However, violations can occur, due to -

- thin trading of currencies & underdeveloped credit markets.
- biases in forward forecast errors
  - Forward parity (which is $F_t^{d/f} = E[S_t^{d/f}]$) is a noisy predictor, and there are persistent (positive or negative) forecast errors.

Valuing a U.S. investment in China

- ➢ A parent firm wants a return in its own functional currency



Valuation when IPC does not hold

A. The project's (local or foreign) perspective

– Let $V_0^d|i^f$ represent the value of a foreign project when discounted in the foreign currency

B. The parent's (domestic) perspective

– Let $V_0^d|i^d$ represent the value of a foreign project when discounted in the domestic currency

A and B may not be equal when the IPC does not hold

When parity doesn't hold…

$V_0^d|i^f > 0$

$\Rightarrow$ The project has value from the perspective of a foreign investor

(that is, relative to alternatives in local financial markets)

$V_0^d|i^d > 0$

$\Rightarrow$ The project has value from the perspective of the parent

Clear losers…

		Parent's perspective	
		$V_0^d \mid i^d < 0$	$V_0^d \mid i^d > 0$
Project's perspective	$V_0^d \mid i^f < 0$	Reject	
	$V_0^d \mid i^f > 0$		

Local losers:
There must be something better…

		Parent's perspective	
		$V_0^d \vert i^d < 0$	$V_0^d \vert i^d > 0$
Project's perspective	$V_0^d \vert i^f < 0$	Reject	Look for better projects in the foreign currency
	$V_0^d \vert i^f > 0$		

Local winners:
Somebody's gotta want this…

Parent's perspective

Project's perspective		$V_0^d\|i^d < 0$	$V_0^d\|i^d > 0$
	$V_0^d\|i^f < 0$	Reject	Look for better projects in the foreign currency
	$V_0^d\|i^f > 0$	Try to lock in the time 0 value of the project	

Alternatives for capturing the time t=0 value of a foreign project

- In the asset markets
  - Sell the project to a local investor
  - Bring in a joint venture partner
- In the financial markets
  - Hedge the cash flows from the project against currency risk
  - Finance the project with local currency debt or equity

If foreign cash flows are certain,
then you can create a perfect hedge.
If foreign cash flows are uncertain,
then forward hedges are imperfect hedges.

Winners at home and abroad: Structuring the deal…

Project's perspective	Parent's perspective: $V_0^d|i^d < 0$	Parent's perspective: $V_0^d|i^d > 0$
$V_0^d|i^f < 0$	Reject	Look for better projects in the foreign currency
$V_0^d|i^f > 0$	Try to lock in the time 0 value of the project	Accept, then structure the deal

$$V_0^d|i^f > V_0^d|i^d > 0$$

The project has more value locally than it does from the parent's perspective

$\Rightarrow$ You should hedge

Hedging provides the parent with higher expected value and lower exposure to currency risk

$$V_0^d|i^d > V_0^d|i^f > 0$$

The project has more value from the parent's perspective than it does to local investors

$\Rightarrow$ Whether you hedge or not will depend on the firm's hedging policy

Hedging lowers exposure to fx risk, but also lowers the expected NPV of the project

MNC’s Capital structure and Cost of Capital

- Shapiro: Chapter 14
- Eiteman et all: Chapter 13
- Butler: Chapter 15

Learning objectives

- Capital structure and cost of capital – basics
- The MNC’s optimal capital structure
- Project valuation and the cost of capital
  - The impact of market imperfections
  - WACC versus APV
- Sources of funds for multinational operations

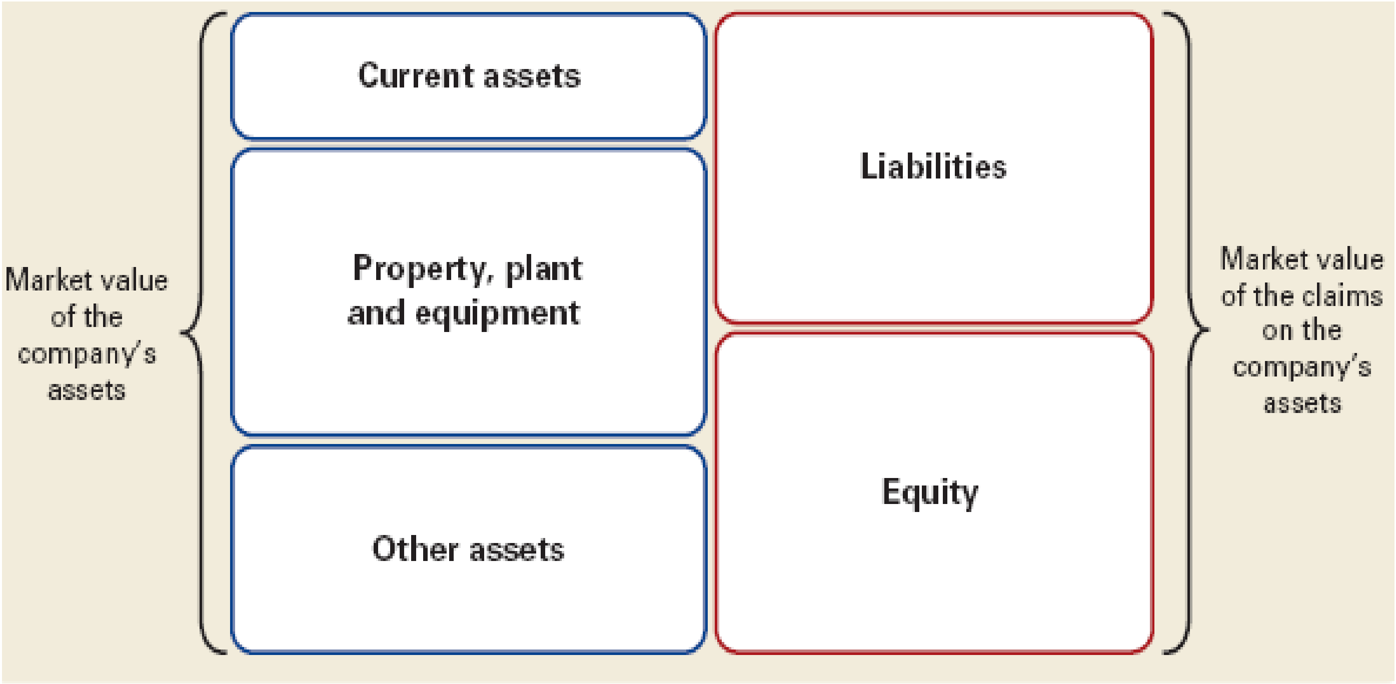
Capital structure and cost of capital – basics

- Capital structure refers to the proportion of (long-term) debt and equity capital (and/or hybrid securities) and the particular forms of capital chosen to finance the assets of the firm
- Cost of Capital - the return the firm's investors (bond and share holders) could expect to earn if they invested in securities with comparable degrees of risk
- COC of a firm with no debt is simply the cost of equity (required return on equity). Firm loses the benefits of tax deductibility of interest
- COC of a firm with debt could be lower than the COC of a firm with no debt due to tax benefit of using debt finance (higher leverage)
- The fraction of debt financing is a measure of the financial leverage/gearing
- Higher leverage can increase the distress cost and hence, does not guarantee to lower the COC (Lehman brothers failure)

Capital structure and cost of capital – basics

- Firms strive to choose *a financing mix* that minimizes the COC or WACC (weighted average cost of capital)
- WACC is the weighted average of returns (weighted by market value) on debt, equity and hybrid securities
  - Therefore, the cost of capital or WACC is equal to the expected rate of return on the portfolio of assets.
- The cost of capital/WACC is used as the discount/hurdle rate for evaluating investments in new assets of *similar* risk.

- Remember! Cost of capital is based on market values not the book values, i.e., MV of Assets = MV of liabilities + MV of equity.
- Firms need to calculate the costs involved with the right-hand side components of the balance sheet



Managers must choose…

- The proportions of debt, equity, and hybrid securities
- Features of the instruments
  - Debt: fixed or floating rate interest payments, indenture provisions: conversion features, callability, seniority, and maturity
  - Equity: Voting rights, share classes
- The location(s) where securities are issued and traded
- The currency of denomination

Optimal capital structure: Is there one?

Far better an approximate answer to the
right question, which is often vague,
than an exact answer to a wrong question,
which can always be made precise.

John W. Tukey

MM's Irrelevance Proposition (1958) - MM Proposition I

The perfect market assumptions were developed and applied in the following paper

Franco Modigliani and Merton Miller, "The Cost of Capital, Corporation Finance and the Theory of Investment," *American Economic Review* 48, June 1958, pages 261-297.

(paper can be downloaded from Deakin library site)

MM Proposition I…cont.

The perfect market assumptions

- Frictionless markets
  - No transaction costs, taxes, government intervention, agency costs, or costs of financial distress
- Equal access to market prices
  - Everyone is a "price taker"
- Rational investors
  - Return is good and risk is bad
- Equal access to costless information

MM Proposition I..cont..

With equal access to perfect financial markets, individuals can replicate any financial action that the firm can take.

This leads to Modigliani-Miller's famous irrelevance proposition:

If financial markets are perfect, then corporate financial policy is irrelevant, i.e., market value of a company is not affected by its capital structure

Thus, $V_L = V_U$ (V_L and V_U are the market values of the levered and the unlevered firm, respectively)

MM's Relevance Proposition (1963) - MM Proposition II

MM-II includes the impact of

- leverage and taxes,
- financial distress,
- agency costs, and
- asymmetric information

on a company's cost of equity, cost of capital, and optimal capital structure

Modigliani and Miller, "Corporate Income Taxes and the Cost of Capital: A Correction," *American Economic Review* 53, June 1963. (paper can be downloaded from Deakin library site)

MM Proposition – II…cont.

When taxes are introduced, the value of the firm is enhanced by the tax shield provided by the interest deduction. The tax shield:

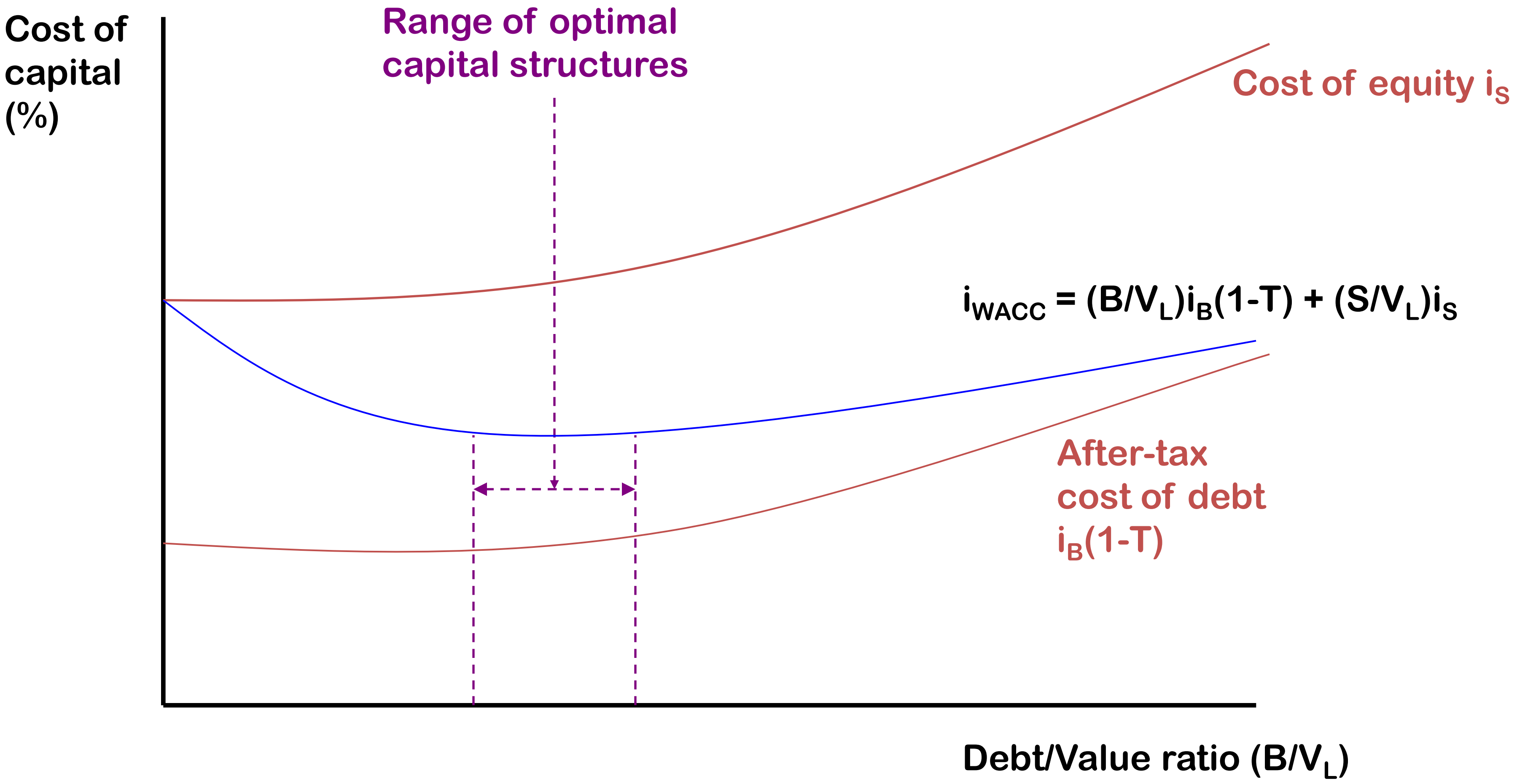
- Lowers the cost of debt.
- Lowers the WACC as more debt is used.
- Increases the value of the firm by *tD* (that is, marginal tax rate times debt)

To summarize, value of a firm is given by:

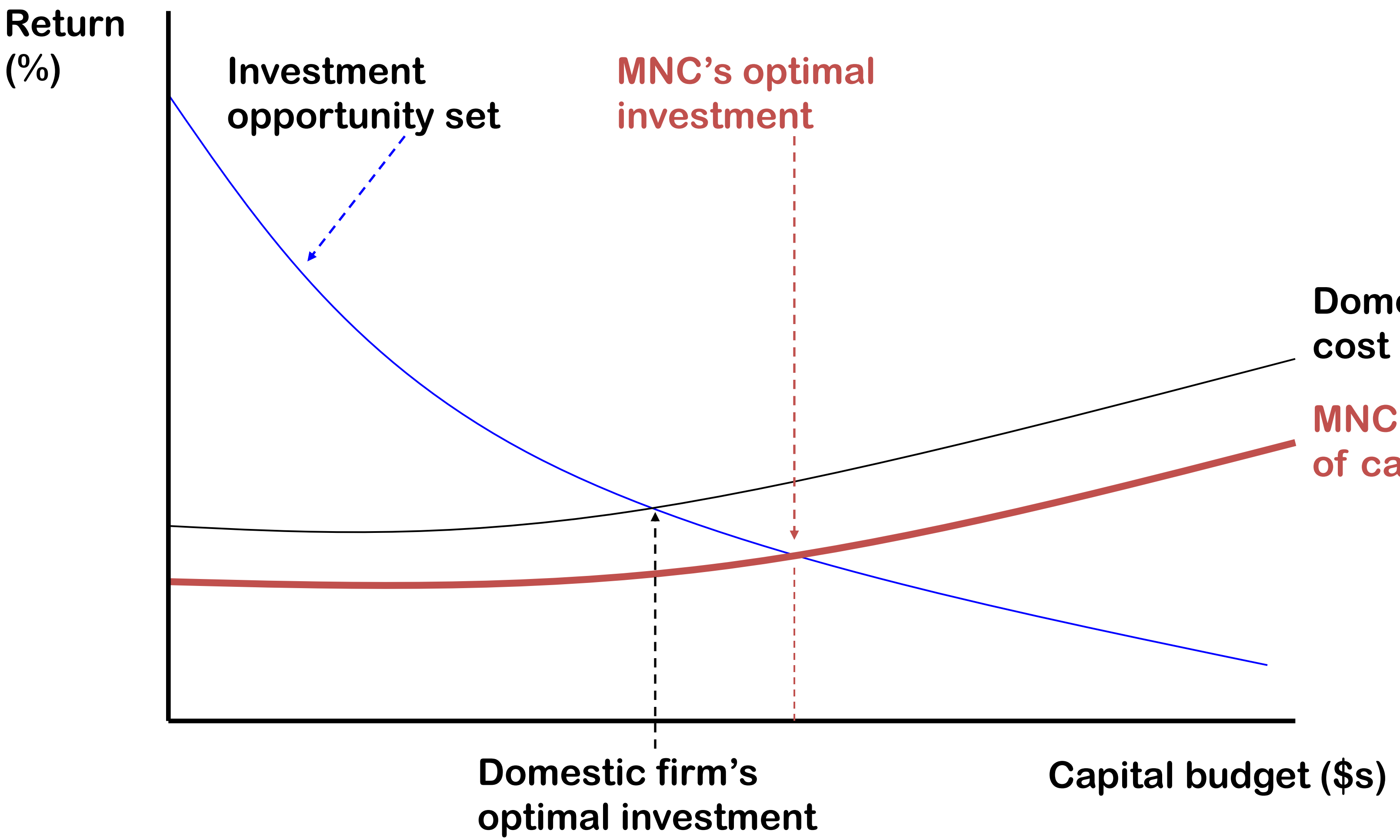
$V = (B/V)i_B(1\text{-}Tc) + (S/V)i_S$, i.e., $V_L \neq V_U$

- Tc is the corporate income tax rate
- B and S are the market values of debt (bonds) and equity (stock)
- i_S, and i_B are the required returns on the equity, and bond, respectively
- Value of levered firm (V_L) may not be equal to the value of unlevered firm (V_U)

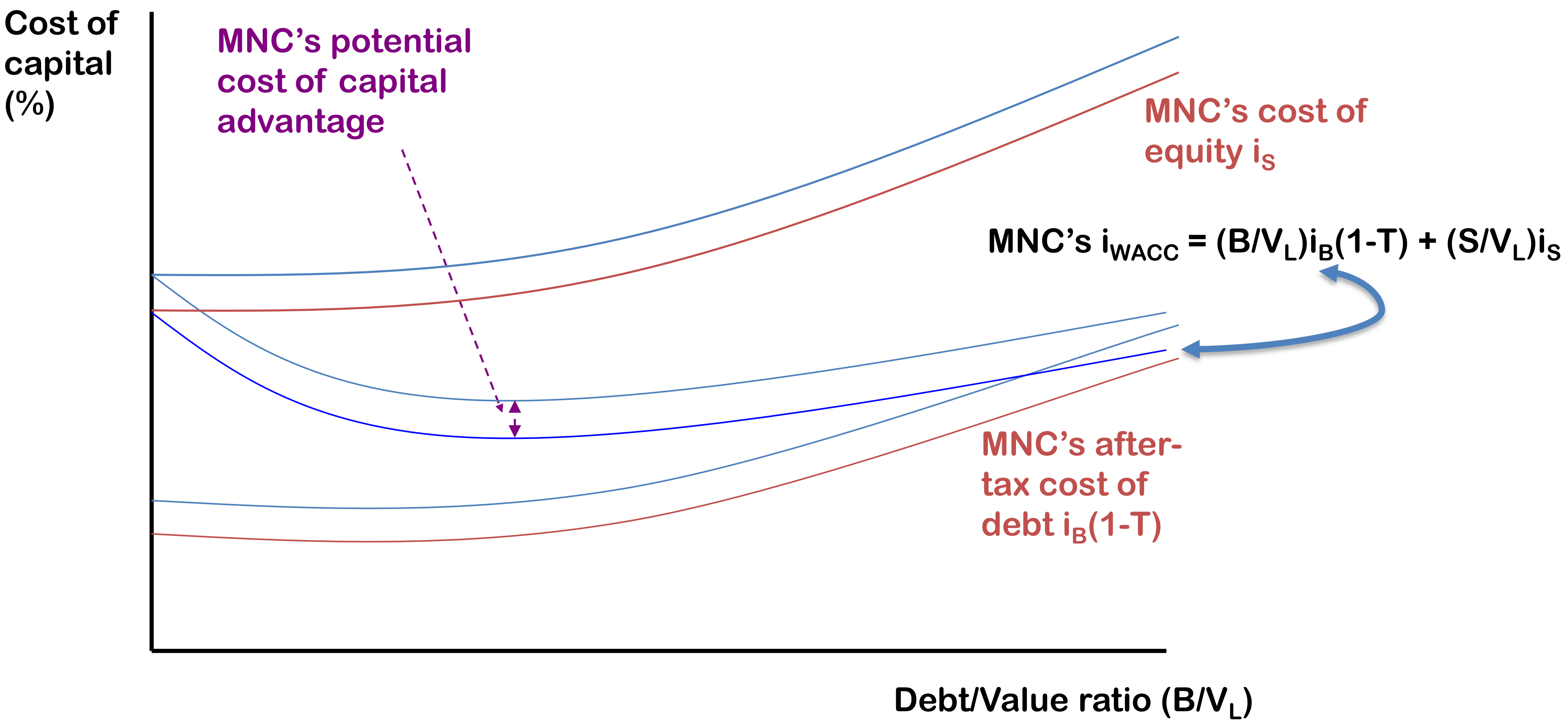
Implications of MM's propositions in WACC



The MNC's advantage from external financing



The MNC's advantage of cost of capital



MNC's advantage of cost of capital...

A. MNC Advantage

uses more debt due to diversification

B. What is proper capital structure?

1. Borrowing in local currency helps to reduce exchange rate risk
2. Allows subsidiary to exceed parent capitalization norm if local market has lower costs

Sources of funds for multinational operations

- Internal debt (bond, debentures) and equity (share)
- External debt and equity (bond, debentures, commercial papers) and equity (share)
- Foreign debt and equity (euro bond, foreign bond, debentures, commercial papers, euro loans) and equity (share)
- Other dimensions:
  - Crowdfunding
  - Business angels
  - Venture capital
  - Letter of credit and Back-to-back L/C

The financial “pecking order”

- Internal sources of funds are preferred by most managers
- External sources of funds are accessed only after internal sources are exhausted
  - External debt is the preferred external funding source
  - External equity is used only as a last resort

Approaches to project valuation and CoC

The two most popular approaches …

- 1. WACC = Weighted average cost of capital, need to know CAPM and bond valuation
- 2. APV = Adjusted present value (same as NPV; exception-APV uses Cost of equity rather than WACC)

- Always use an asset-specific discount rate
  - Discount nominal cash flows at a nominal discount rate
    Discount real cash flows at a real discount rate
  - Discount domestic currency cash flows at a domestic currency discount rate
    Discount foreign currency cash flows at a foreign currency discount rate

An Example of WACC

$$NPV = \Sigma_t \left[E[CF_t] / (1+i_{WACC})^t \right]$$

$$i_{WACC} = [(B/(B+S))\, i_B\, (1-T_C)] + [(S/(B+S))i_S]$$

B = the market value of corporate bonds

S = the market value of corporate stock

i_B = required return on corporate bonds

i_S = required return on corporate stock

T_C = marginal corporate tax rate

WACC of an MNC

Sharp is an MNC, which produces household electronic items. Sharp wants to determine its WACC. It issued bonds at international markets @8.65% p.a. with a total value of $9.4265million. Sharp also has preference shares outstanding in the international market. The value of the preference share is $8.00 million paying 12.5% dividend p.a. In addition, Sharp has 5 million ordinary shares on issue in the domestic market. The share has a current market price of $5.00 each. Assume that the current risk-free rate is 7% and the return on market portfolio is 12%. Sharp has a beta which has been recently estimated at 1.2. Sharp enjoys a tax rate of 30%.

Calculate Sharp's weighted average cost of capital.

Answer

- i_B (given) = 8.65% (if nothing is mentioned and the tax rate is given, assume that this is pre-tax cost of debt)
- i_p (given) = 12.5%
- i_s not given. Using CAPM, the cost of equity,

$$i_s = R_F + [\beta(k_M - R_F)]$$

$$= 0.07 + 1.2\ (0.12 - 0.07) = 0.13\text{, or } 13.0\%$$

Source	Cost % (k_i)	Value	Weight (W_i)
Bonds (B)	8.65	9,426,500	0.222
Equity (S)	13	25,000,000	0.589
Pref. Share (P)	12.5	8,000,000	0.189
		42,426,500	1

Answer….cont

$$i_{WACC} = \left(\frac{B}{B+S+P}\right) i_B \times (1-T) + \left(\frac{S}{B+S+P}\right) i_S + \left(\frac{P}{B+S+P}\right) i_P$$

$$= (0.222)0.0865 \times (1-0.3) + (0.589)0.13 + (0.189)0.125$$
$$= 0.1136 \text{ or } 11.36\%$$

B = Bond; S = Ordinary Share; P = Preference Share; T = Tax rate;

i_B = cost of bond; i_S = cost of ordinary share; i_P= cost of preference share

Next week

- Revision lecture
- Any suggestions to revise anything?

MAF306: Topic 6
Foreign Exchange Markets

Dr Sohel Azad

Learning objectives & Resources

- to get an overview of foreign exchange (FX) market
- to examine the difference between spot and forward exchange rates
- to understand FX quotations and conventions
- to consider FX arbitrage
- to calculate forward premiums and discounts

Shapiro and Hanouna: 11th Ed - Chapter 6; 10th Ed - Ch 7
Eiteman et al (2011) chapter 5
Butler (2008) chapter 4 pp. 90-101

Is this where you might like to be??

An international holiday?......What role do FX markets play?

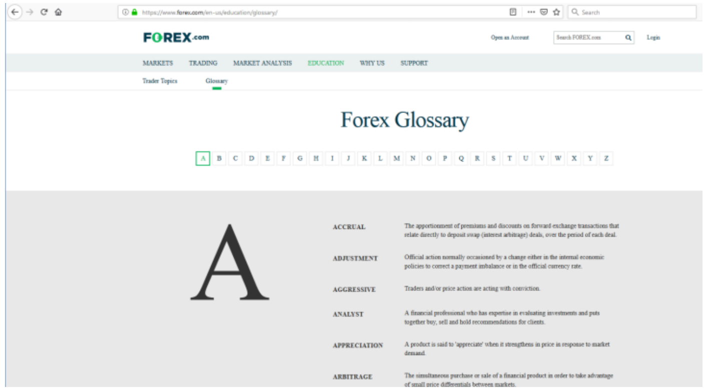
Overview of Foreign Exchange Market

The Foreign Exchange Market

What: Where money denominated in one currency is bought and sold with money denominated in another currency

Examples

- An Australian tourist is going to Hawaii…..s/he needs to exchange AUD for USD for transactions undertaken in Hawaii
- Volkswagon sells cars to car dealers in Melbourne
- An Australian Bank places its excess funds in the Eurodollar market rather than the domestic market. The bank needs to buy USD and sells AUD
- https://www.forex.com/en-us/education/glossary/



The Foreign Exchange Market

Why do we need an FE market (Use)?

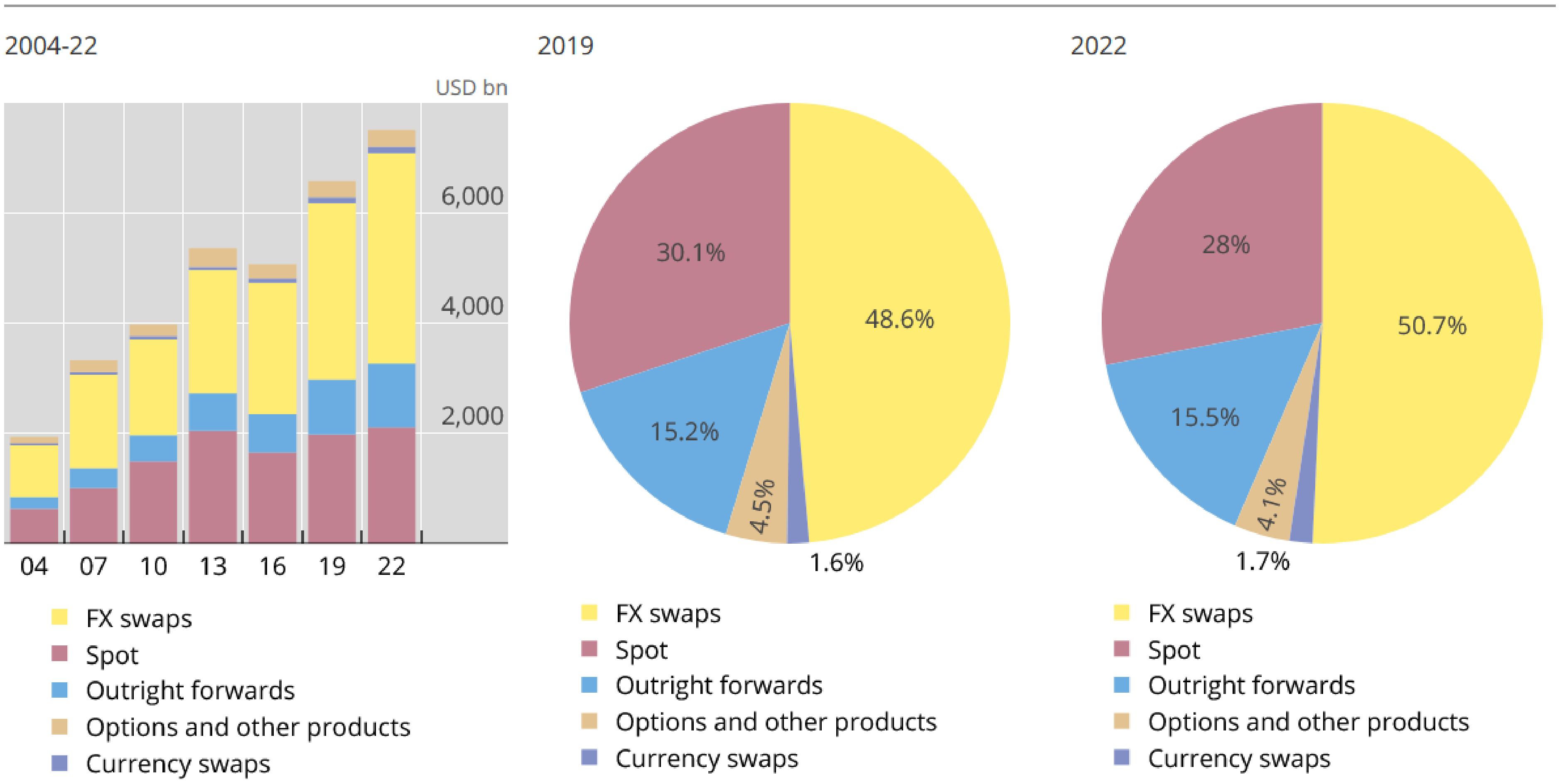
- **Examples**
  - Import payments/export receipts
  - Transfer of purchasing power across nations
  - Facilitate international trade …..i.e., buy/sell stuff, FDI!
  - Determine ER (i.e., Demand & Supply of forex in the market)
  - Facilitate FE stability
  - Facilitate FE risk management
  - Give assistance or foreign aid (e.g., Third world countries, nations affected by natural disasters)
  - Remittances
  - Hedging, arbitraging, trading

What makes Foreign Exchange Markets Unique?

Key Features

- The largest financial market
  - The average daily turnover in the FX market was over US$7.5 trillion by the April 2022.
- the extreme liquidity of the market; wholesale transactions: 95%
- the large number of, and variety of, traders in the market
- geographically dispersed
- no fixed opening and closing time. In fact, some foreign exchange dealers in certain banks maintain 24 hours a day operation.
- no centralized meeting place like stock exchange market
- transactions are done via telephones, or computer dealing systems, or through brokers. there is no physical transfer of one currency for another currency. For example:
  - Malaysian Ringgit for Swiss Francs: It is simply a bookkeeping entry whereby the bank in Malaysia and the bank in Switzerland credit each other's account for the currency. The Swiss bank keeps an account with the Malaysian bank and vice versa.

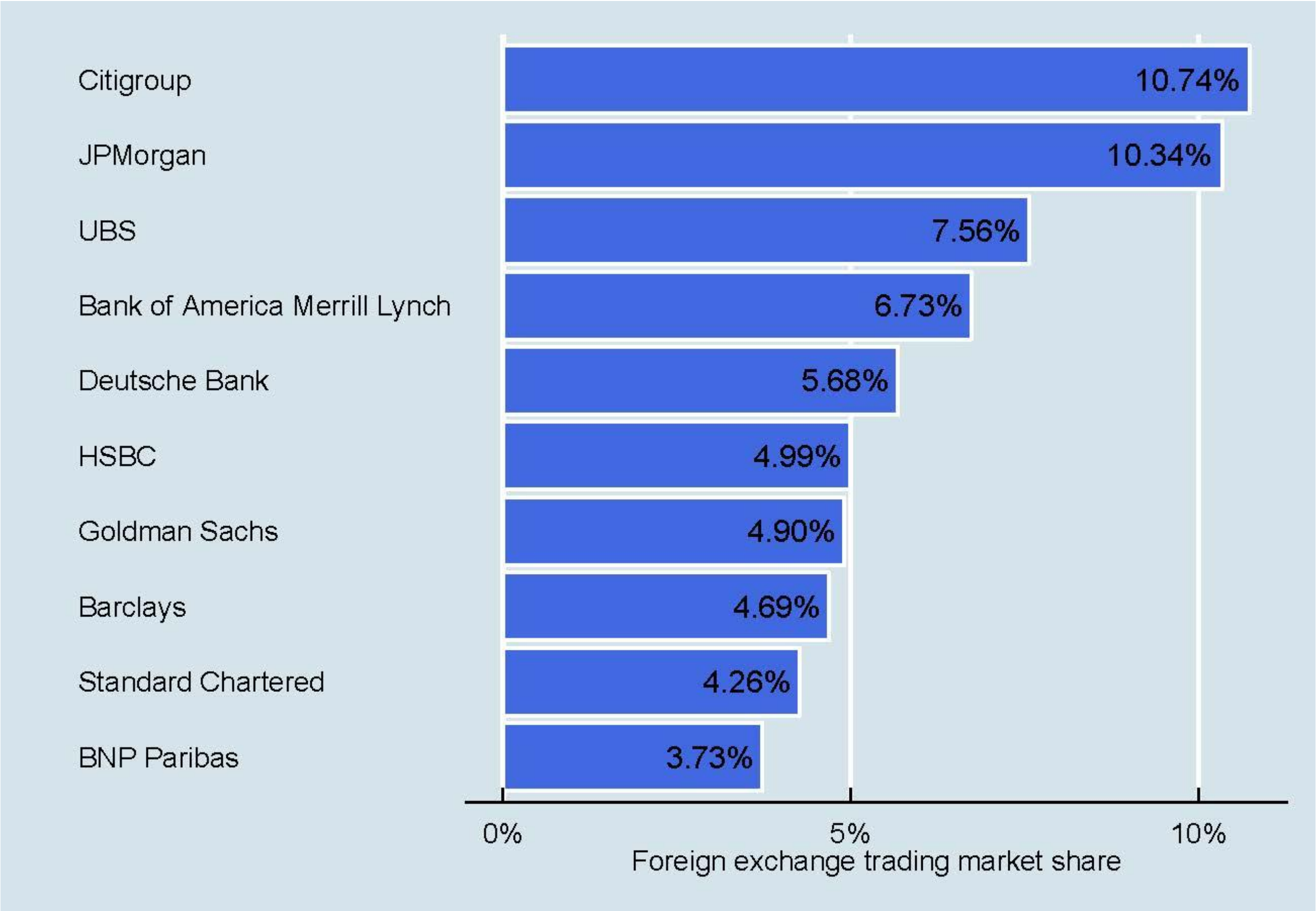
Foreign exchange market daily turnover by instrument



Source: BIS Triennial Central Bank Survey, Oct 2022

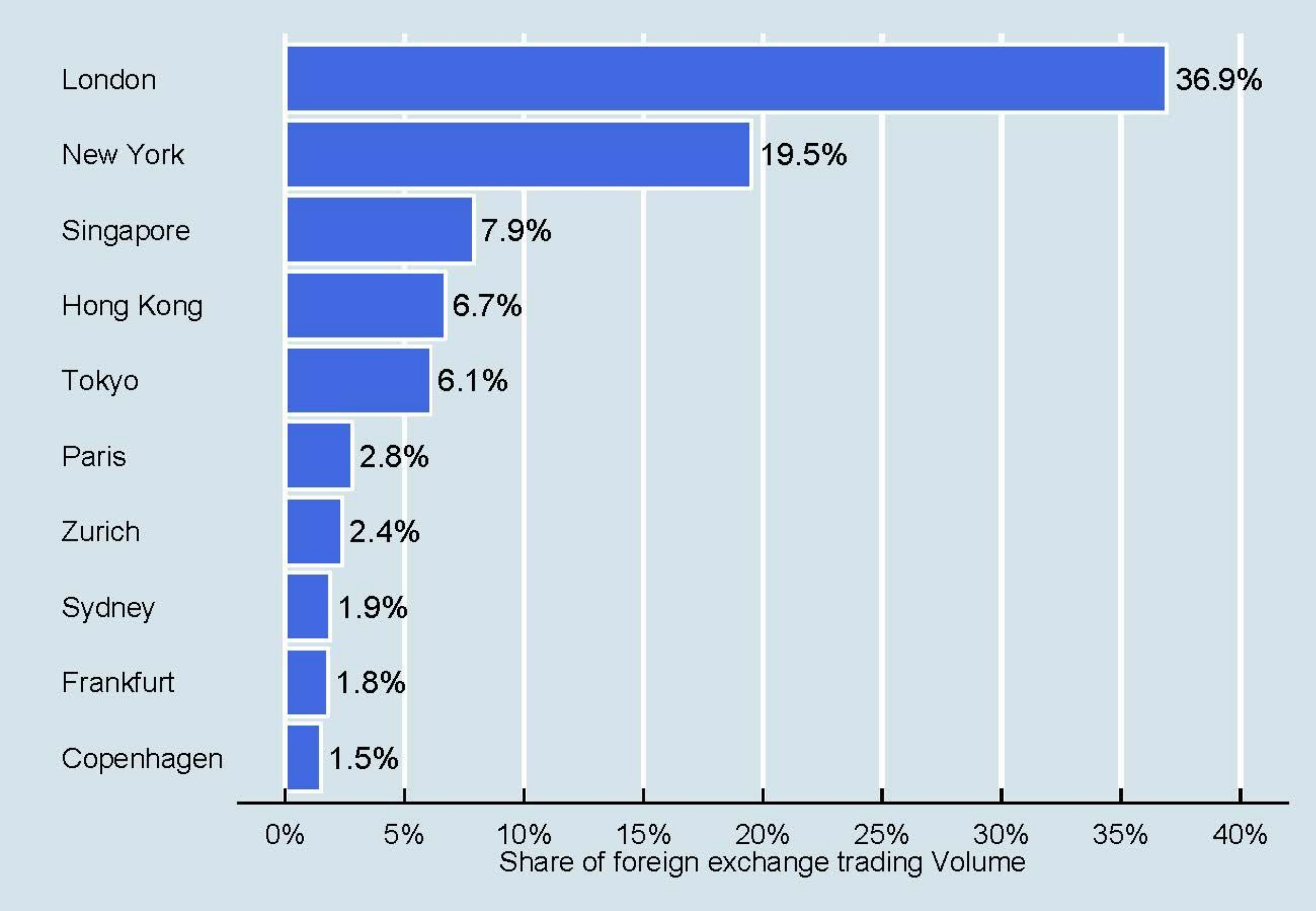
10 Largest FX traders

- There are roughly 20 large banks that account for most FX trading.
- The 10 Largest banks account for over 60% of all FX trades.



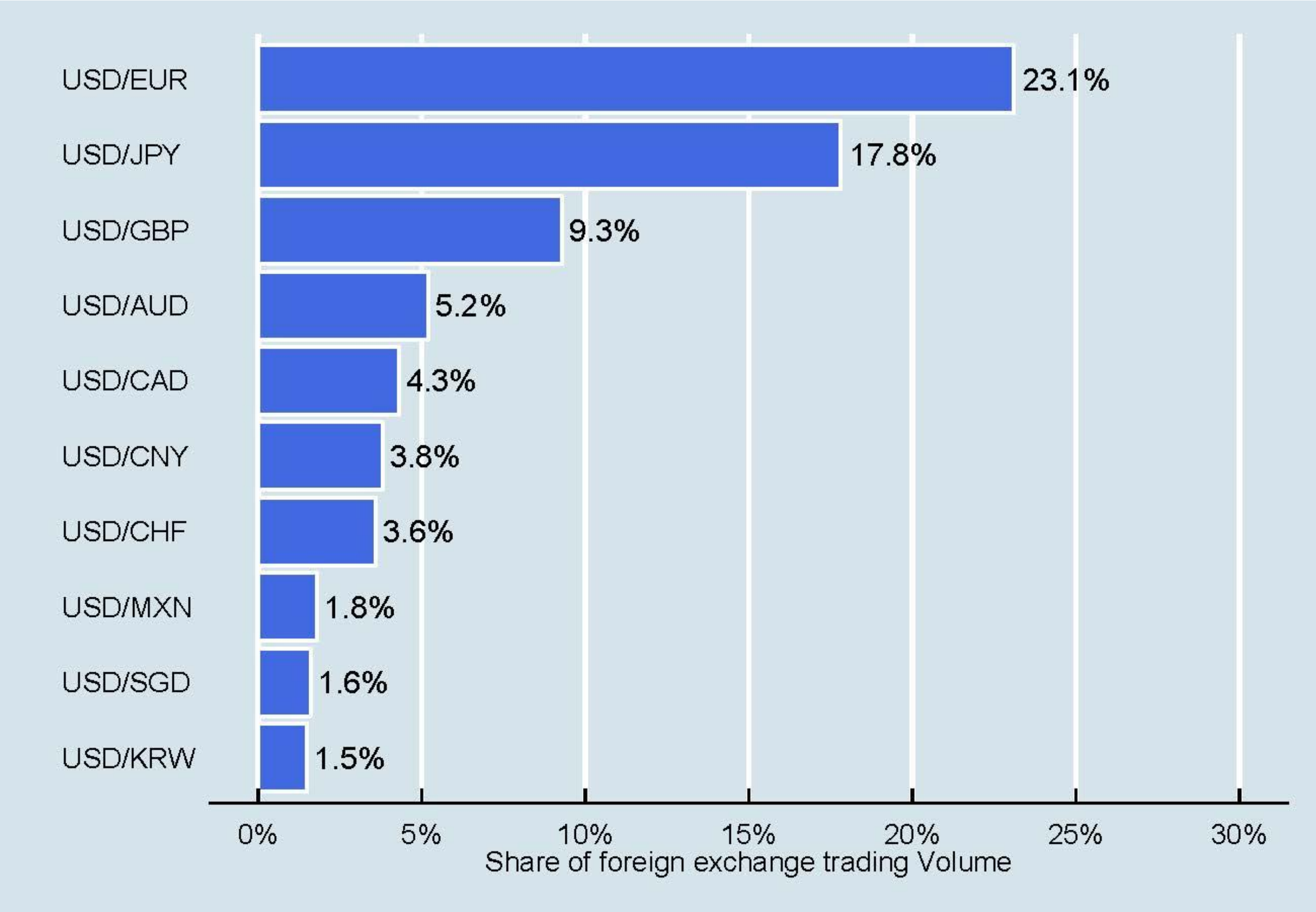
FX Financial Centers

- London with 1/3rd of all trading volume is the largest FX financial center
- The top 10 FX cities account for over 80% of all trades.



Most traded currency pairs

- Not surprisingly, the most traded currency pair is the USD/EUR since it ties the world's two largest economies.
- Before the Deutsche Mark (DEM) was superseded by the Euro in 2002 the USD/DEM was the most traded currency pair.



Market timing

The Foreign Exchange Market

Trades type

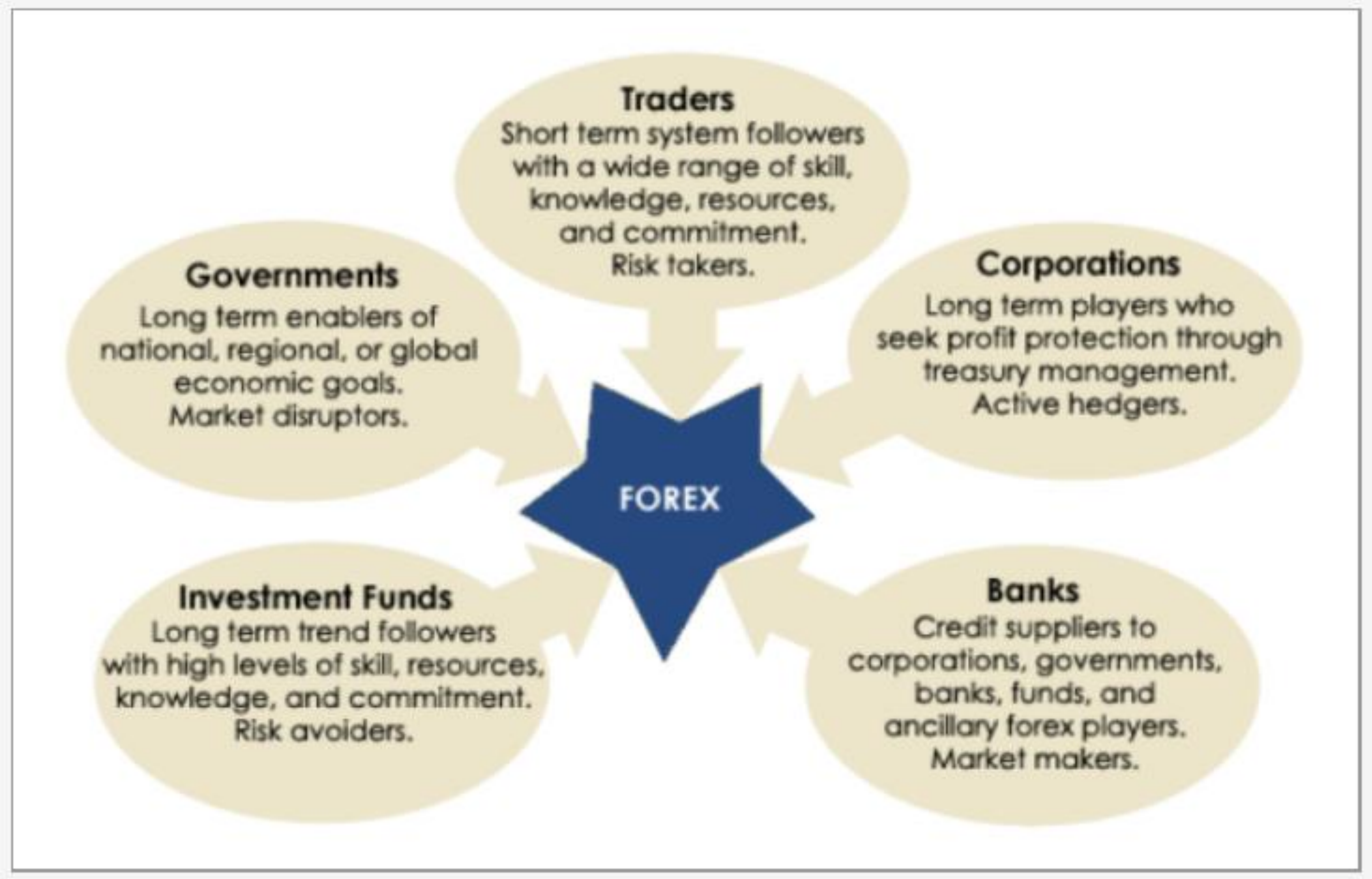
- Historically, most trades by phone, telex, or SWIFT

(SWIFT: Society for Worldwide Interbank Financial Telecommunications)

- More recently, electronic trading
    - Increases liquidity
    - Reduces trading cost
    - Makes information more accessible

The Foreign Exchange Market

Players/participants

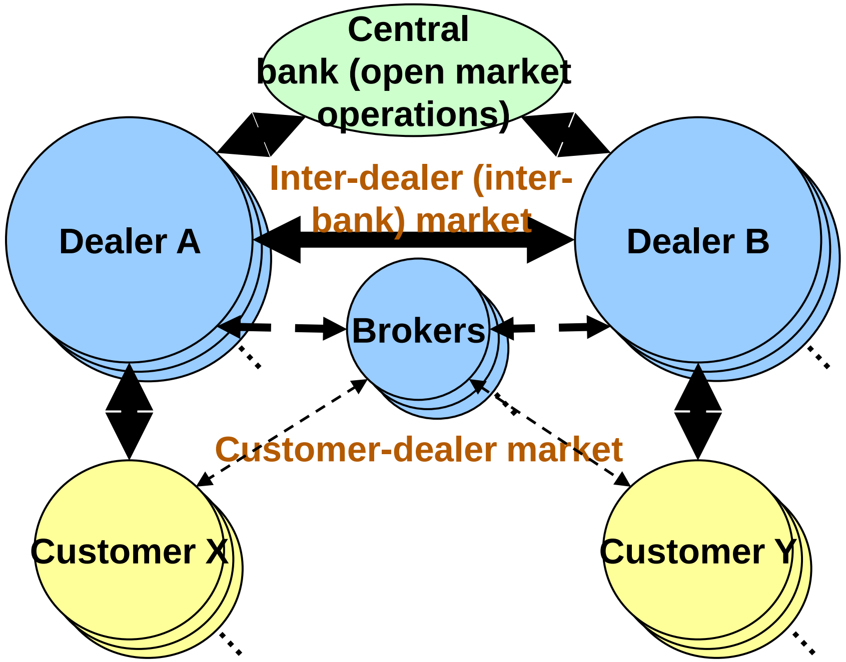


Source: http://learningcenter.fxstreet.com

Market is mainly dominated by OTC: Generic OTC participants

Participants:

- Customers (buyers and sellers)
  - Sellers want to sell instrument
  - Buyers have need for it
  - Instrument ultimately delivered by seller to buyer at agreed price
  - Dealers (speculators/arbitragers)
    - Often do not need instrument
    - Make market, often trade for themselves (buy and resell for profit - maintain bid/ask spread)
  - Brokers
    - Don’t trade for themselves
    - Act for buyers and sellers
    - Earn fees/brokerage



Exchange Rate

- The price of one currency expressed in terms of another currency
- Most currencies are quoted against the USD and it is also known as the base/commodity currency.
- In the currency market convention, all currencies are quoted in the following manner:-

  1 unit of a currency = x units of another currency

  e.g., 1 unit of US$ = JPY122.65
- This quotation means that it costs JPY122.65 to buy one US$ or if one wishes to convert/sell 1 US$ into JPY one would receive JPY122.65.

Depreciation/Appreciation

When we say that a currency is appreciating against another currency, it means that the currency is now worth more than the other currency.

- For example:
  - before 1 US$ = 1.4680 SFr
  - now 1 US$ = 1.4710 SFr

We say that US$ is appreciating against SFr because it costs 1.4710 SFr to buy 1 US$ now which is more than before by 0.0030 SFr

Types of FOREX transactions and quotations

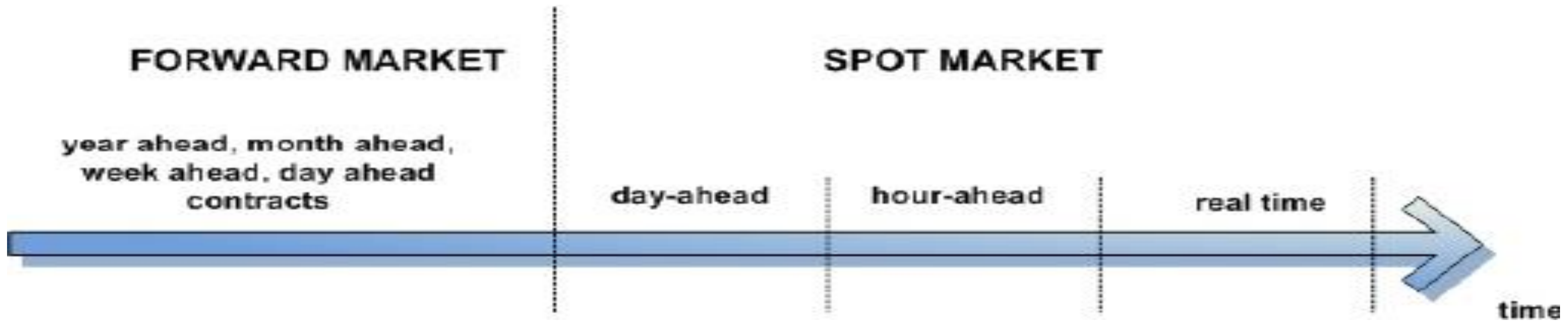
Types of Foreign Exchange Transactions

Spot Market:

- immediate transaction – cash market
- settled on the 2nd business day after the date of transaction

Forward Market:

- transactions at a pre-specified price take place on a specified future date



Spot FX Transactions

- These transactions involved the exchange of one currency for another at an agreed rate for delivery two working days after the deal date/contract date.
- The deal/contract date is the date on which the transaction is concluded.
- The value/delivery date is the date on which the cash flows occur.
- Example: Spot rate for AU$1 is US$ 1.0170:
  - if deal/contract date: 4th June 2024
  - then value/delivery date: 6th June 2024

Forward FX Transactions

Forward quotes:

- 30-day forward
- 90-day forward
- 180-day forward

- The value/delivery date of these forward FX transactions is later than the spot rate i.e., later than 2 working days.
- Example:
  - 1 month forward over spot for US$ /A$1 = 1.0150
- If the deal date is : 3rd June '24
- Then the value date is : 5th July '24

Foreign Exchange Rates and Quotations

Direct quote (or American terms*)* and *indirect quote* (or European terms)

- A *direct* quote is a home currency (DC) price of a unit of foreign currency (FC). (1 FC = x units DC); e.g., example: $1.21/€
- An *indirect* quote is a foreign currency price of a unit of home currency.

  (1 DC = x units FC); e.g., example: Peso1.713/$
- The form of the quote depends on what the speaker regards as "home."

Most interbank quotations around the world are stated in European terms/indirect quotes

Foreign Exchange Rates and Quotations

- Direct quotes place the domestic currency in the numerator and the foreign currency in the denominator (DC/FC);

  e.g., AUD1.2500/USD for a resident of Australia or
  €0.0090/¥ for a resident of Europe

- Indirect quotes place the domestic currency in the denominator and the foreign currency in the numerator (FC/DC);

  e.g., USD0.8000/AUD for a resident of Australia
  ¥110.95/€ for a resident of Europe

  - Direct and indirect quotes are reciprocals (next slide)

Reciprocal Rates

The reciprocal rate is the inverse/reciprocal of a conventional quote.

Examples:

- 1 US$ = SGD 1.7650 (SGD 1.7650/US$1)

Reciprocal rate is:

  - ➢1 SGD = US$ 1/1.7650=US$ 0.5666

- 1 AU$ = US$ 0.7650 (US$0.7650/AU$1)

Reciprocal rate is:

  - ➢1 US$ = AU$ 1.3072

Test your understanding of quotations

a. \$1.35/£

b. €1.41/£

c. \$1.35/ €

d. £0.55/€

e. £0.50/\$

f. SG\$0.50/AU\$

g. AU\$1.50/CHF

Commodity and Term Currency

In every foreign exchange transaction, there are 2 currencies involved. The commodity currency is expressed in terms of a number of units of the other currency. The other currency is known as the term currency.

Commodity currency is also called as a base currency, the currency that is bought or sold; while term currency is also referred to as the quoted currency.

Commodity and Term Currency: Example

(a) 1 AU$ = US$ 0.7850
(b) 1 US$ = SGD 1.7725

In (a) the *Australian dollar is the commodity (or base) currency*. One Australian dollar is expressed as 0.7850 units of US dollars and *the US$ is the term/quoted currency*.

In (b) *US$ is the commodity (or base) currency* and *SGD is the term currency (or term/quoted currency)*.

BID, ASK (or, buy and sell) and mid-rate

- In the interbank spot Fx market, the spot rate is always quoted as a 2-way price
- Buying price or the bid price on the left-hand side and the ASK/selling price or the offer price on the right-hand side.
- The average of bid-ask is called mid-rate.
- Example: the dealer quotes spot US$/AU$1 as follows

0.7850/0.7860 or 0.7850-60

(BID) (ASK or Sell) (BID) - (ASK)

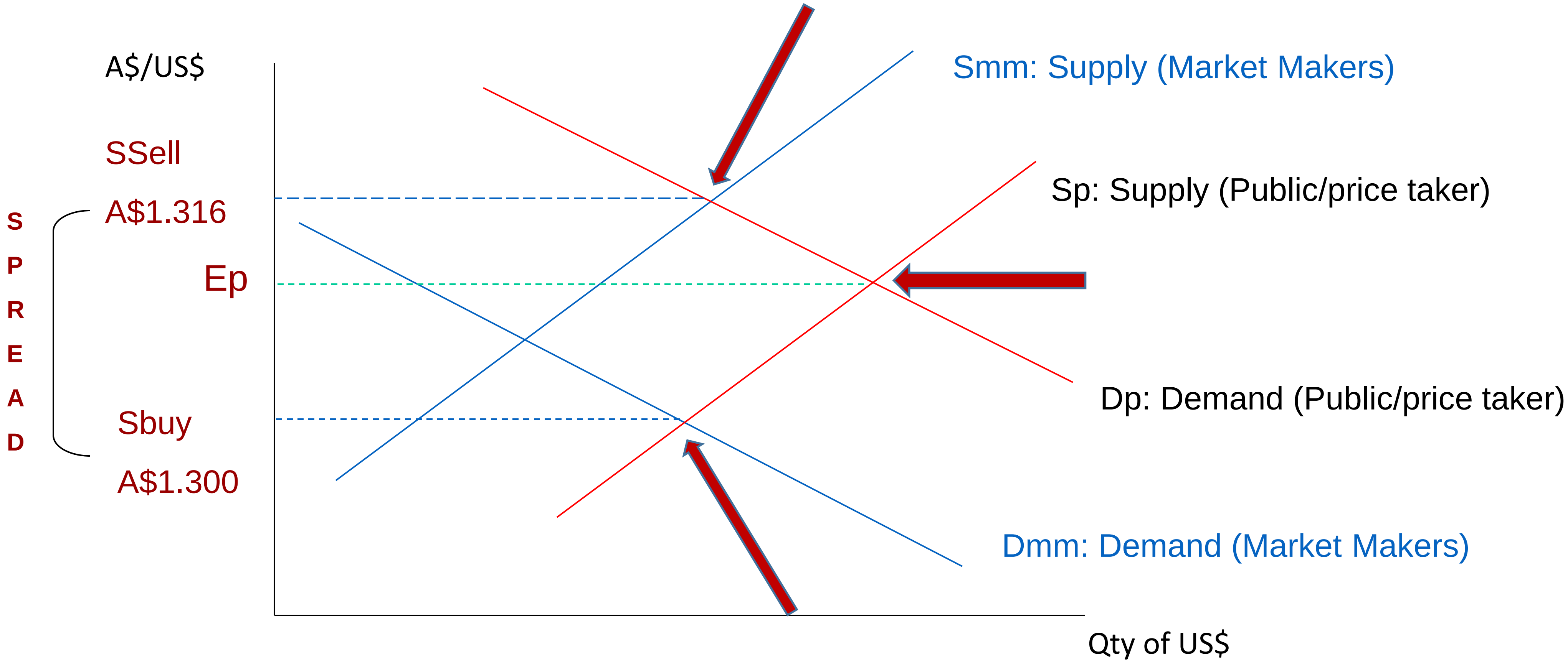
Spread/Transaction costs

- The spread/transaction cost is the difference between bid rate and offer rate.
- For example, the spread is 10 points or 10 pips in the following quotation:

  i.e., 0.7860 - 0.7850 = 0.0010 or 10 pips
- A pip/point refers to one unit in the final decimal place to which a given exchange rate is conventionally quoted.
- In the above example, the spread is = 10000 x (0.7860 – 0.7850) = US$10
- Bid-Ask spread in percentage = [(Ask-Bid)/Ask]*100
- In the real world, spread is typically 2-5 pips
- Why the interbank market spread is narrow than the retail market? (Answer: Large volume, highly competitive, see Topic: International financial markets)

Spread/Transaction costs

Buy-sell or Bid-ask spread using direct quotes for AUD vs USD



A$1.316/US$ = US$0.76/AU$
A$1.300/US$ = US$0.77/AU$

Spread/Transaction costs

- Assume initially the public/price taker's demand and supply functions in the spot market are Dp and Sp
- If public's "bid" and "ask" tenders could be revealed to each other $\Rightarrow$ Equilibrium at "Ep" would occur
- Dmm (market marker's demand curve) - dealers willingness to <u>purchase</u> US$ (with $A) from the public
- Smm – (market maker's supply curve) – dealers willingness to <u>sell</u> US$ (for $A) to the public
- *Intersection of Dmm with Sp* determines the exchange rate at which US$ are bought from the public, i.e. Sbuy
- *Intersection of Dp with Smm* determines the exchange rate at which US$ are sold to the public, i.e. Ssell

Interbank market: Quoting Bank (Price/market maker) and Calling Bank (Price taker)

- In the interbank FX market, there are 2 parties involved - the calling bank (price maker) and the quoting bank (market maker).
- The quoting bank is the bank that quotes the 2-way price while the calling bank is the bank that calls to ask the quoting bank for its 2-way price.
- When a price is quoted, the calling bank can either accept it in which case a deal is contracted or reject it with no transaction occurring

Exercise

- Bank of Tokyo (the market maker) quotes you the following rates:

  JPY/US$1 = 81.55/81.58 or 81.55-58

- If you are a buyer of USD from BoT, which rate does apply to you?
- If you are a seller of USD to BoT, which rate does apply to you?

Cross Rates

A cross rate is a rate which derives from two other exchange rates. The rates given are as follows:

- US$/AU$1 = 0.7250
- US$/SFr1 = 0.6845

Which rate do you need to calculate?

Answer: Cross rate between AU$/SFr

Cross rates

- Monday, September 10, 2022, listed the yen–dollar and euro–dollar rates as ¥78.56 and €0.7802, respectively. Suppose we need to know euro–yen exchange rate, i.e.

$$\frac{€}{¥}$$

- How to get this rate?
- Because the dollar is the common currency here, cancel it. Think in terms of the currencies involved: When you divide the euro–dollar exchange rate by the yen–dollar exchange rate, the dollars cancel and you get the euro–yen exchange rate:

$$\frac{€}{\$} \div \frac{¥}{\$} = \frac{€}{\cancel{\$}} \times \frac{\cancel{\$}}{¥} = \frac{€}{¥}$$

Cross rates

Calculating Cross Rates (American terms/direct quote)

Suppose you are given the following quotations/rates:

AUD\$1.30/€ and AUD\$1.60/£

1) If you are asked to calculate the £/€ cross rate, then:

$$\frac{AUD}{EUR} \div \frac{AUD}{GBP} = \frac{\cancel{AUD}}{EUR} x \frac{GBP}{\cancel{AUD}} = \frac{GBP}{EUR}$$

£/€ rate = AUD\$1.30 /€ ÷ AUD\$1.60 /£ = £0.8125/€

2) If you are asked to calculate the €/£ cross rate, then:

$$\frac{AUD}{GBP} \div \frac{AUD}{EUR} = \frac{\cancel{AUD}}{GBP} x \frac{EUR}{\cancel{AUD}} = \frac{EUR}{GBP}$$

£/€ rate = AUD\$1.60 /£ ÷ AUD\$1.30 /€ = €1.2308/£

Another way:
making the direct into indirect
and then cross multiply

£/AUD=1/1.60 = £0.625/AUD;
so, £/€ =
0.625*1.30=£0.8125/€

€/AUD=1/1.30 = €0.7692/AUD;
so, €/£ =
0.7692*1.60=€1.2308/£

Cross rates

Calculating Cross Rates (European terms/indirect quote)

Suppose you are given the following rates:

€0.7692/AUD$ and £0.6250/AUD$

1) If you are asked to calculate the £/€ cross rate, then:

$$\frac{GBP}{AUD} \div \frac{EUR}{AUD} = \frac{GBP}{\cancel{AUD}} x \frac{\cancel{AUD}}{EUR} = \frac{GBP}{EUR}$$

£/€ rate = £0.6250/AUD$ ÷ €0.7692/AUD$ = £0.8125/€

2) If you are asked to calculate the €/£ cross rate, then:

$$\frac{EUR}{AUD} \div \frac{GBP}{AUD} = \frac{EUR}{\cancel{AUD}} x \frac{\cancel{AUD}}{GBP} = \frac{EUR}{GBP}$$

€/£ rate = €0.7692/AUD$ ÷ £0.6250/AUD$ = €1.2307/£

Cross rates

From mixed quotes (direct and indirect quotes)

Given US$/AU$1 = 0.7250

and SFr/US$1 = 1.4610 (SFr or CHF = Swiss Franc)

What rate is not given?

Q: What is the cross rate for AU$ in terms of SFr? or

Q: What is the cross rate for SFr in terms of AU$

First get the reciprocal of AU$1=0.7250 of US (i.e., 1 US$ = 1.3793 AU$)

– **SFr/AU$1 =** 1.4610/1.3793 = 1.0592 SFr/AU$1

– **AU$/SFr1 = 1/1.0592 =** 0.9441 AU$/SFr1 (i.e., reciprocal rate of SFr/A$1).

Or, just multiply for a mixed quote

Exercise

- Given SGD/US$1 1.2835
  US$/€ 1 1.2165

What is the cross rate for € in terms of SGD (SGD/€1) and its reciprocal, i.e., €/SGD$1?

Or, just multiply for a mixed quote

First get the reciprocal of US$/€1; €/US$1 =1/1.2165 =0.8220

Then, get the cross-rate **SGD/€1**: 1.2835/0.8220 = SGD$1.5614/ €1

Spot market cross rates and FX arbitrage

Cross rates and Arbitrage

- If cross rates differ from one financial centre to another, then profit opportunities exist
- Arbitragers take this cross-rate difference and make profit
- Buy cheap in one market, sell at a higher price in another

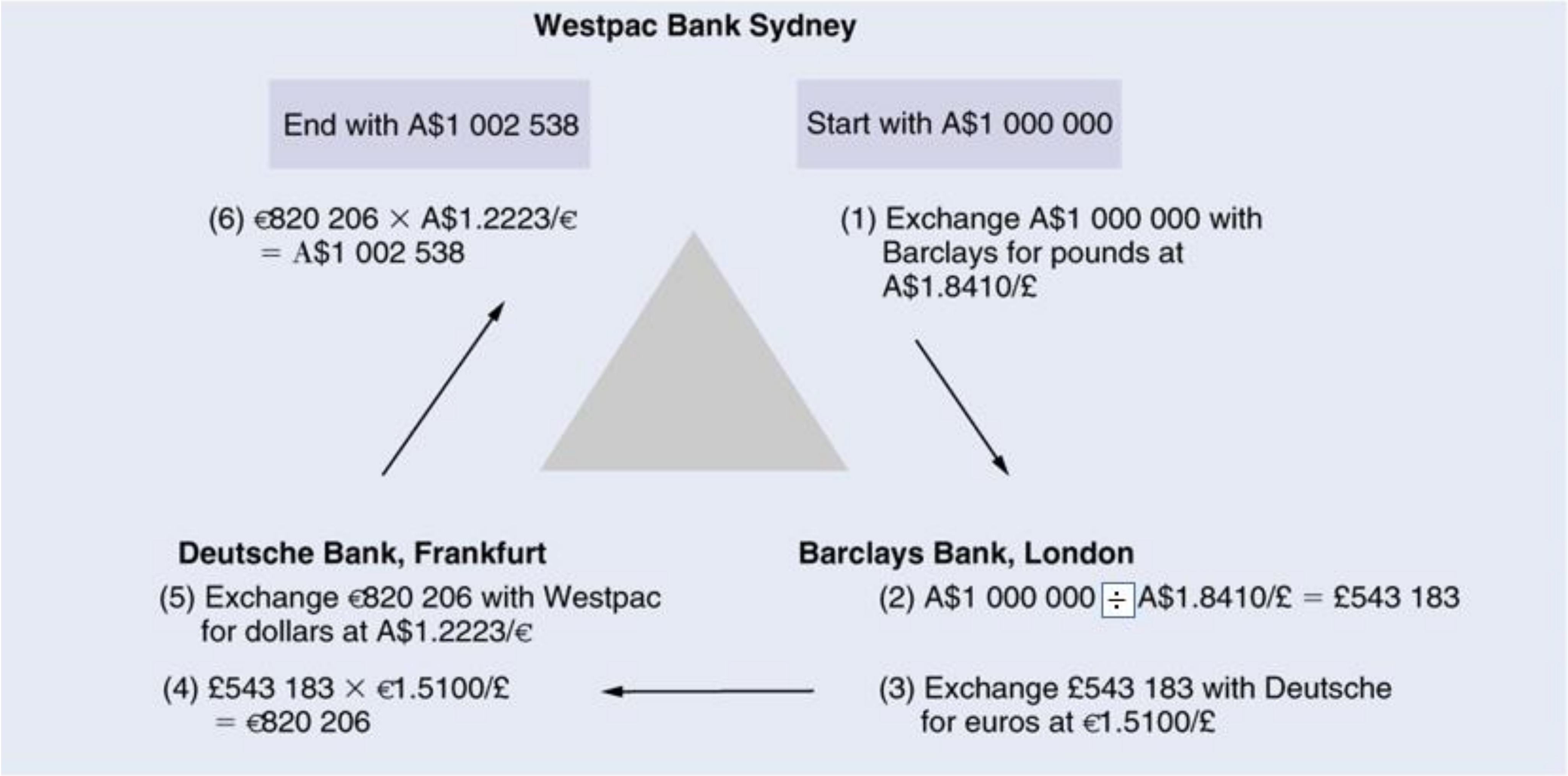
Currency arbitrage: An example

Barclays Bank quotes Australian dollars per pound sterling:	A\$1.8410/£
Deutsche Bank quotes euros per pound sterling:	€1.5100/£
Westpac Bank quotes Australian dollars per euro:	A\$1.2223/€

The cross rate between Westpac and Barclays will yield €/£

$$\frac{A\$1.8410/£}{A\$1.2223/€} = €1.5062/£$$

Triangular currency arbitrage



Note: transaction (2) in the textbook has an error, it should be divided: A$1 000 000/1.8410= £543 183

Forward markets

The Forward Market

- Definition of a Forward Contract:

an agreement between two parties for the delivery, on a specified future date, of a specified amount of currency against another currency at a fixed exchange rate

- Use of a Forward:
  - **Hedging**: reducing/eliminating exchange rate risk for existing FX position: Example >> if you have a AU$ long position and worry AU$ to fall, then sell a forward AU$ contract; if you have a short AU$ position and worry AU$ to rise, then buy a forward AU$ contract
  - **Speculation**: taking open positions in the FX market: Example >> If you think AU$ is going to appreciate, then buy a forward contract; If you think AU$ is going to depreciate then sell a forward contract

- Forward premium
  - A currency's value in the forward exchange market is *higher* than in the spot exchange market
- Forward discount
  - A currency' value in the forward exchange market is *lower* than in the spot exchange market

The Forward Market

Calculating the Forward Premium or Discount on ***foreign*** currency (Direct formula)

$$\frac{f_1 - e_0}{e_0} * \frac{360}{n}$$

where f_1 = the 1-period forward rate (direct quote)

e_0 = the current spot rate (direct quote)

n = the number of days in the forward contract

The Forward Market

Calculating the Forward Premium or Discount on ***foreign*** currency (Indirect formula)

$$\frac{e_0' - f_1'}{f_1'} * \frac{360}{n}$$

where f_1' = the 1-period forward rate (indirect quote)

e_0' = the current spot rate (indirect quote)

n = the number of days in the forward contract

% Forward Premium/Discounts

- Note: both formulas calculate the forward premium/discount of the <u>foreign</u> currency relative to the domestic currency.
- Regardless of using direct or indirect quotes, the calculation of the premium should be the same, provided that the currency quote is consistent with the formula used, i.e., for direct quote use direct formula, for indirect quote use indirect formula.

Indirect Quotes % Forward Premium/Discounts

- An example, assume the following indirect quotes (i.e. AUD is the home currency)

Spot rate e_0' = SFr1.7126/AUD

Forward rate f_1' = SFr 1.7373/AUD

Forward contract number of days

= 180 (6 months)

Indirect Quotes and % Forward Premium/Discount

- SFr is selling at a forward discount relative to AUD
- AUD is selling at a forward premium relative to SFr
- Using indirect formula, the forward premium/discount of SFr is calculated as:

$$\frac{e_0' - f_1'}{f_1'} * \frac{360}{n}$$
$$= \frac{1.7126 - 1.7373}{1.7373} * \frac{360}{180}$$
$$= -0.0284 \; or -2.84\%$$

Therefore, SFr (AUD) is selling at a forward discount (premium) of -2.84%

Direct Quotes and % Forward Premium/Discount

In order to use the DIRECT formula, convert indirect quotes to direct quotes:

Spot rate e_0 = 1/1.7126= AUD0.5839/SFr

Forward rate f_1= 1/1.7373=AUD 0.5756/SFr

Using direct formula, the forward premium/discount of SFr is calculated as:

$$\frac{f_1 - e_0}{e_0} * \frac{360}{n}$$
$$= \frac{0.5756 - 0.5839}{0.5839} * \frac{360}{180}$$
$$= -2.84\%$$

Again, Therefore, SFr (AUD) is selling at a forward discount (premium) of -2.84%

Forward rate and interest rate parity (IRP)

- IRP links the FX markets and the international money markets
- The interest rate (i/r) differential should be approximately equal to the forward premium/discount
  - e.g., if i/r in Australia is 3% higher than i/r in the U.S., the AUD should be selling at (approximately) a 3% forward discount against the USD
- When the above condition is met, <u>the spot and forward rates are said to be at IRP, and</u> the covered i/r differential equals to zero, i.e., there are no CIA (covered interest arbitrage) opportunities

Forward rate and interest rate parity (IRP)

- The no-arbitrage condition is

Direct quote	Indirect quote
$\frac{f1}{eo} = \frac{(1+r_h)}{(1+r_f)}$	$\frac{eo}{f1} = \frac{(1+r_h)}{(1+r_f)}$

- To preclude arbitrage, forward rate should be calculated as:

Direct quote	Indirect quote
$f1 = eo\frac{(1+r_h)}{(1+r_f)}$	$f1 = eo\frac{(1+r_f)}{(1+r_h)}$

$f1$ (eo) is forward (spot) rate
r_h = nominal i/r in home country; r_f = nominal i/r in foreign country
Interest rates need to be adjusted for the time periods

Forward rate and interest rate parity (IRP)

The 90-day interest rates **(annualized)** in the U.S. and Japan are, respectively, 10% and 7%, while the spot quote for the yen in New York is $0.004300. At what 90-day forward rate would interest rate parity hold?

Answer

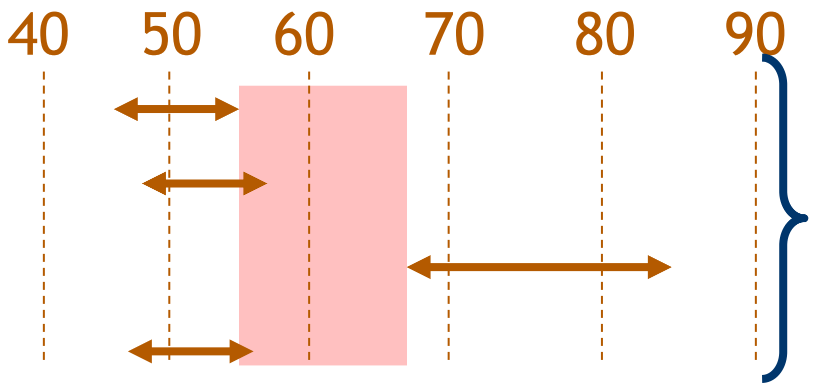
$$f_1 = e_0 \frac{1 + r_h}{1 + r_f}$$

$$= 0.0043 \frac{1 + \frac{0.1}{4}}{1 + \frac{0.07}{4}}$$

$$= 0.004332$$

1 problem with multiple MCQs

	USD/AUD
• Bank A	0.9045/55
• Bank B	0.9047/57
• Bank C	0.9067/87
• Bank D	0.9046/56



An introduction to Forex Trading

See additional slides (it is non-assessable) uploaded in the cloudsite

MAF306: Week 7_Topic 7

Foreign exchange derivatives: Forwards, Futures, Options and Swap

Dr Sohel Azad

DEAKIN BUSINESS SCHOOL

AACSB ACCREDITED

EFMD EPAS ACCREDITED

Foreign exchange derivatives

Objectives

- To explain what currency futures, forwards, options and swap contracts are
- To compare and contrast between currency forward, futures, option and swap contracts
- To identify the basic factors that determine the value of a currency option

- **Reading:**
- Shapiro: Ch 7 and 8

Derivatives at a glance

History of currency derivatives

- While currency derivatives have existed for a long time, their popularity soared in the 1970's. Why?
- Recall that in 1971, the Bretton Woods system of fixed exchange rates ended, and exchange rates became highly volatile.
- Such volatility meant that under the floating exchange rate system MNCs started to face higher foreign exchange rate risk and use currency derivatives.
- Changes in exchange rates could make foreign competitors more (or less) attractive (both at home and abroad), increase (or decrease) the MNCs costs from foreign suppliers, decrease (or increase) the value of foreign assets, etc.

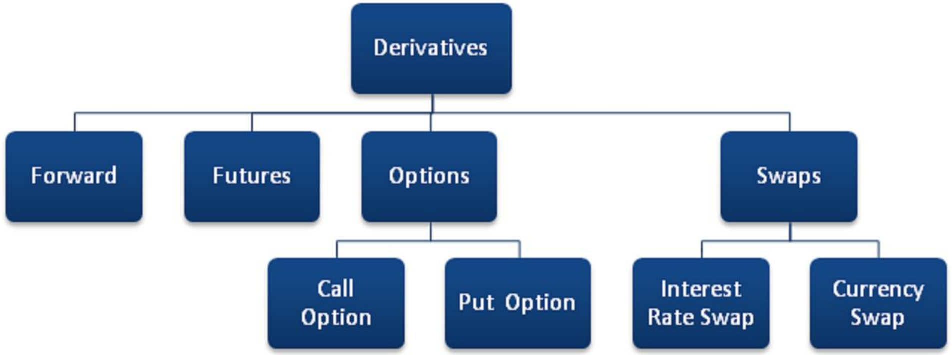


Photo by finideas

Derivative Securities

- Derivatives are contracts that derive their value from some underlying instruments
- "Derivatives are financial weapons of mass destruction" - Warren Buffett
- Sons of Gwalia (SoG) was a Perth based mining company established in 1981. It was Australia's 3rd-largest gold miner and the world's largest miner of tantalum, a key input in computer and mobile phone production. SoG was a listed company on the ASX, and its market capitalization exceeded AU$1 billion during its peak time in 2001. In August 2004, the company was placed into voluntary administration (filing for bankruptcy).
- What went wrong?
  - In order to achieve high profits, SoG purchased exotic options (on Gold) called Indexed Gold Put Options, which allowed SoG to sell gold at a price higher than the market price. This exposed SoG to enormous risk since the company was effectively holding 5 sold calls for each bought put. When the spot gold price increased later, the embedded call options became in-the-money, and SoG suffered huge losses.

Other failures of derivatives

Date	Company	Description
1991	Allied-Lyons (UK)	Losses of £165 million related to speculation on currency options
1993	Shell Showa Sekiyu (Japan)	Over ¥1.5 billion in losses arising from recognition of cumulative losses on forward contracts continually rolled over between 1989 and 1993
1993	Metallgesellschaft (Germany)	A flawed petroleum futures hedging strategy essentially caused the collapse of the organisation
1994	Codelco (Chile)	A copper futures trader for the national copper company of Chile, Codelco, loses approximately 0.5% of Chile's gross domestic product for 1994 through speculative futures trading
1994	Kashima Oil (Japan)	Hundreds of millions of yen are lost on a failed forward speculation on the Japanese yen
1994	Procter & Gamble (US), Gibson Greeting Cards (US), Air Products (US), Dharmala (Indonesia)	All suffer material losses in the millions of dollars on leveraged swap agreements with Bankers Trust of the United States
1995	Barings Brothers (UK)	The oldest investment bank in London fails as a result of the losses on futures trading suffered by one trader in its Singapore office, a Mr Nicholas Leeson (see the case study)
2002	Allied Irish Bank (US/UK)	A rogue currency trader in the Baltimore offices of Allied Irish Bank is credited with losing more than USD691 million

Page 218, Eiteman/2nd Ed, 2011

Foreign exchange derivatives

- The value of foreign exchange derivatives depends on the value of the currency underlying the contract
- Major types of foreign exchange derivatives:
  - Futures
  - Forwards
  - Options and
  - Swaps
- Users of derivatives:
  - Hedgers
  - Speculators
  - Arbitrageurs

Currency futures and forwards

Currency Futures and Forwards

- Futures are more common than forwards
- A futures contract is *like* a forward contract done in advance for future cash flows. However, a futures contract is *different from* a forward contract (see the table below):

Forwards	Futures
Over the counter	Exchange Traded
Self-regulating	Regulated by the Commodity and Futures Trading Commission
Over 90% are settled by actual delivery.	Less than 1% have actual delivery
Customized Contract Size	Standardized Contract Size
Any delivery date	Can only be delivered on specific dates a year
Settlement occurs on actual date	Settlement is made daily via Mark-to-Market
Quoted in units of foreign currency per $	Quoted in $ for 1 unit of foreign currency
Transaction costs are based on bid-ask spread	Transaction costs are based on broker fees
Margins are not required	Margins are required
Counterparty risk exist for both sides.	The Exchange assumes the counterparty risk

Advantages and disadvantages of futures

Advantages	Disadvantages/limitations
1. Smaller size 2. Easy liquidation 3. Well-organized market	1. Available for limited number of currencies 2. Limited dates of delivery 3. Rigid contract sizes 4. Hedging mismatch ▪ Size ▪ Maturity 5. Margin calls 6. No upside

Currency Futures - Hedging

- Australian Importer worried that AU dollar will depreciate (USD payment increases):
  - Sell AUD futures - upon delivery, sell AUD for USD at the predetermined exchange rate specified in the futures contract

- Australian Exporter worried that AU dollar will appreciate (USD return decreases)
  - Buy AUD futures – upon delivery, sell USD for AUD at the pre-specified exchange rate

Global futures exchanges

1. Chicago Mercantile Exchange (CME)
2. London International Financial Futures Exchange (LIFFE)
3. Chicago Board of Trade (CBOT)
4. Singapore International Monetary Exchange (SIMEX)
5. Deutsche Termin Bourse (DTB)
6. Hong Kong Futures Exchange (HKFE)
7. Sydney Futures Exchange (SFE)

Standardized Contracts

Some Futures Contracts traded in Chicago Mercantile Exchange: Currency & Contract Size

1. Australian dollar / A$100,000
2. British pound / £62,500
3. Canadian dollar /C$100,000
4. Euro / €125,000
5. Japanese yen / ¥12.5 million
6. Mexican peso / P500,000
7. Swiss franc / SFr125,000

Standardized Contracts

TABLE 7.1

Contract specification for selected major currency futures

Source: **CME Group.**

	Australian Dollar	**British Pound**	**Canadian Dollar**	**Euro**	**Japanese Yen**	**Swiss Franc**
Contract Size	AUD100,000	GBP62,500	CAD100,000	EUR125,000	JPY12,500,000	CHF125,000
Margin requirements						
Initial (USD)	1,375	2,420	1,265	2,530	1,980	2,860
Maintenance (USD)	1,250	2,200	1,150	2,300	1,800	2,600
Minimum Price (USD)	0.0001	0.0001	0.00005	0.00005	0.0000005	0.0001
Change	1 pt.	1 pt.	1/2 pt.	1/2 pt.	1/2 pt.	1 pt.
Value of 1 point (USD)	10.00	6.25	5.00	6.25	6.25	12.50
Months traded		March, June, September, December				

The Operation of Margins

- Investors are required to hold a margin account (MA) with their broker
  - Initial margin (IM) - the deposit made by the investor when a futures contract is first entered into
  - Maintenance margin (MM) – minimum balance required for the margin account.

- Daily settlement
  - The gain/loss made on the futures contract is calculated and added to/subtracted from the margin account at the end of each trading day
  - The investor receives a *margin call* if the MA balance falls below the MM. The investor is required to top up the MA to the IM level upon receiving the margin call
  - This settlement process is also known as marking to market

The Operation of Margins

- Exchange clearing house acts as an intermediary in futures transactions, it is the counterparty in each futures contract

- Brokers maintain similar margin accounts with the clearing house

- The operation of the margin system significantly reduces counterparty default risk in the futures market

Daily Settlement: An Example

- Consider a long position in the CME (Chicago Mercantile Exchange) Euro futures contract:
  - Long (short) position = Enter the contract to buy (sell) the underlying currency
- The contract is written on SFr125,000
- The delivery price specified in the contract is \$0.95/SFr
- Maturity is 3 months
- The initial margin is \$2,700
- The maintenance margin is \$4,000

Daily Settlement: An Example

An investor with a long position in the contract agrees to BUY SFr125,000 at $0.95 per SFr on Tuesday morning to mature in three days time. However, this position is marked to market on a daily basis as illustrated below:

TABLE 7.2

An example of daily settlement with a futures contract

Time	Action	Cash Flow
Tuesday morning	• Investor buys CHF futures contract that matures in two days. Price is $0.95	• None
Tuesday close	• Futures price rises to $0.955 • Position is marked to market	• Investor receives: 125,000 × (0.955 – 0.95) = $625
Wednesday close	• Futures price drops to $0.943 • Position is marked-to market	• Investor pays: 125,000 × (0.955 – 0.943) = $1,500
Thursday close	• Future price drops to $0.94 • (1) Contract is marked to market • (2) Investor takes delivery of CHF125,000	• Investor pays: 125,000 × (0.943 – 0.94) = $375 • (2) Investor pays: 125,000 × 0.94 = $117,500 Net loss: $1,250

Forward-Futures Arbitrage

Keeps futures rates in line with forward rates. If they are not in unison, arbitrage will take place

Example:

- Bank forward bid for Oct 14/2020 on £ is $1.8927
- Price of £ futures for delivery on Oct 14/2020 is $1.8915
- An arbitrager can simultaneously buy £ futures, sell £ to the bank at the forward bid rate and pocket the difference
- Contract size for £ futures is £1,000,000. Suppose the arbitrager simultaneously bought a £ futures contract and sold £1,000,000 forward.
- Profit = (1.8927-1.8915)* 1,000,000 =$1200

Currency options

Currency Options

- Definition:
  - a contract giving the holder (the buyer) the right BUT not the obligation to buy or sell a given amount of currency at a fixed exchange rate by a certain date

- Basic types of options
  - Call Options – give the holder the right to buy currency
  - Put Options – give the holder the right to sell currency

- Trading:
  - Unlike futures, options are traded both on exchanges and in the OTC markets
  - Exchange traded currency options - first offered on Philadelphia Exchange (PHLX) in 1983

Currency Options: Usage

- Hedging
  - An ***importer*** with accounts payable in foreign currency can ***buy call*** on foreign currency.
  - An ***exporter*** with accounts receivable in foreign currency can ***buy put*** on foreign currency
- Speculation
  - If speculating on the ***appreciation (depreciation)*** of foreign currency, ***buy call (put)*** on the foreign currency
  - More risky strategy is to write options – example of Sons of Gwalia
- Arbitrage
  - Relies heavily on option pricing models to identify arbitrage opportunities
    - Binomial option pricing model
    - Black-Scholes option pricing model
    - Monte Carlo simulation

Option Terminology

- *Exercise price* or *strike price*
  - The price specified in an option contract at which the underlying currency can be bought or sold
- *Option premium*
  - The value or price of the option
- *Expiration date* or *maturity date*
  - The date specified in the option contract
- *European option*
  - Can be exercised only on the maturity date
- *American option*
  - Can be exercised at any time during the life of the option

Currency Options

- Status of an option
  - in-the-money
    - Call: Spot > strike
    - Put: Spot < strike
  - out-of-the-money
    - Call: Spot < strike
    - Put: Spot > strike
  - at-the-money
    - Spot = strike

If the underlying fx spot price does not move above (lower) the strike price before the option expiration date, the call (put) option will expire worthless

Currency Options

- Some notations:

  S – current spot exchange rate

  S_T – spot rate at expiration date of the option

  X – exercise or strike price

  c – call premium

  p – put premium

- Four types of option positions
  - A long position in a call option (option buyer)
  - A long position in a put option (option buyer)
  - A short position in a call option (option seller/writer)
  - A short position in a put option (option seller/writer)

Currency Options

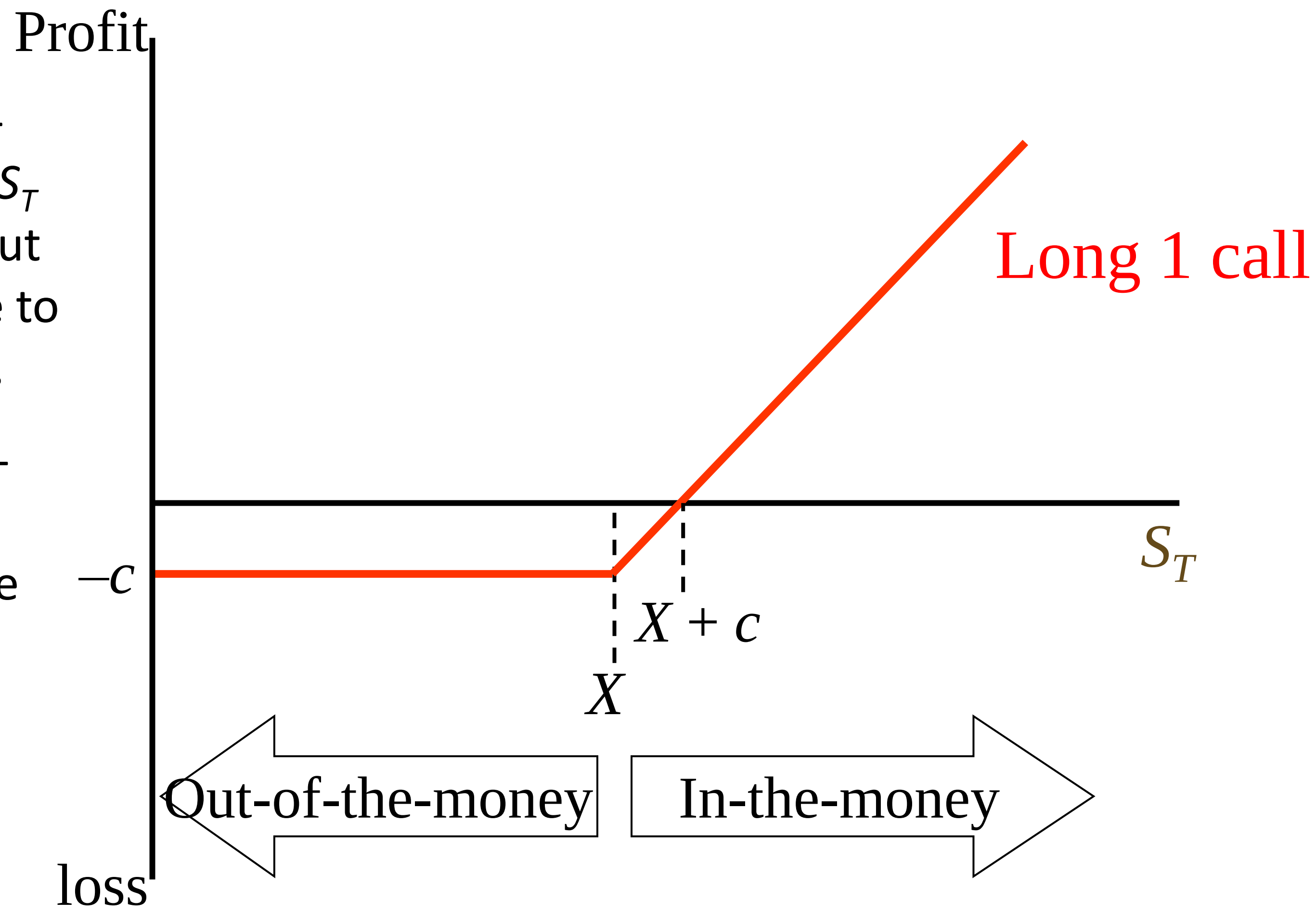
Assume all options are European (Standard)

- Option payoff:
  - From a long call: max $(S_T - X, 0)$
  - From a long put: max $(X - S_T, 0)$
  - Options are zero-sum games, therefore payoff from short call (put) is the opposite to the payoff from long call (put)
- Option profit
  - For option buyer (long position): profit = payoff – premium
  - For option seller (short position): profit = payoff + premium

Call Option Profit at maturity - Buyer

If the call is in-the-money, it pays off $S_T - X$ *to the buyer*. But profit < payoff due to the premium paid.

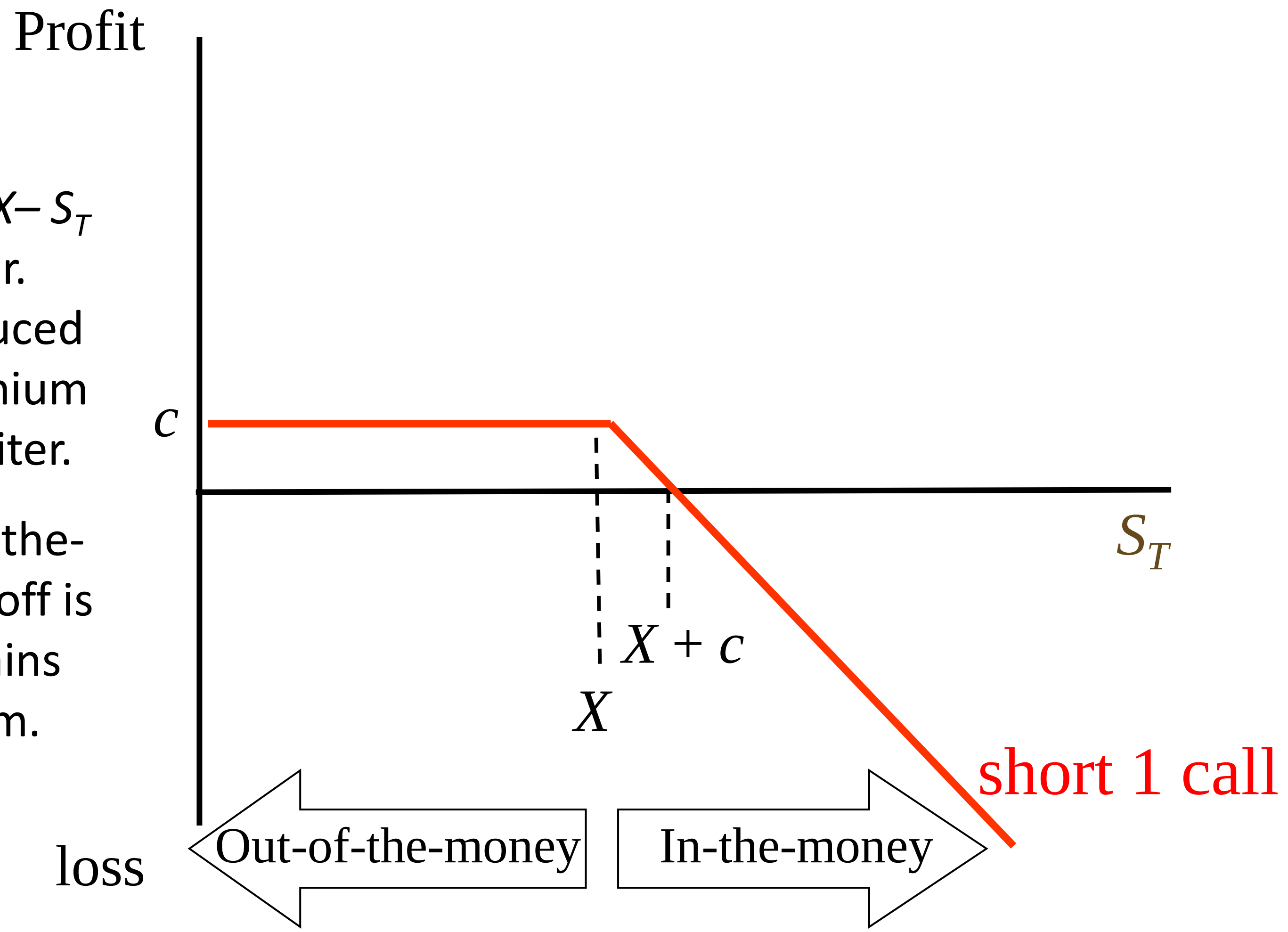
If the call is out-of-the-money, it pays off nothing and the buyer of the call loses the initial investment c (call premium).



Call Option Profit at maturity - Seller

If the call is in-the-money, it pays off $X - S_T$ to the option writer. But this loss is reduced by the option premium received by the writer.

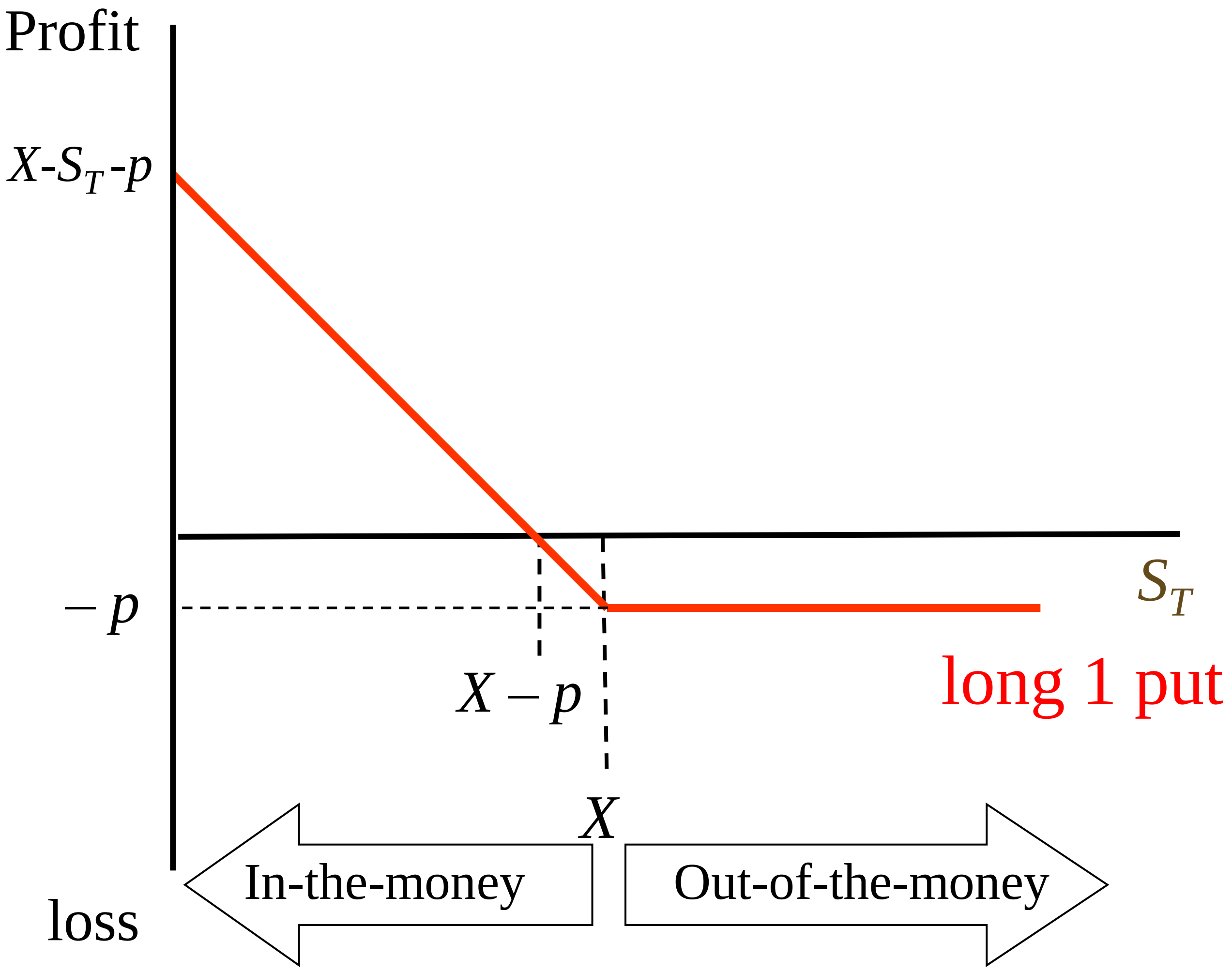
If the call is out-of-the-money, option payoff is 0 and the writer gains the option premium.



Put Option Profit at maturity - buyer

If the put is in-the-money, it pays off $X - S_T$ to the buyer. The profit is the difference between payoff and premium paid.

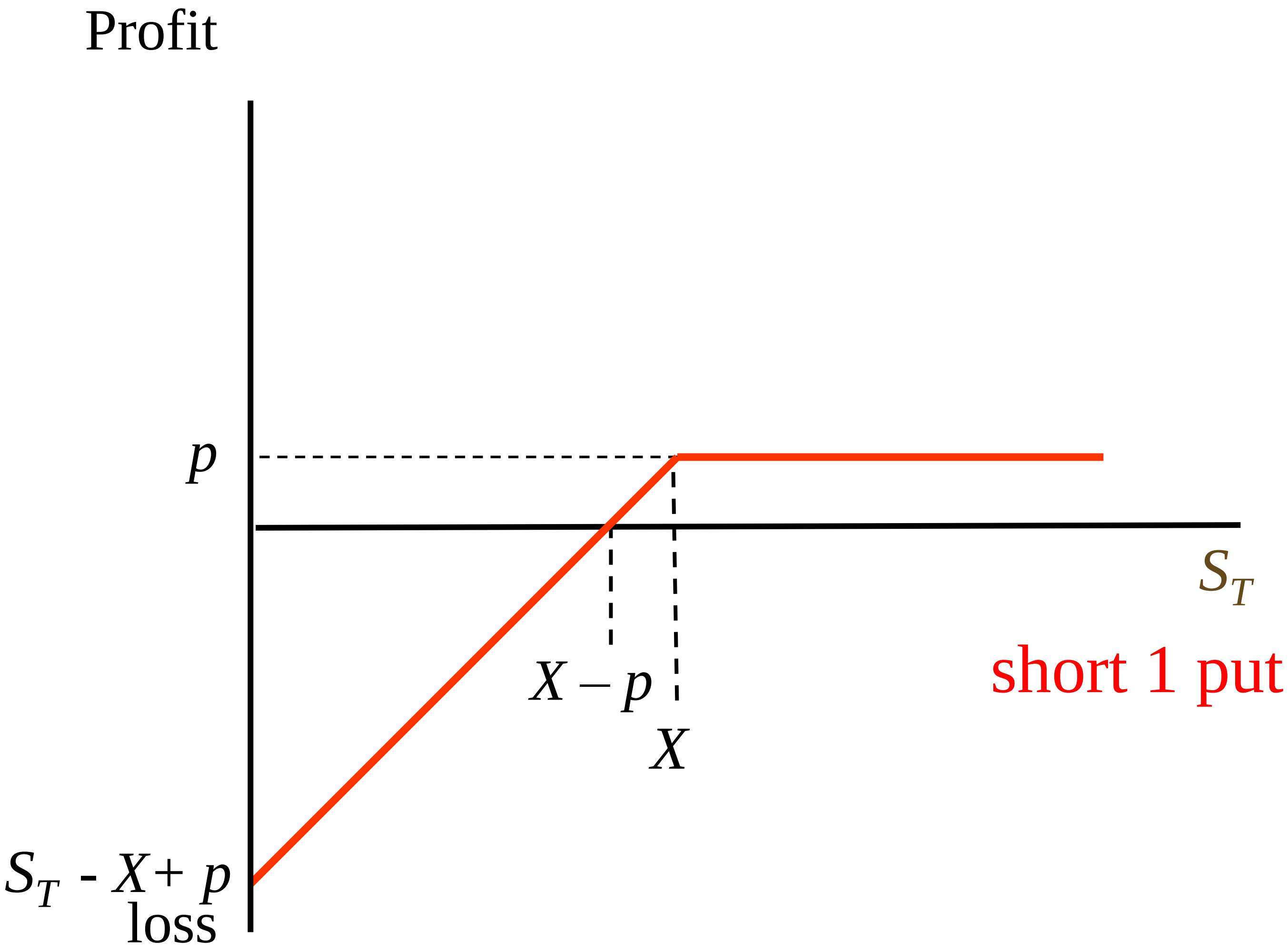
If the put is out-of-the-money, it is worthless and the buyer of the put loses the initial investment of *p*.



Put Option Profit at maturity - seller

If the put is in-the-money, the payoff to option writer is S_T-X . This loss is reduced by the premium received.

If the put is out-of-the-money, it is worthless and the seller of the put gains the option premium.



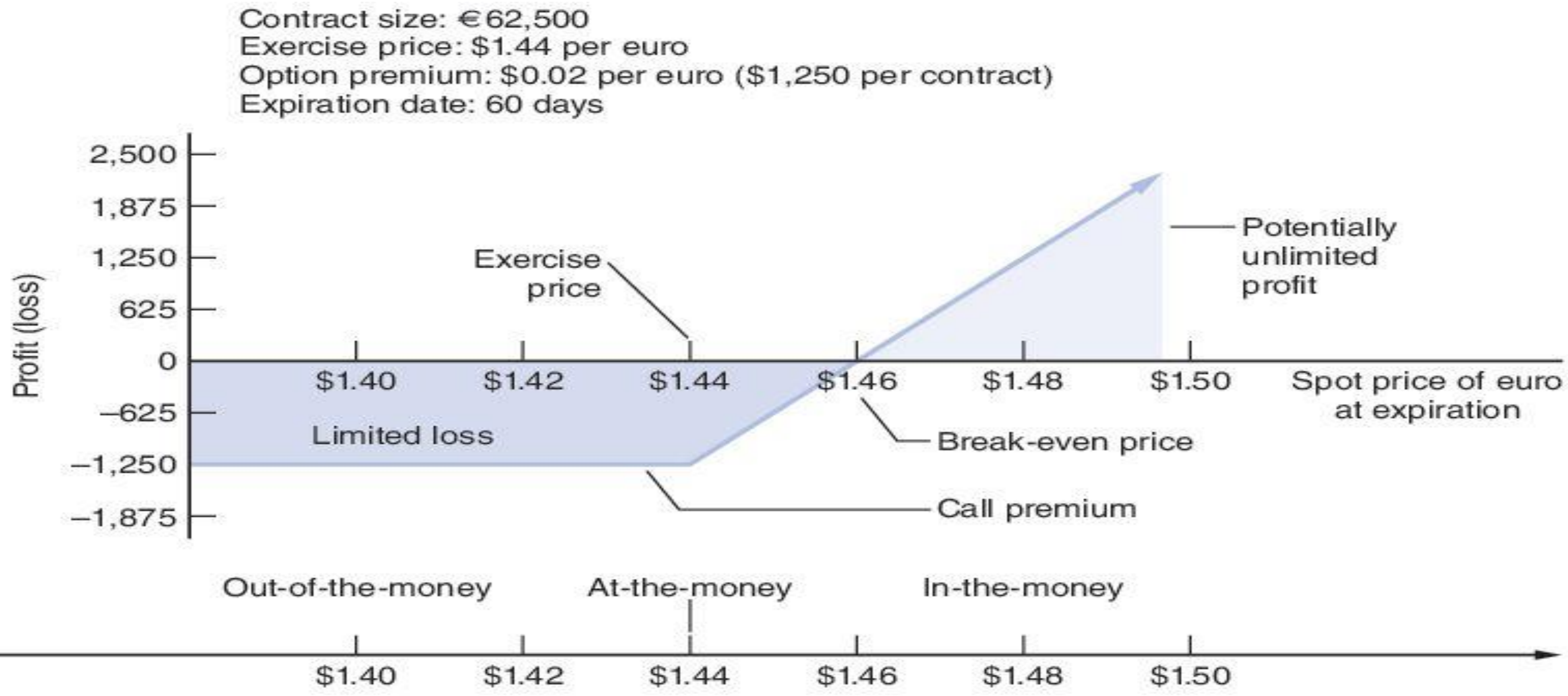
Currency Options

Call option example:

Suppose a US importer has to make a €62,500 payment to a German exporter in 60 days. The importer could purchase a European call option to have the euros delivered to him at a specified exchange rate (the strike price) on the due date. Suppose further that the option premium is USD0.02 per euro and the exercise price is USD1.44

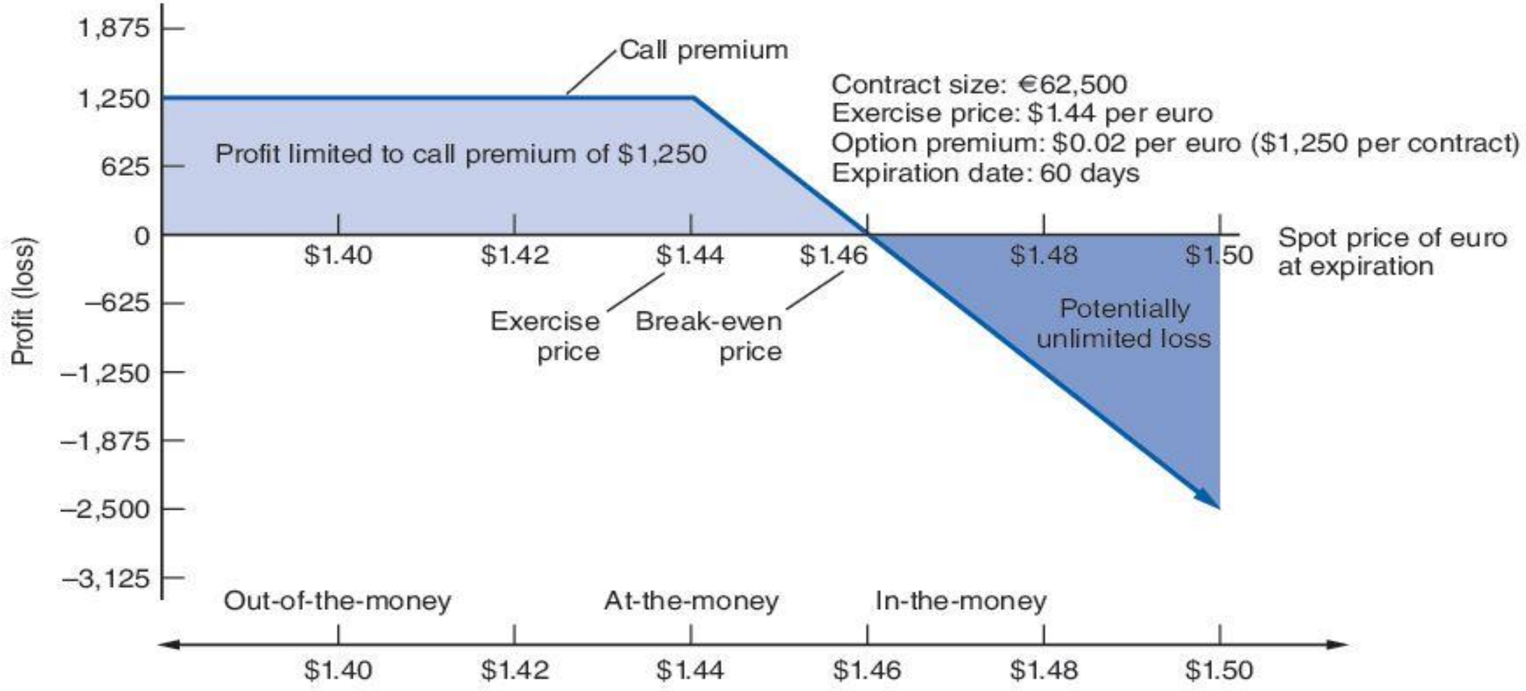
Next slide illustrates the importer's gains or losses on the call option.

Currency Options: Profit/loss from buying a call option



Inflows						
Spot sale of euros	—	—	—	91,250	92,500	93,750
Outflows						
Call premium	–1,250	–1,250	–1,250	–1,250	–1,250	–1,250
Exercise of option	—	—	—	–90,000	–90,000	–90,000
Profit (loss)	–$1,250	–$1,250	–$1,250	$0	$1,250	$2,500

Currency Options: Profit/loss from selling a call option



	$1.40	$1.42	$1.44	$1.46	$1.48	$1.50
Inflows						
Call premium	1,250	1,250	1,250	1,250	1,250	1,250
Exercise of option	—	—	—	90,000	90,000	90,000
Outflows						
Spot purchase of euros	—	—	—	–91,250	–92,500	–93,750
Profit (loss)	$1,250	$1,250	$1,250	$0	–$1,250	–$2,500

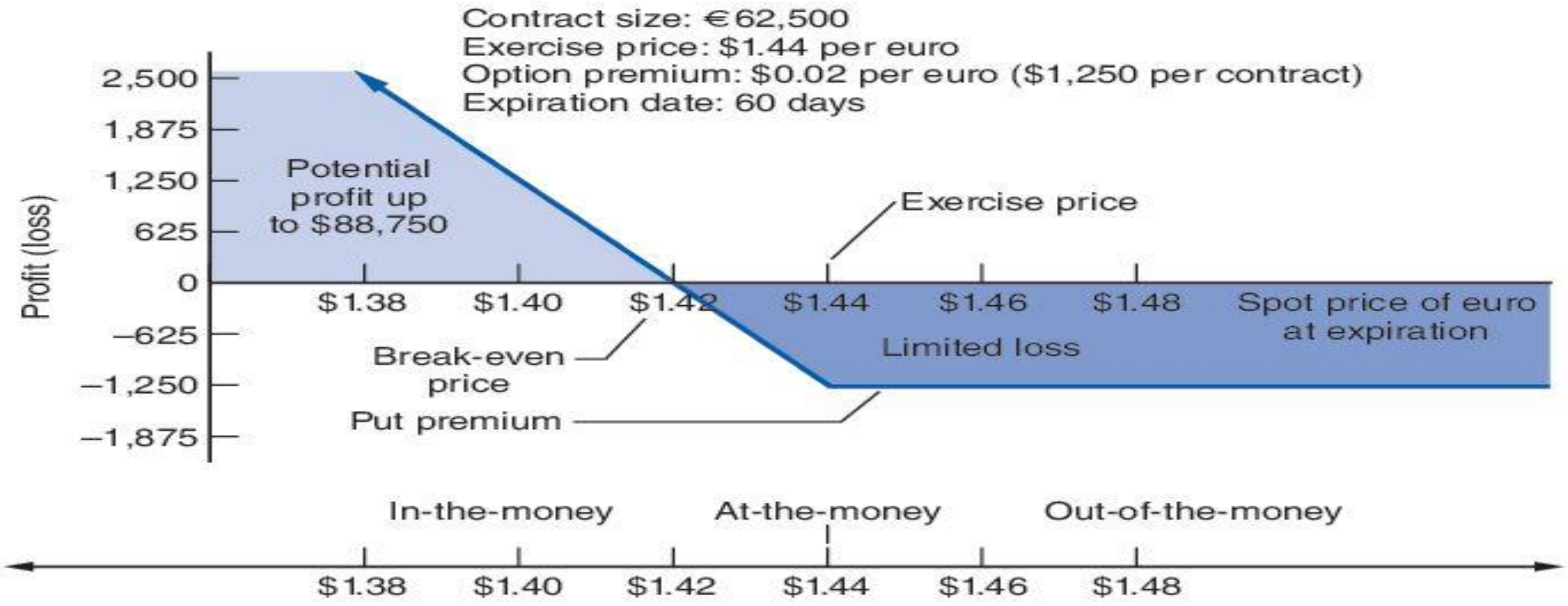
Currency Options

Put option example:

Suppose a US exporter will receive €62,500 from an European importer in 60 days. The exporter could purchase a European put option to have the euros delivered to him at a specified exchange rate (the strike price) on the due date. Suppose further that the option premium is USD0.02 per euro and the exercise price is USD1.44.

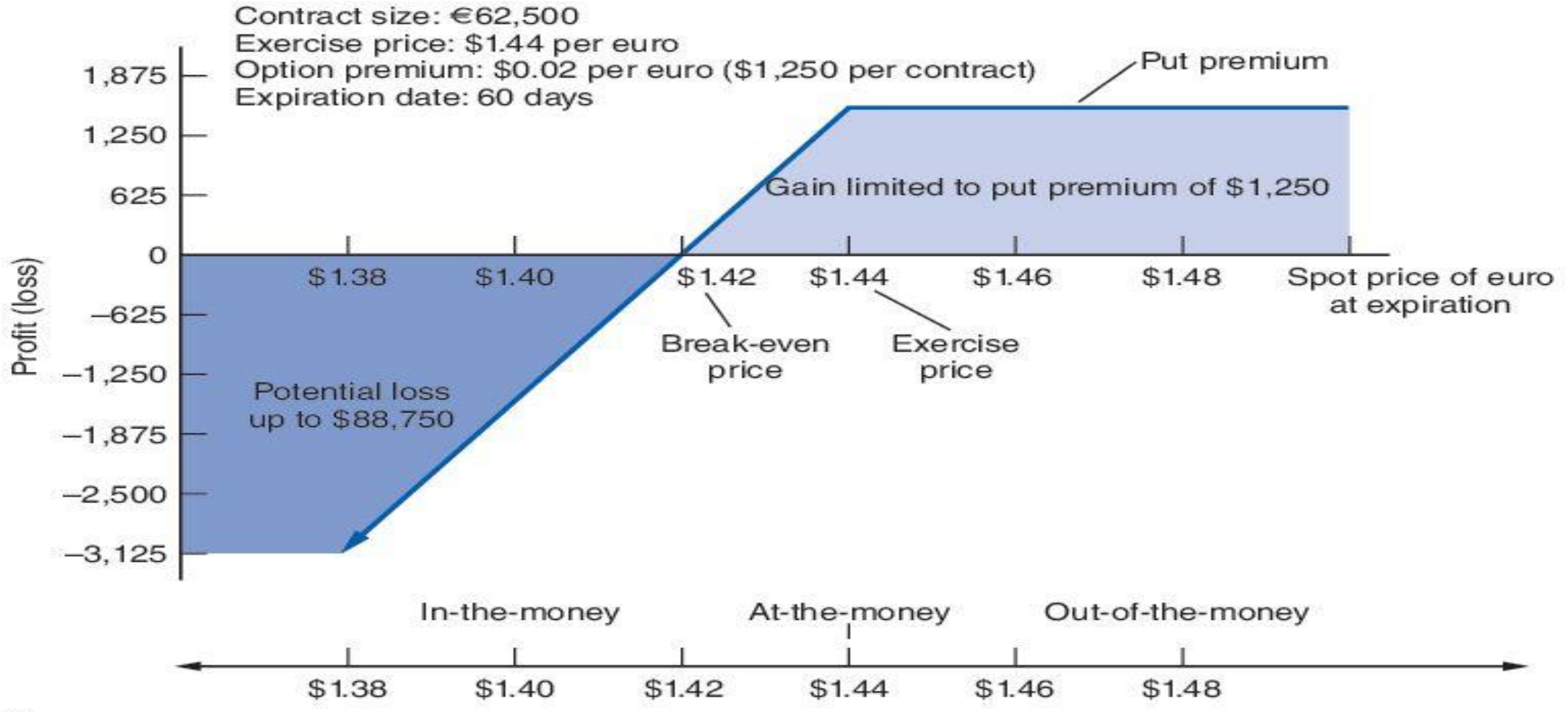
The payoff for this put option is shown in the next two slides for the buyer and the seller (writer), respectively

Currency Options: Profit/loss from buying a put option



	$1.38	$1.40	$1.42	$1.44	$1.46	$1.48
Inflows						
Exercise of option	90,000	90,000	90,000	—	—	—
Outflows						
Put premium	–1,250	–1,250	–1,250	–1,250	–1,250	–1,250
Spot purchase of euros	–86,250	–87,500	–88,750	—	—	—
Profit (loss)	$2,500	$1,250	$0	–$1,250	–$1,250	–$1,250

Currency Options: Profit/loss from selling a put option



	$1.38	$1.40	$1.42	$1.44	$1.46	$1.48
Inflows						
Put premium	1,250	1,250	1,250	1,250	1,250	1,250
Spot sale of euros	86,250	87,500	88,750	—	—	—
Outflows						
Exercise of option	–90,000	–90,000	–90,000	—	—	—
Profit (loss)	–$2,500	–$1,250	$0	$1,250	$1,250	$1,250

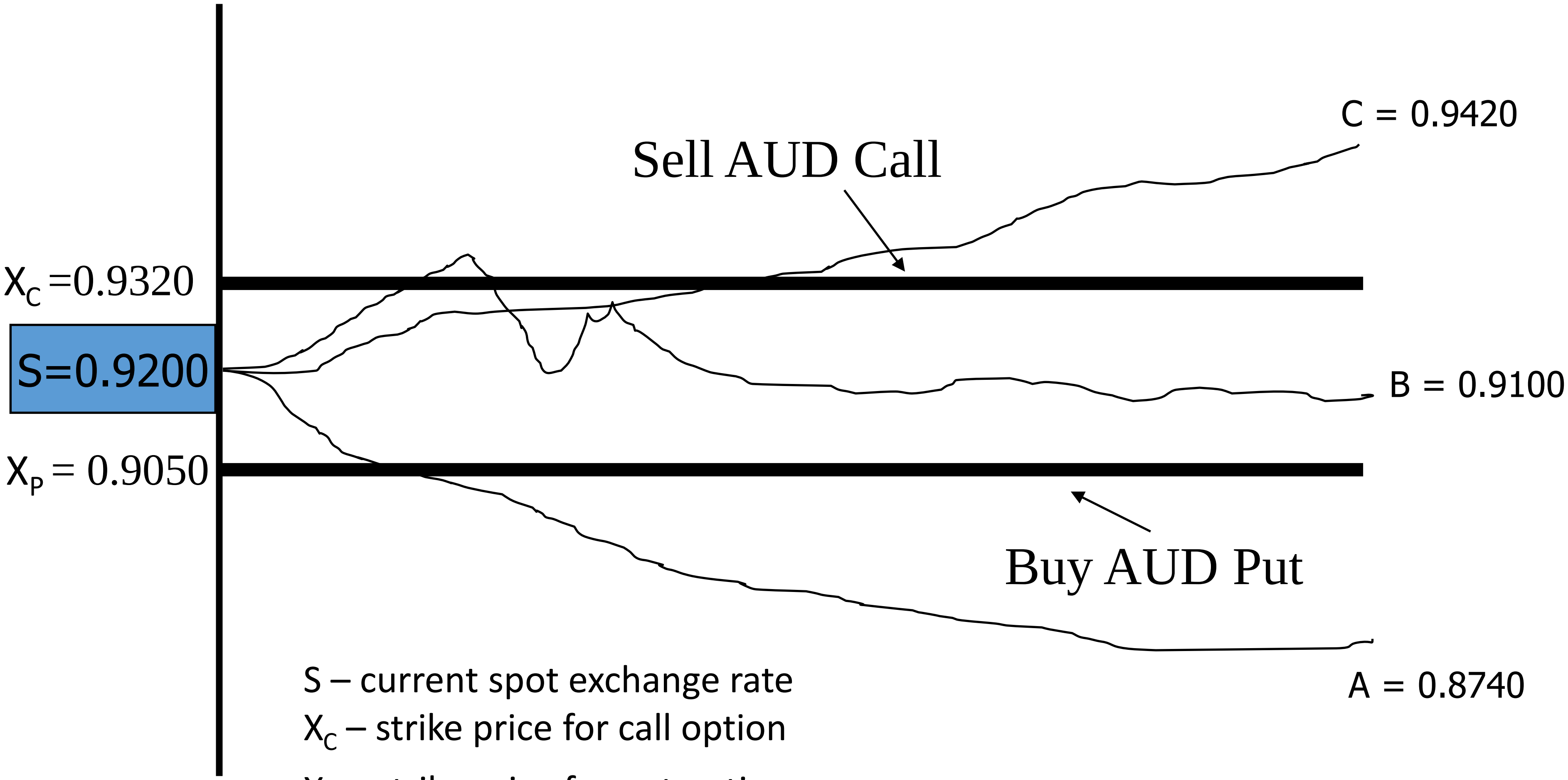
Zero Premium Collar

- A type of options collar strategy to minimize the cost of hedging by (e.g.) buying (sell) a call and selling (buy) a put that cancel each other out. Premium in call (put) is covered by the premium in sold put (call), forcing premium to be zero.
- Assume an Australian importer has to buy USD 10ml in 60 days.
  - Either buy USD call or buy an AUD put

Key points:

- Buy the AUD put (or USD call) for forex protection. Have to pay a premium – a % of the face value
- Sell the AUD call (or USD put) to offset the premium on the bought call
- Possible outcomes are in the range between the two strikes, i.e., can be zero cost
- Variants:
  - Zero premium with barrier options, such as knock-in and knock-out
  - Dynamic hedging (rolling up the strike), zero premium with barrier options

Zero Premium Collar



S – current spot exchange rate

X_C – strike price for call option

X_P – strike price for put option

A, B & C = possible spot prices at the expiration date

Option pricing and valuation

- Option value (also called 'market value' or 'value' of the option) = Intrinsic value + time value
  - Intrinsic value – amount by which option is in the money (the financial gain when the option is exercised)
    - Intrinsic value of call = S_T – X if S_T > X (the intrinsic value is positive), 0 if X > S_T
    - Intrinsic value of put: X – S_T if X > S_T, 0 if X < S_T
  - Time value - the possibility that option could finish in-the-money before it expires and depends on
    - Volatility of underlying assets – more volatile a currency is, the more likely it is to finish in-the-money, therefore it has more time value
    - Time until option expires – more time there is until expiration, the more time it has to finish in-the-money; time value of the option equals option premium minus intrinsic value.
- **The value of the option with no intrinsic value is just the premium**

Example

- Suppose the British pound is at \$1.30. A 6-month call option with a strike price of \$1.25 has a price of \$0.07 (option premium per GBP).
- What are the intrinsic and time values?
- Intrinsic value: max(1.30 – 1.25, 0) = \$0.05
- Time value: \$0.02 (option premium minus intrinsic value)

Factors determining Option pricing and valuation

Value of a currency option depends on

- Underlying exchange rate
- Exercise price
- Riskless rate of interest in domestic currency
- Riskless rate of interest in foreign currency
- Time to expiration
- Volatility in the underlying exchange rate

FX Options vs. Forwards

	Options	Forwards
1	Right but not obligation to buy/sell	Obligation to buy/sell
2	Cost - upfront premium	No cost to enter
3	Cash flow depends on whether option exercised	Cash flow known in advance
4	Opportunity to gain from favourable exchange rate movements	No opportunity to gain from favourable exchange rate movements

Forward, futures and options

	Forward	Futures	Option
Regulation	OTC (self-regulated)	Exchange (EX)	EX & OTC
Delivery	>90% physical delivery	<1% physical delivery	Lapse or exercised
Size/delivery date	Customer-tailored	Standardized	Depending on EX or OTC
Margins?	No	Yes (clearing house)	Premium paid upfront
Credit risk	Yes	No (clearing house)	Depending on EX or OTC
Transaction costs	No	Yes	N/A

Currency swaps

Currency Swaps

- A *currency swap* resembles a back-to-back loan except that it does not appear on a firm's balance sheet.
- A firm and a swap dealer or swap bank agree to exchange an equivalent amount of two different currencies for a specified amount of time at periodic intervals
- Most often used when companies make cross-border capital investments or projects
  - An Aussie parent company wants to finance a project undertaken by its subsidiary in France
  - Project proceeds would be used to pay interest and principal
  - Options:
    - Borrow A$ and convert to € – exposes company to exchange rate risk
    - Borrow in France – rate available may not be as good as that in Australia if the subsidiary is relatively unknown
    - So, find a counterparty and set up a currency swap

Currency Swaps

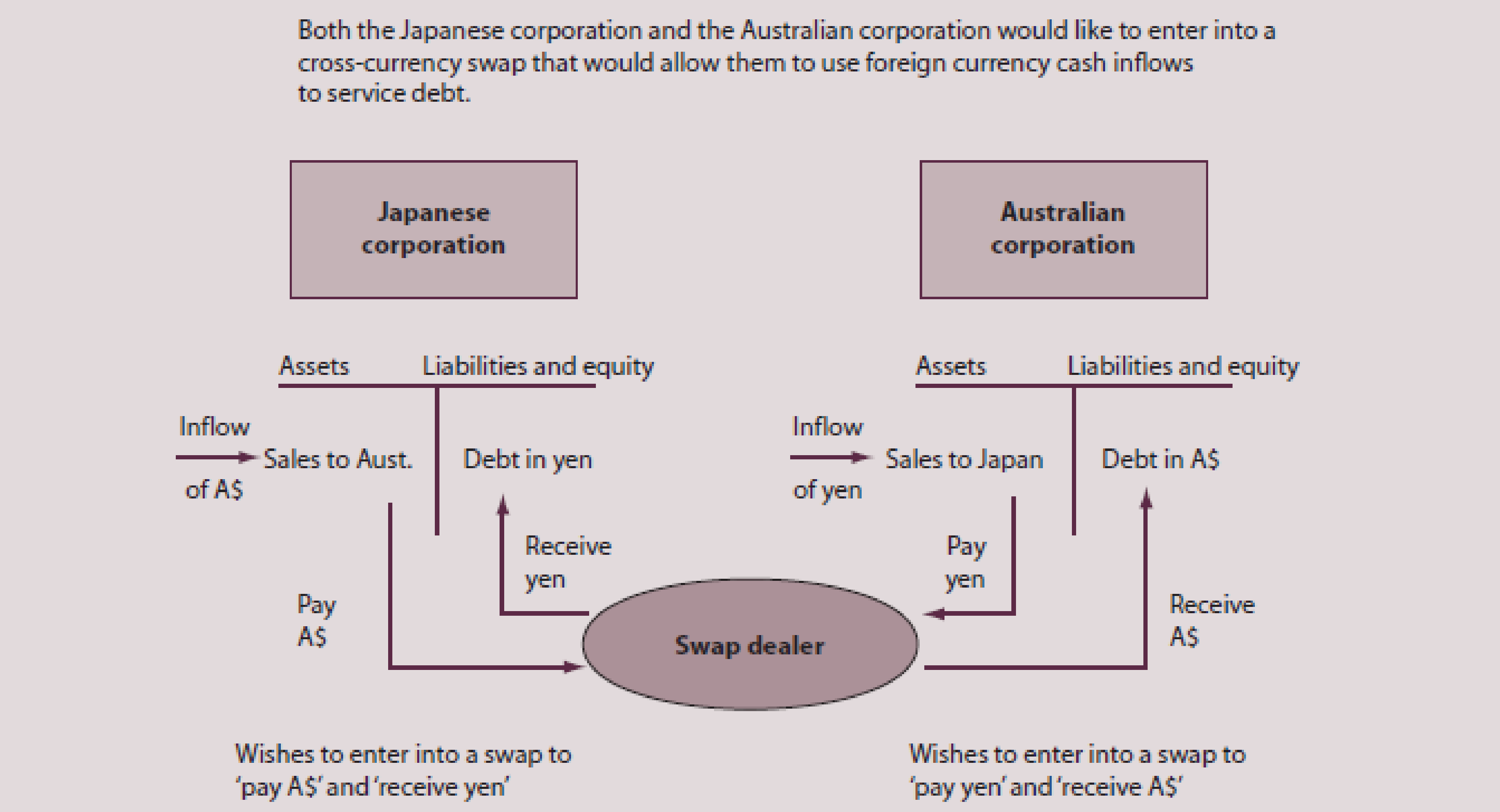
- In a fixed-for-fixed currency swap, the interest payments that the two parties exchange are both fixed rates. The main purpose a fixed-for-fixed currency swap is to manage exchange rate risk.
- In a fixed-for-floating currency swap, one counterparty exchanges a fixed rate payment denominated in one currency for a floating rate denominated in another currency. A fixed-for-floating currency swap can help manage both interest rate and exchange rate risk.
- In the case of currency swaps there is an exchange of principal amounts at maturity but there is no exchange of principal in an interest rate swap.

Use/reasons/benefits of Currency Swaps

- The usual motivation for a currency swap is to replace cash flows scheduled in an undesired currency with flows in a desired currency.
- The desired currency is usually the currency in which the firm's future operating revenues (inflows) will be generated.
- Firms often raise capital in currencies in which they do not possess significant revenues or other natural cash flows (a significant reason for this being cost, i.e., interest costs).

Example of a currency swap

FIGURE 11.5 Using a cross-currency swap to hedge currency exposure



(Eiteman et al, 2nd Ed., Page 341)

Japanese corporation, which expects to receive AU$ in future, is worried that future AU$ price will drop, whereas AU corporation, which expects to pay ¥, is worried that ¥ will go up. Hence both Japanese and AU corporations enter into a swap contract with a dealer.

MAF306: Week 8

Foreign Exchange Risk Management

Dr Sohel Azad

Foreign exchange derivatives

Learning outcomes for topics 5-8

> Critically analyze and evaluate exchange rate determination and....
risk exposure management of business financing and investment activity.....

> Consider these impacts on multi-national companies and countries, in order to review and implement appropriate risk management strategies.

FX risk management

Lecture objectives:

- To understand FX risk exposure and hedging
- To identify FX risks originating from
  - Transaction exposure
  - Translation exposure
  - Operating exposure
- To analyse and evaluate various FX risk management strategies

Learning resources:

- Shapiro Chapters 9 & 10
- Eiteman et al Chapters 10, 11 & 12
- Butler Chapters 8, 10 & 11

FX Exposure and FX Risk

- Foreign exchange exposure refers to the amount of foreign currency one holds or is expected to hold
- Foreign exchange risk refers to the potential loss due to adverse movement in exchange rate

Hedging/not hedging: Why/why not?

- Hedging protects the owner of an asset (future stream of cash flows) from loss (but do not over hedge, like SoG or other rogue traders)
- However, it also eliminates any gain from an increase in the value of the asset hedged against (limiting upside potential)
- Since the *value of a firm* is the *net present value* of all *expected future cash flows*, it is important to realize that variances in these future cash flows will affect the value of the firm and that at least some components of risk (i.e., currency risk) can be hedged against

Partial Hedging

Sometimes, corporations, exporter/importer may choose to hedge half of their exposure and leave the rest unhedged.

In this case it would only *reduce* rather than *eliminate* the foreign exchange exposure.

FX Risk Management

It’s all about ***hedging*** and ***managing*** FX risk exposures arising from

- Transaction Exposure
- Translation Exposure
- Economic Exposure or Operation Exposure

Transaction Exposure

Transaction Exposure

- Transaction exposure arises if payables and receivables (cash inflows and outflows) are denominated in foreign currencies.
- The exposure arises with trade flows and capital flows
- Examples
  - Trade: Import costs; export revenues, any other cross border payments/receipts.
  - Capital: Foreign investment, foreign loan or deposit.
- It is important to recognize the exposure at the point the *exposure is initiated*.

Transaction exposure: Trade example

Receipt of Yen

At Start

At Maturity

AU$1/JPY 75.00; Unit price JPY 400.
If 1 million units were exported, then
budgeting for 1 million units @¥400:
= ¥ 400,000,000 or
= AU$ 5,333,333.33 @ ¥ 75/A$1

Receipt of Yen: ¥ 400,000,000; A$1/JPY1=?
-if AUD goes up ¥ 80/AUD = $5,000,000 ↓
-If AUD goes down ¥70/AUD =$5,714,285.71 ↑

Loss for the AU exporter if AU$ goes up; Profit for the Japanese importer if JPY goes up

Transaction exposure: Foreign currency loan example

Items	Values
Principal	JPY 1,700 million
Commencement date	1 January 2020
Exchange rate at commencement	AU$1 = JPY 95.50
Maturity date	30 September 2020
Floating interest rate	7% p.a. (payable quarterly)
Exchange rate at maturity	AU$1 = JPY 80.50

Transaction exposure: FX Loan Example - Principal

- At commencement:
  - JPY 1,700 million @ ¥95.50/AU$1 = A$17,801,047
- At maturity:
  - JPY 1,700 million @ ¥80.50/AU$1 = A$21,118,012
- Additional cost:
  - A$21,118,012 - A$17,801,047 = A$3,316,965 (An increase of 18.63%)

Transaction exposure:
FX Loan Example – Interest & total

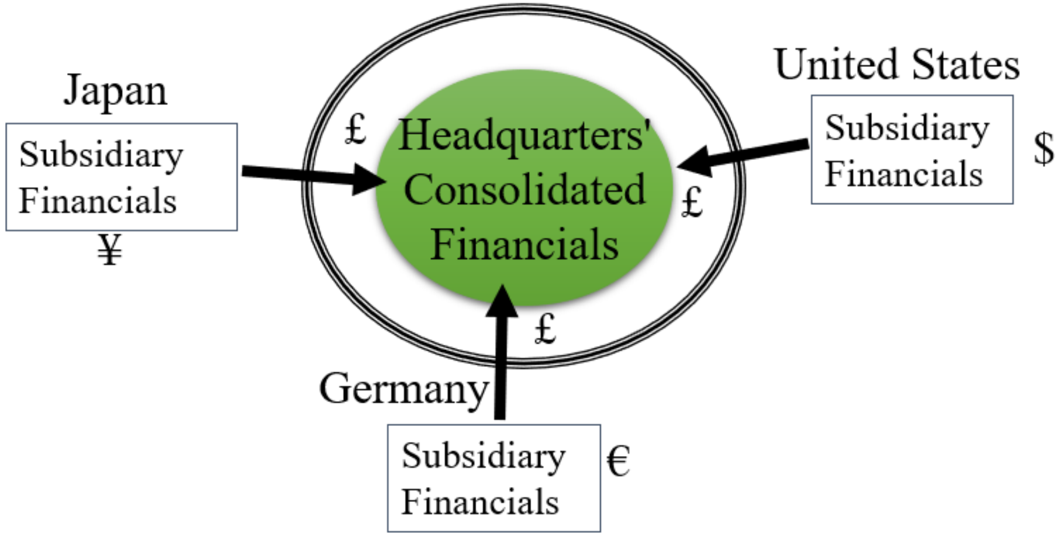
- Interest payable quarterly @ 7% p.a.
- JPY 1,700 m x 7%/4 = JPY 29.75m (quarterly)
- Interest payable for 3 quarters = ¥29.75m*3 = ¥89.25m
  - In AU dollar terms = ¥89.25m/80.50 = A$1,108,695.65
- Principal + interest = A$21,118,012 + A$1,108,695.65

  = A$ 22,226,707.65
- Borrowing at floating rates exposes the company to rises in JPY interest rates

Translation Exposure
Economic Exposure

Translation Exposure

- Translation exposure, also called accounting exposure, arises from the consolidation of foreign currency assets, liabilities, net income and other items in the process of preparing consolidated financial statements
- Exposure arises at the time of translating foreign currency cash flows in the home currency
- *Functional Currency* - the dominant currency used by a foreign subsidiary in its day-to-day operations, where it generates cash flows/debt/equity.



Methods for Translation exposure

- Three principal methods for the translation of foreign subsidiary financial statements are available:
  - *The current/noncurrent method- current ER when acquired*
  - *The monetary/nonmonetary method – money/physical assets*
  - *The temporal method – historical rates*
- The *current rate method* is the most prevalent in the world
  - *Assets & liabilities are translated at the current EX rate, e.g., Cash, Short Term loans*
  - *Income statement items are translated at the exchange rate on the dates they were recorded or an appropriately weighted average rate for the period*
  - *Dividends (distributions) are translated at the rate in effect on the date of payment*
  - *Common stocks and paid-in capital accounts are translated at historical rates*

Managing Translation Exposure

Before we need to move funds!

- ❖ increase hard-currency (more stable, less volatile and likely to appreciate) assets
- ❖ decrease soft-currency (highly volatile, unstable and likely to depreciate) assets

If **devaluation** of local (host country) currency appears likely, then:

- ❖ reduce the level of cash
- ❖ tighten credit terms to decrease accounts receivable (less outstanding)
- ❖ increase Local Currency borrowing
- ❖ delay accounts payable (will be cheaper to pay in future)
- ❖ sell the weak currency forward

If **appreciation** appears likely – use the opposite tactics above

Economic/Operating Exposure

- Affects long term cash flows or future earning power as a result of changes in exchange rates.
- Two types of economic exposure:
- *Direct economic Exposure*
  - Arises as a result of a company's long-term commitment in foreign currency transactions, e.g., selling a product in the US with an AUD cost base.
- *Indirect economic Exposure*
  - Relates to the loss of sales or erosion of profit margin as a result of competitors achieving an advantage from favorable currency movements.

Managing Operating Exposure

- *Diversifying operations:* World-wide diversification in effect pre-positions a firm to make a quick response to any loss from operating exposure.
- *Diversifying financing:* Unexpected devaluations change the cost of the several components of capital—in particular, the cost of debt in one market relative to another.
- **Proactive management of operating exposure (also used for other exposure)**
  - *Matching currency cash flows*
  - *Risk-sharing agreements*
  - *Back-to-back loans*
  - *Currency swap*

Managing FX Risk Exposure

Management of FX risks exposure

Some strategies for managing FX risk exposure:

1. Using ***Contractual hedges***
   a) Forward and futures market hedge
   b) Money market hedge
   c) Option market hedge
   d) Currency swap
2. Using ***Natural hedges***
   a) Risk sharing
   b) Risk shifting
   c) Matching/Exposure netting
   d) Lead and Lag

Hedging via Forward

Hedging via Forward Market

- An importer (exporter) is concerned his/her/their foreign currency denominated payable (receivable), if unhedged, would have adverse effect on his/her/their cash flow
- If A$ depreciates against foreign currency on settlement date:
  - importer would have to pay more in home currency for the foreign currency amount
- If A$ appreciates on settlement date:
  - exporter would receive less in home currency
- Examples used:
  - Hedging import payables
  - Hedging foreign borrowings
  - Hedging export revenues

Example: Hedging Import Payables

Suppose an Australian car dealer has to pay €2.0ml to their German supplier in 3 month's time. Therefore, the car-dealer has a foreign currency exposure of €2.0ml.

– The risk/concern? A$ depreciates or € appreciates

– Spot rate = €1.2030/AU$1

– 3-month AU$ int. rate 10.50%

– 3-month € int. rate 7.20%

– 3-month forward rate = €1.1940/AU$1

- The car-dealer can use forward exchange market to hedge the exposure before AU$ falls. Once hedged, the dealer will not be affected by the future spot rate, FX risk is eliminated

Example: Hedging Foreign Borrowings

An Aust. company borrows SG$10ml for 6 months @ 3.50% p.a.

The risk/concern? SG$ appreciates or A$ depreciates

- Spot rate = SG$/AU$1 = 2.0000
- 6-month forward rate SG$/AU$1 = 1.9500 Lock in

- Total due in 6 month's time in SG$ =
  - Principal + Interest for 6-month (3.5%/2=1.75%)
  - (10ml) + (0.175ml) = SG$10.175ml
  - AU$ cost if hedge = 10.175ml/1.9500 = AU$5,217,948.72
- If AU$ appreciates against SG$, the cost of buying SG$10.175m to repay the loan will fall.

Example: Hedging Export Revenues

An Aust. manufacturer exports Lobster daily to the USA. S/he expects to receive USD 3.5 mil in 3 months

- The risk/concern? A$ appreciates or US$ depreciates
- Spot AUD/USD = A$1.2950/US$
- 3 month US int. rate = 5.50%
- 3 month AUST int. rate = 4.50%
- 3 month forward rate = A$1.2932 /US$ Lock in

The lobster exporter can use forward exchange market to hedge the exposure before USD falls. Once hedged, the dealer will not be affected by the future spot rate, FX risk is eliminated

Hedging via Futures

Hedging with Futures Market

- Short Hedge
  - Selling a currency futures contract to protect future foreign currency exposure assuming that the underlying currency might fall/depreciate due to unfavorable exchange rate movements

- Long Hedge
  - Buying a currency futures contract to protect future foreign currency exposure assuming that the underlying currency might appreciate due to unfavorable exchange rate movements.

Short Hedge (A$ futures)

<u>Physical Position</u>

- future payment in foreign currency
- loses if A$ depreciates

<u>Futures Position</u>

- sold/short futures position
- gains if A$ depreciates
- the gain from the futures transaction offsetting the loss in foreign currency payment

Long Hedge (A$ futures)

<u>Physical Position</u>

- future receivable in foreign currency
- loses if A$ appreciates

<u>Futures Position</u>

- bought/long futures position
- gains if A$ appreciates
- the gain from the futures transaction offsetting the loss in physical market

Hedging using A$ futures (exporter)

- Suppose an AU exporter expects to receive US$80,000 in 6 months' time. The exporter is concerned that US$ (A$) may depreciate (appreciate) from its current rate of U$0.78/A$1. Assume that 6-month future rate is U$0.80/A$1. However, spot rate in 6-months' time turns out to be U$0.82/A$1

- The risk/concern? A$ appreciates (or US$ depreciates)

- Explain how AU exporter could use currency futures to hedge US$ exposure?

If unhedged, then the exporter will lose US 2 cents per A$.

Hedging using A$ futures…cont

So,

- At t_0, the exporter buys a 6-month A$ futures contract of A$ 100,000 at the future exchange rate of U$0.80 (i.e., pays U$80,000).

- At t_1, 6 months later, the exporter sells (closes out) the 6-month contract @ U$0.82/A$1

Hedging using A$ futures…cont.

Spot Market

- Expect to receive U$80,000 in 6 month's time
- 6 months later -

  receive U$80,000 and sells @ spot rate U$0.82/A$1

  = A$97,561 (80,000/0.82)

➢ If not hedged, exporter would make a loss of A$2439 (=100,000 – 97,561)

Futures market

- Buy a 6-month futures contract of A$100,000 @ U$0.80/A$1 = U$80,000
- 6 months later -

  Sells (closes out) 6-month contract @ U$0.82, i.e.,

  0.82*100,000 = U$82,000

  Profit in futures=U$2,000

➢ If hedged, loss of spot is compensated by the profit from futures

Hedging using A$ futures...cont.

Results	US$	AU$
Export earnings @U$0.80/A$1	80,000	97,561
Futures profit = 100000@U$0.02	2,000	2,439*
Total receipts	82,000	100,000

*At current spot at the expiry, i.e., U$ 0.82=2000/2439

Money market hedge

Money market hedge

- Similar to using Forward markets: Involves simultaneous borrowing and lending activities in two different currencies to lock in the home currency value of a future foreign currency cash flow
- To hedge foreign currency payable (receivable) via money markets, we need to do the following:
  - Borrow in home (foreign) currency equivalent to the present value of the foreign currency
  - Sell the home (foreign) currency in the spot market
  - Invest this foreign (home) currency amount in money market
  - Repay the home (foreign) currency borrowing

Money market hedge

- Susan will borrow euro (€) in Frankfurt today (say, 30th April 2024) and immediately convert € into A$ to lend in Australia. Susan will later repay the loan in November using the sale proceeds from A$.
- If IRP (interest rate parity) holds, forward market hedge and money market hedge will generate the same hedging result:

$$1 + r_h = \frac{(1 + r_f)f_1}{e_0} \longrightarrow f_1 = e_0 \frac{1 + r_h}{1 + r_f} \quad \text{or} \quad f1 = eo \frac{(1 + r_f)}{(1 + r_h)}$$

(Direct quote) (indirect quote)

- Money market hedge effectively creates a "synthetic" forward contract

Hedging via Options

Hedging with Options

- Option is the most effective way (compared to forwards or futures) in hedging uncertain exposures to foreign currency

- Protects against *downside risk* while keeping the *upside potential*

- The choice of option ***strike price*** is important as both ***option premium*** and payoff pattern are affected by the strike price

- Types of option: covered in week 7.

Hedging with Options, cont.

- AUD Put (Foreign currency Call) option
  - *Used by an importer* to hedge against a weakening of the AUD
  - Buy from a bank
  - Ability to take advantage of favourable movements i.e., AUD strengthens
- AUD Call (Foreign currency Put) option
  - *Used by an exporter* to hedge against a strengthening of the AUD
  - Buy from a bank
  - Ability to take advantage of favourable movements i.e., AUD weakens

Hedging with Options, cont.: Example

- An AU company has just bought US$1 million worth of goods. It has to pay the money in 3 months' time. It is concerned that US$ may appreciate but it does not want to hedge this exposure with a forward or futures contract, however, decides to buy US$ call option and gets a quote of US$0.012 from JPC Bank, with the strike price is 0.7400/A$ and the current exchange rate is 0.7500/A$.
- If exchange rate is 0.7200, 0.7400 or 0.7600 at maturity, how does this compare with being remained unhedged?
- Cost of premium = 0.012 * 1 million = US$ 12,000
  - In AU$ term = 12,000/0.75 = A$ 16,000

Hedging with Options, cont.: Example

If spot rate is 0.7200, option will be exercised (The US$ appreciates, so its call option expires **in-the-money**)

- If unhedged, the cost is US$1m/0.72 = A$ 1,388,889
- If hedged, (buy a US$ call option), AU company saves A$ 21,538 (see below)

1	Spot price = US$1 mil/0.7200	A$ 1,388,889
2	Premium US$0.012 * US$1 mil = US$12000, which in AU@0.75 (upfront at spot rate)	A$ 16,000
3	Strike price = US$1m/0.74	A$ 1,351,351
4	Profit from option/savings if hedged (1-2-3)	A$ 21,538

Options will not be exercised if spot exchange rates are 0.7400 and 0.7600 at the maturity

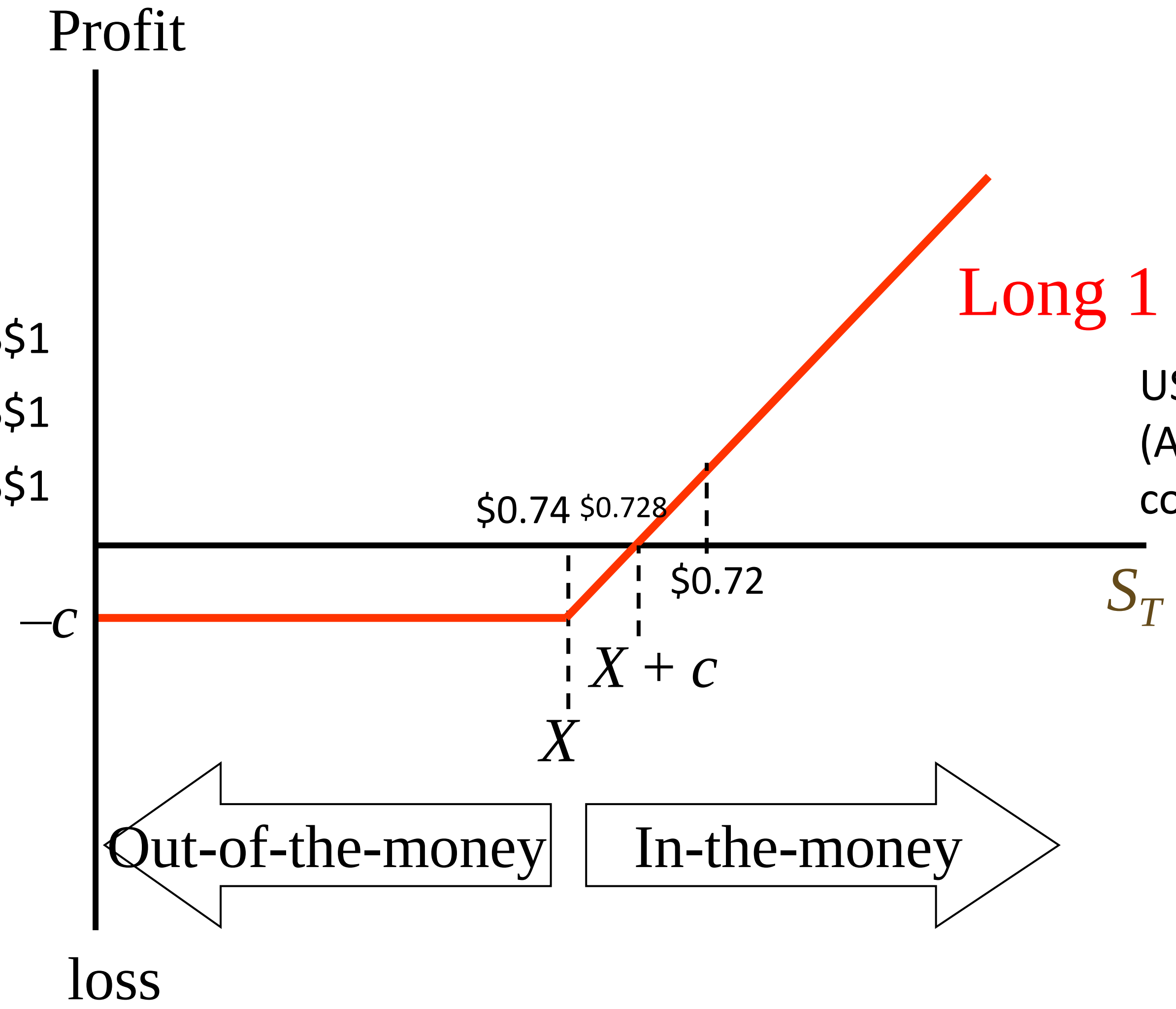
Hedging with Options…cont.

Direct-Indirect conversion

US$0.72/A$1=A$1.3889/US$1

US$0.74/A$1=A$1.3514/US$1

US$0.76/A$1=A$1.3158/US$1



US should appreciate to $0.728 (A$1.i.e., A$1.3736/US$1) to cover cost of premium

Hedging via Currency swaps

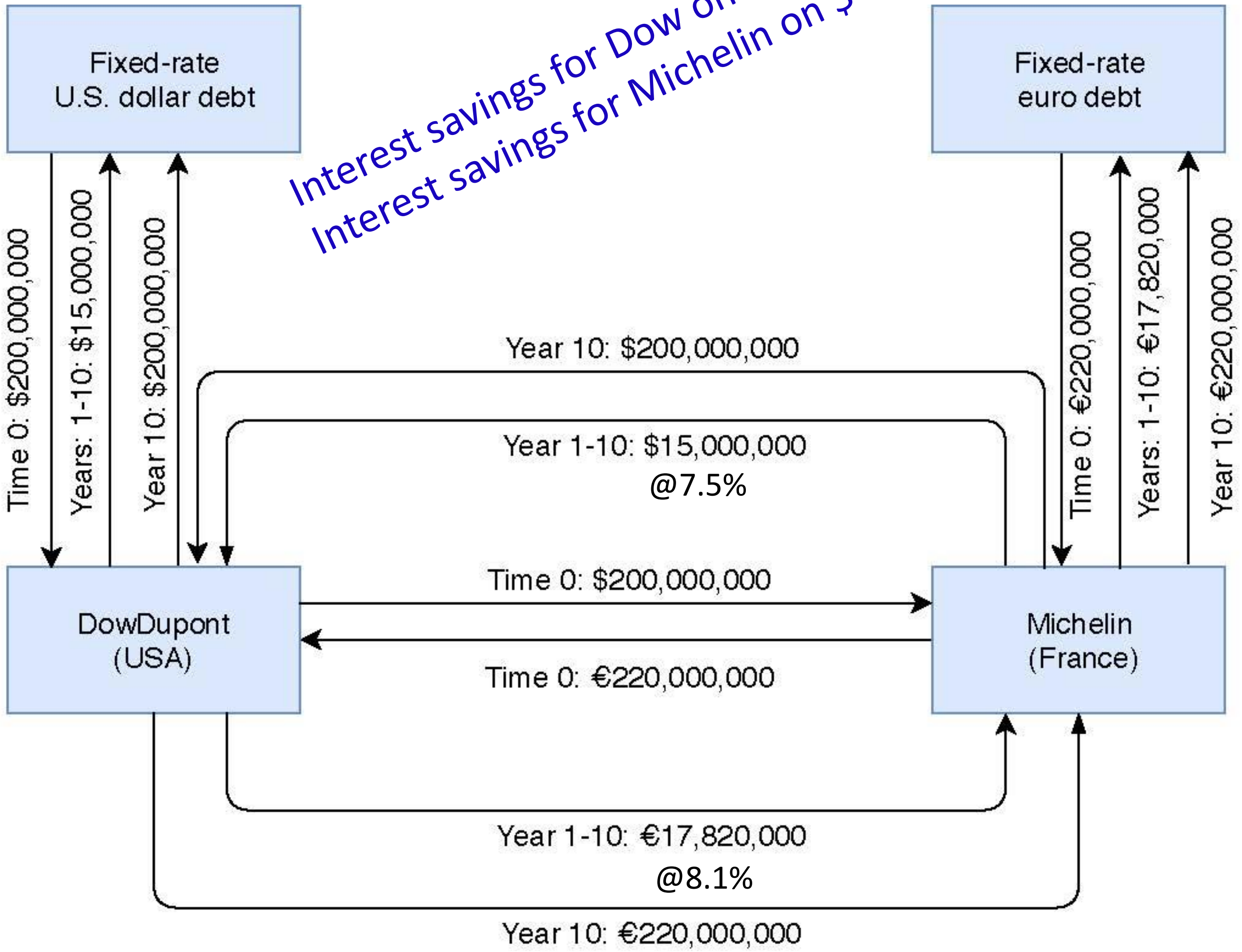
Currency Swap

- A currency swap is an exchange between two parties of interest payments that are denominated in different currencies.
- In a fixed-for-fixed currency swap the interest payments that the two parties exchange are both fixed rates. The main purpose a fixed-for-fixed currency swap is to manage exchange rate risk.
- In a fixed-for-floating currency swap one counterparty exchanges a fixed rate payment denominated in one currency for a floating rate denominated in another currency. A fixed-for-floating currency swap can help manage both interest rate and exchange rate risk.
- In the case of currency swaps there is an exchange of principal amounts at maturity but there is no exchange of principal in an interest rate swap.

Currency Swap

- Suppose the US-based chemical company DowDupont is looking to borrow $200 million in euros and the France-based tire maker Michelin is looking to borrow $200 million. Both would like a fixed rate and a maturity of 10 years.
- DowDupont can issue $ debt at 7.5% and € debt at 8.25%.
- Michelin can issue $ debt at 7.7% and € debt at 8.1%.
- The spot exchange rate is €1.1/$1.
- Michelin has a comparative advantage at borrowing in euros and DowDupont in dollars.
- They can then swap the proceeds and the cash flows.
- Both parties benefit from a lower cost of borrowing

Interest savings for Dow on €= 0.15% (=8.25%-8.10%)
Interest savings for Michelin on $= 0.20% (=7.7%-7.5%)



Natural hedging

Risk Sharing

- Involves developing a customized contract
- The contract typically takes the form of a *Price Adjustment Clause,* whereby a base price is adjusted to reflect certain exchange rate changes
- Parties would share the currency risk *beyond* a neutral zone of exchange rate changes.
- The *neutral zone* represents the currency range in which *risk is <u>not</u> shared*.

Risk Shifting

An agreement between the importer/buyer and exporter/seller to share or split the impact of currency movement on payments)

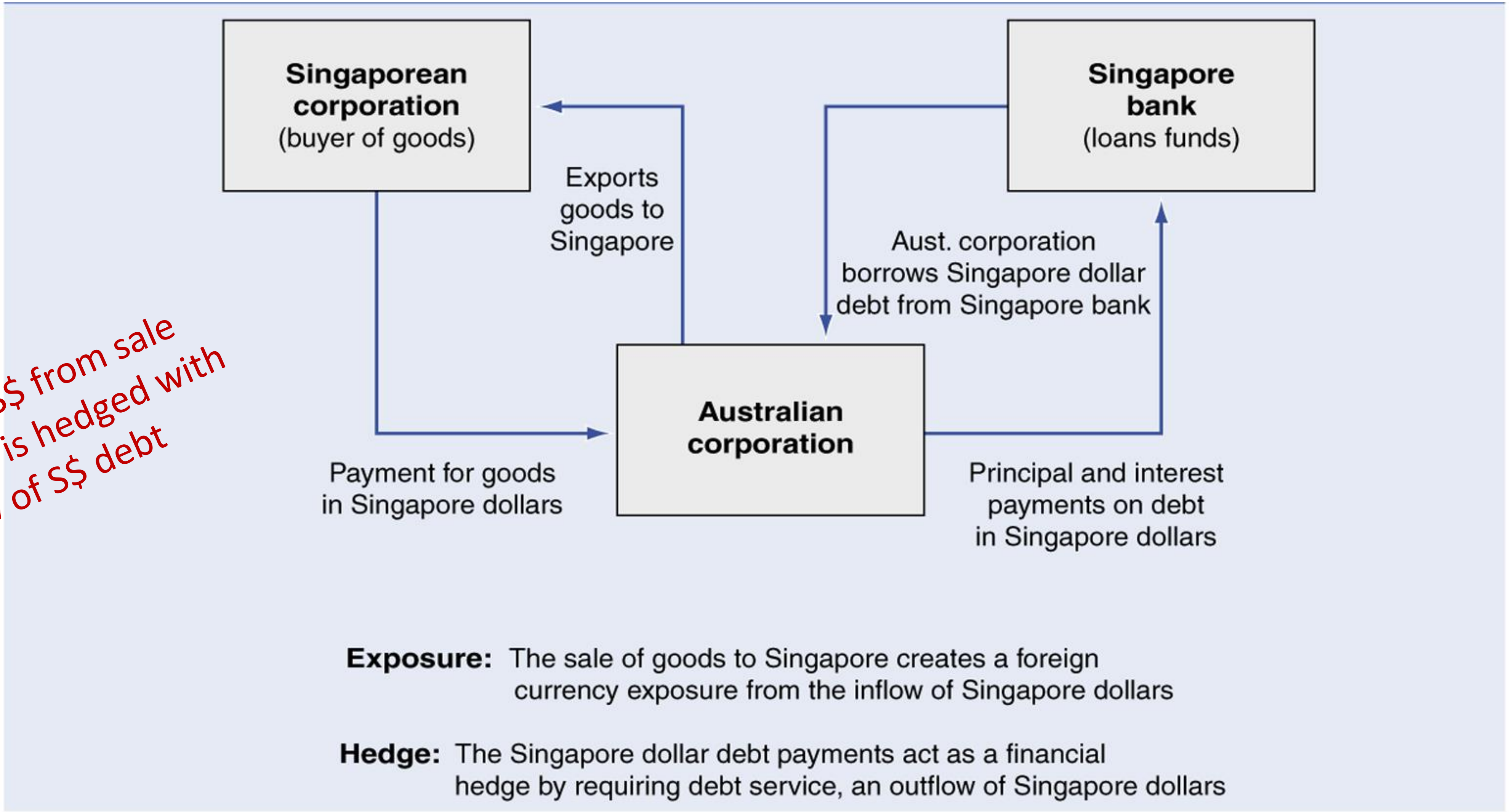
- Importer’s home currency invoicing (risk shifting)
  - A zero-sum game as currency risk is shifted to the counterparty/exporter
- Price adjustment clause, where a base price is adjusted to reflect ER movement (risk sharing)
- Limitations:
  - Firms will try to invoice exports in strong currency and imports in weak currency
  - This is not possible with informed customers or suppliers

Matching/Exposure Netting

Involves offsetting exposures in one currency with exposures in the same or another correlated currency

- An Australian firm has continuing export sales to Singapore
- In order to compete effectively in Singaporean markets, the firm invoices all export sales in Singapore dollar
- This policy results in a continuing receipt of Singapore dollars month after month
- This exposure can be hedged by taking out loans denominated in Singapore dollar, which requires periodic payments in Singapore dollar
- The Singapore dollar cash inflows from export sales can be used to service the principal and interest payments on Singapore dollar debt

Matching/Exposure Netting



Inflow of S$ from sale of goods is hedged with outflow of S$ debt

Eiteman Figure 8.3

Lead and lag

Retiming the transfer of funds

- Firms can reduce both operating and transaction exposure by accelerating or decelerating the timing of payments that must be made or received in foreign currencies.
- LEAD, bring forward - when expecting A$ to depreciate (Importer)
- LAG, delay - when expecting A$ to appreciate (Importer) – opposite for Exporters!
- Intracompany leads and lags is more feasible as common set of goals for the consolidated group.
- Intercompany leads and lags requires the time preference of one independent firm to be imposed on another!

Questions and comments

MAF306: WEEK 9

International Parity Conditions

Dr Sohel Azad

Reminder: Assignment submission deadline

- When? 8pm, **12th May, 2024**
- Submit both group report and individual reflective part
- One copy of group report in Word format to be submitted by a team member
- Individual reflective part must not be shared among team members
- Include evidences of team communications, if any, in the reflective part
- Check for Turnitin score and other Academic Integrity breaches

Learning objectives and resources

Objectives

- Understand the international parity conditions (IPC) and 'the law of one price (LOP)'
- Understand the difference between *absolute* and *relative* purchasing power parity (PPP)
- Examine the Fisher and International Fisher Effects
- Understand the theory of Interest Rate Parity
- Explore the process of interest rate arbitrage and how to make riskless profits in Foreign Exchange markets
- Covered Interest rate Arbitrage (CIA)

Study resources

❖ Shapiro Chapter 4

❖ Eiteman et al Chapter 6

❖ Butler Chapter 5

International Parity Conditions (IPC): Definition

- The economic theories that link exchange rates (ER), price levels, and interest rates together are called *international parity conditions.*
- These parity conditions form the core of the financial theory that is unique to international finance.
- The IPC sets equilibrium relationships
  - Inflation – price levels and ERs
  - Interest rates and ERs
  - Spot rates and forward rates

International Parity Conditions (IPC)

- Parity theories help to observe in the real world, but they are central to understanding of how multinational business is conducted and funded in the world today

- The mistake is often not with the theory itself, but with the *interpretation* and *application* of the theory

Law of One Price (LOP): Definition

If the identical product or service can be sold in two different markets; and no restrictions exist on the sale; and transportation costs of moving the product between markets are equal, then the prices of identical goods traded in different markets should be the *same* in both markets.

LOP and arbitrage

- If 2 identical goods are priced differently in two countries, then arbitrage will take place, How?
  - Everyone will buy the cheaper one, which leads to price increase
  - No one will buy the expensive one, which leads to price decrease
  - Eventually prices will converge, and arbitrage opportunity disappears
- What is arbitrage?

– Simultaneous purchase and sale of an asset or commodity in different markets to profit from price discrepancies.

– Arbitrage is attractive because a profit can be made with no risk and no net investment!

LOP and arbitrage

- Suppose the price of wheat is US$ 4 per bushel in the U.S and A$ 6.4 in Australia. According to the Law of One Price, Spot exchange rate (US$/A$1) must be

$$US\$4 / A\$6.4 = US\$0.625 / A\$$$

- However, if the spot rate is US$ 0.65/A$ in the FX market, then the US$ price of wheat per bushel in Australia is

A$ 6.4* (US$0.65/A$) = US$4.16 (> US$4)

- This will lead to *arbitrage*, because according to LOP the price of wheat per bushel is cheaper in the US

LOP and arbitrage

- Buy wheat in the U.S. for US$4/bushel
- Ship to Australia and sell for US$4.16
- Arbitrage profit will be made if shipping cost is less than US$0.16

LOP and Arbitrage

Five parity conditions result from international arbitrage activities:

1. Purchasing Power Parity (PPP)

   (a) Absolute PPP (b) Relative PPP
2. The Fisher Effect (FE)
3. International Fisher Effect (IFE)
4. Interest Rate Parity (IRP)
5. Unbiased Forward Rate (UFR)

Purchasing Power Parity (PPP)

(a) Absolute PPP (b) Relative PPP

Purchasing Power Parity (PPP)

- PPP deals with the relationship between the prices of goods and services, and the exchange rates

- PPP says that the exchange rates would adjust to offset the difference in inflation rates between countries.

Absolute PPP

- According to Absolute PPP
  - Exchange rates would adjust to offset the difference in price levels between countries
  - One unit of home currency has the same purchasing power globally
  - Law of One Price holds for every good and service
- If we put together two identical baskets of goods and services, the law of one price should also hold for the baskets;

i.e.,

$$P_{US} = S(US\$ / A\$).P_{A\$}$$

where

- ***P*** represents the total price of the basket of goods and services.
- ***S*** is the spot price

Absolute PPP

- Rearranging, we get

$$S(US\$/A\$) = \frac{P_{US}}{P_{Aust.}}$$

- The spot rate *S(US$/AU$)* represents absolute PPP
- The spot rate *S(US$/AU$)* changes to adjust the pricing differences (hence inflation rates) between the two countries

Absolute PPP:
The Big Mac standard

Big Mac Index

- It serves as one form for comparison of whether currencies are currently trading at market rates that are close to the exchange rate implied by Big Macs in local currencies.
- The index works by dividing the local price of a Big Mac by the US price of a Big Mac. This gives the PPP exchange rate.

The Big Mac standard: Is the Yuan under/overvalued ??

- The Yuan exchange rate on 8th July, 2019 was Yuan 7.58/US$ in the FE markets. (Actual ER)
- On the same day, a Big Mac costs 11.0 Yuan in China & $3.41 in the USA.
- The rate needed to equalise the burger's price in two countries is
  - PPP ER = Yu 11/$3.41 = Yu 3.23/$
- Hence – according to the Big Mac Standard – the Yuan is drastically undervalued (7.58/US$ vs 3.23/$)!
  - Undervaluation in percentage = [(Yu 3.23/$ - Yu7.58/$)/Yu7.58/$]= -0.57 or approx. 57%

The Big Mac standard: Flaws and success

- Flaws in the Big Mac Index
  - Big Macs are not exchanged across country borders
  - Included in the cost of a Big Mac are real estate, local taxes, and local services
  - These differ worldwide and are non-traded services
- Success with the Big Mac Index
  - 1999 – signaled correctly that the Euro was overvalued at its launch
  - 2002 – signaled that U.S. $ was overvalued, soon after US$ plummeted in value

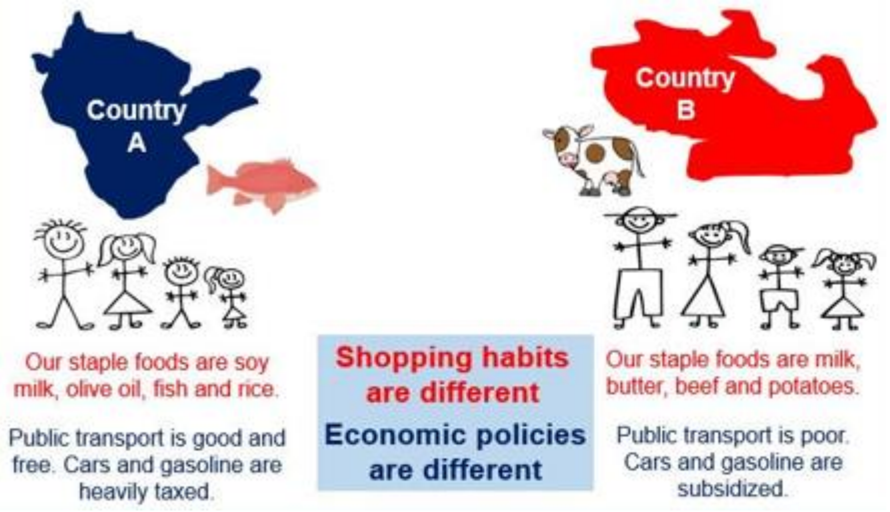
The CommSec iPod index:

- Designed as a modern-day variant of the long-running Big Mac index
    - Compare the iPod prices across boarders
    - The difference with the Big Mac index is that iPods can be exchanged across the globe
    - If they are particularly cheap in one country, it could cause tourists to buy their iPods when on holiday, prompt people to buy iPods online in other countries or encourage some companies to take advantage of arbitrage opportunities
- CommSec iPod index: http://www.adviservoice.com.au/wp-content/uploads/2012/08/CommSec-iPad-index.pdf

The CommSec iPod index: Is the Yuan under/overvalued?

- The Apple iPod nano MP3 players are manufactured in China and sold across the globe.
- The retail price of the MP3 player in China was 1198 yuan in March 2010, or just over USD 175 using the then prevailing market EX rate of Yuan 6.8365/US$.
- The same device can be purchased for USD 149 in the U.S.!
- The EX rate would actually need to weaken to Yu 8.04/US$ (instead of Yu 6.8365/US$) for iPods to be equivalently priced in China and the U.S.
- Hence, according to the iPod index, the Yuan is overvalued!

Absolute PPP



- Weakness
  - difficult to test because:
    - baskets may vary between countries
    - not all goods are traded internationally
    - thus, prices may not conform to PPP, even if the law of one price holds.
  - ignores differences in trade restrictions e.g., Tariffs, quotas, transportation costs, transaction costs, etc.
- Value/strength
  - Used by central banks as a guide to establish new par values for their currencies
  - Long term forecasting of future ER

RELATIVE PPP

- *Relative purchasing power parity* (RPPP) relaxed the assumptions of the absolute version of the PPP theory
- According to the RPPP, the relative change in prices between two countries over a period of time determines the change in the exchange rate over that period
- The spot exchange rate of one currency against another will adjust to reflect changes in the price levels of the two countries
  – if inflation in the home country exceeds inflation in the foreign country, the home currency(A$) will depreciate relative to foreign currency(U$)
  – currency with high inflation rates should depreciate relative to currency with low inflation rates
  – if inflation rates of both home country and foreign country are the same, there should be no change in exchange rate

RELATIVE PPP…cont.

In mathematical terms :

Direct quote

$$\frac{e_t}{e_0} = \frac{(1+i_h)^t}{(1+i_f)^t}$$

Indirect quote

$$\frac{e_t}{e_0} = \frac{(1+i_f)^t}{(1+i_h)^t}$$

where e_t = future spot rate

e_0 = current spot rate

i_h = home inflation

i_f = foreign inflation

t = the time period

RELATIVE PPP…cont.

If purchasing power parity is expected to hold, the *best prediction* for the future spot rate at t should be:

Direct quote

$$e_t = e_0 \frac{(1+i_h)^t}{(1+i_f)^t}$$

Indirect quote

$$e_t = e_0 \frac{(1+i_f)^t}{(1+i_h)^t}$$

RELATIVE PPP…cont.

- Assume annual inflation rates in the US & Australia are 6% and 4% respectively, and the spot rate is US$0.75 per A$. Assume US as home country.
- What is the value of A$ in 1,2,3 years?
- Spot rate at time t implied by PPP

$$e_t = e_0 \left(\frac{1 + i_{US\$}}{1 + i_{A\$}}\right)^t$$

$$e_1 = 0.75\left(\frac{1+0.06}{1+0.04}\right) = US\$0.7644$$

$$e_2 = 0.75\left(\frac{1+0.06}{1+0.04}\right)^2 = US\$0.7791$$

- Inflation rate in the US is greater than Australia, therefore US$ will depreciate in relation to A$
- Evidence of PPP

RELATIVE PPP…cont.

- If PPP holds, domestic competitiveness is unaffected by nominal changes in exchange rates.
- Why?
- If inflation is higher (lower) in Australia than foreign country, then
  - Higher (lower) A$ prices for our goods
  - A$ will depreciate (appreciate) against foreign currency (LOP)
  - Price of our goods in foreign currency remains the same

The Fisher Effect

The Fisher Effect: Interest Rates and Exchange Rates

- The Fisher Effect states that nominal interest rate is equal to the required real rate of return plus compensation for expected inflation.

$$i.e\ \ (1+r)=(1+a)(1+i)$$

- The approximate form of the above equation/theory is:

$$r = a + i$$

where: r = nominal interest rate; a = real interest rate; i = expected inflation

- **Example**:

If the required real return is 5% & expected inflation rate is 10%, then

r = (1.05)(1.1) -1 = 15.5%

or using approximation

r = 5 + 10 = 15% $[r = a + i]$

Generalized Fisher Effect

- Real interest rates should stay relatively constant in the short-run.
- With no government interference, nominal rates vary by inflation differential

$$r_h - r_f = i_h - i_f$$

where

r_h = nominal interest rate (i/r) in home country

r_f = nominal interest rate (i/r) in foreign country

i_h = inflation rate in home country

i_f = inflation rate in foreign country

- According to the generalized Fisher effect, countries with higher inflation rates have higher nominal interest rates.

The International Fisher Effect (IFE)

- IFE takes into account the implications from PPP and the generalized Fisher effect
  a) ERs will move to offset changes in inflation rate differentials (higher inflation is offset by depreciation of ER) – *PPP*
  b) Rise in inflation relative to other countries will be associated with a rise in domestic interest rate (i/r) relative to foreign interest rate (i/r) – *Fisher effect*
- Combining these 2 conditions (of PPP and FE) results in the IFE:

Currencies with high interest rates are *expected* to depreciate relative to currencies with low interest rates

The International Fisher Effect (IFE)

- The *expected* spot rate adjusts the interest rate differential between two countries:
- Direct $\frac{\bar{e}_t}{e_0} = \frac{(1+r_h)^t}{(1+r_f)^t}$ Indirect $\frac{\bar{e}_t}{e_0} = \frac{(1+r_f)^t}{(1+r_h)^t}$
- Where: e_0 ($\bar{e}_t$) is the spot rate (expected future spot at time t)
- The IFE implies that financial market arbitrage should ensure interest rate differential to be an unbiased predictor of future change in spot rate.

Implications:

- If the nominal interest rate is higher in the foreign country relative to the domestic country, the domestic currency is expected to appreciate.
- Currencies with the lower interest rate is expected to appreciate relative to currency with higher interest rate
- In other words, the IFE implies that financial market arbitrage should ensure interest rate differential to be an **unbiased predictor of future change in spot rate**.

Interest Rate Parity (IRP), Forward Rate and Covered interest arbitrage (CIA)

Interest Rate Parity (IRP)

- IRP links the FX markets and the international money markets
- The interest rate (i/r) differential should be approximately equal to the forward premium or discount
  - e.g., if i/r in Australia is 3% higher than i/r in the U.S., the AUD should be selling at (approximately) a 3% forward discount against the USD

$$\boxed{\underbrace{\frac{f_1 - e_0}{e_0}}_{\text{Forward premium or discount}} \approx \underbrace{r_h - r_f}_{\text{Difference in nominal interest rates}}}$$

r_h = nominal i/r in home country
r_f = nominal i/r in foreign country

- *High interest rates* on a currency are offset by forward discounts
- *Low interest* rates on a currency are offset by forward premiums

- When the above condition is met, the spot and forward rates are said to be at IRP, and the covered i/r differential equals to zero (no arbitrage takes place)

Interest Rate Parity

- IRP holds when the covered interest differential is zero and there are no CIA (covered interest arbitrage) opportunities

Interest rate parity conditions	Direct quote formula	Indirect quote formula
Interest rate parity (IRP), i.e., No-arbitrage condition, holds if	$\frac{f_1}{e_0} = \frac{1+r_h}{1+r_f}$	$\frac{f_1}{e_0} = \frac{1+r_f}{1+r_h}$
Home return equals foreign return, when	$1 + r_h = \frac{(1+r_f)f_1}{e_0}$	$1 + r_h = \frac{e_0(1+r_f)}{f_1}$
To preclude arbitrage, forward rate should be	$f_1 = e_0 \frac{1+r_h}{1+r_f}$	$f_1 = e_0 \frac{1+r_f}{1+r_h}$

Comparison of parity conditions

Theorems	Direct quote formula	Indirect quote formula
Purchasing power parity (slides 21-23)	$e_t = e_0 \frac{(1+i_h)^t}{\left(1+i_f\right)^t}$	$e_t = e_0 \frac{\left(1+i_f\right)^t}{(1+i_h)^t}$
International Fisher Effect (slide 30)	$\frac{\bar{e}_t}{e_0} = \frac{(1+r_h)^t}{(1+r_f)^t}$ Re-arranging gives $\bar{e}_t = e_0 \frac{(1+r_h)^t}{(1+r_f)^t}$	$\frac{\bar{e}_t}{e_0} = \frac{(1+r_h)^t}{(1+r_f)^t}$ Re-arranging gives $\bar{e}_t = e_0 \frac{(1+r_f)^t}{(1+r_h)^t}$
Interest rate parity (slides 32-33)	$f_1 = e_0 \frac{1+r_h}{1+r_f}$	$f_1 = e_0 \frac{1+r_f}{1+r_h}$

Covered interest differential

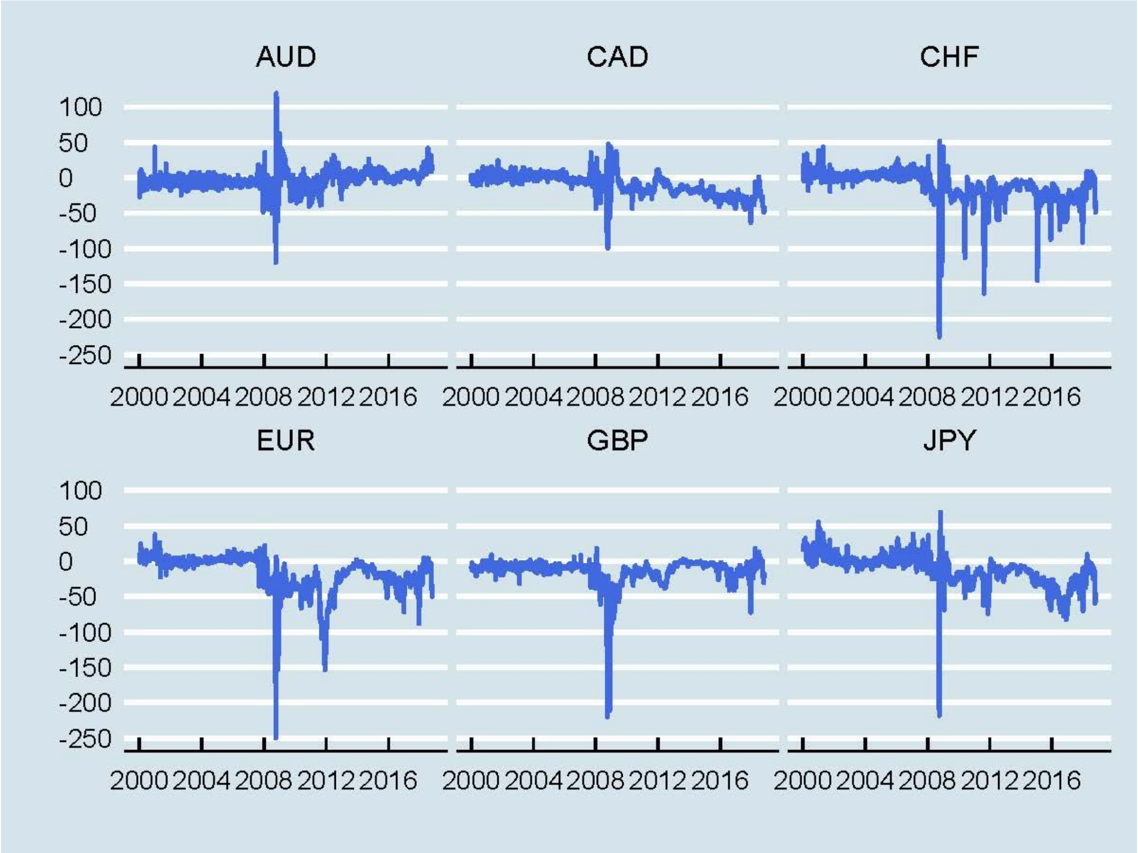
Covered interest arbitrage (CIA)

- The spot and forward exchange markets are not constantly in the state of equilibrium described by interest rate parity.
- The potential for “risk-less” or arbitrage profit exists when the market is not in equilibrium.
- The arbitrager will exploit the imbalance by investing in whichever currency offers the higher return on a covered basis.
- Covered interest arbitrage (CIA) opportunities continue to exist until interest rate parity is re-established

Covered interest differential

Covered interest arbitrage (CIA): empirical evidence

- Deviations from covered interest rate parity are very small prior to the global financial crisis (2008-2010).
- During the crisis there are large deviations as arbitrageurs face a lack of funding.
- But why do large deviations exist after the crisis? Research points to new regulation preventing arbitrage from taking place especially at the end of the quarter when banks need to disclose their capital requirements.



Source: Shapiro and Hanouna: 11th Edition

Covered interest differential

- The difference between domestic i/r and the hedged foreign i/r
- A comparison of investing at home and abroad:

  r_h = nominal i/r in home country

  r_f = nominal i/r in foreign country

  e_0 = current spot rate (direct quote)

  f_1 = 1-period forward rate (direct quote)

- Investing at home yields: $1+r_h$
- Investing abroad yields: $[(1+r_f)/e_0]*f_1$
- Investors are indifferent (no arbitrage) from investing at home or abroad when:

Direct quote formula	Indirect quote formula
$1+r_h=\frac{(1+r_f)f_1}{e_0}$	$1+r_h=\frac{e_0(1+r_f)}{f_1}$

Covered Interest Arbitrage (CIA)

- Using direct quote formula (you can do the same for indirect quote formula) CIA opportunities are available if covered interest differential is nonzero or:

$$1 + r_h \neq \frac{(1 + r_f)f_1}{e_0}$$

- Funds will flow from home to abroad if: $1 + r_h < \frac{(1 + r_f)f_1}{e_0}$

- Funds will flow from abroad to home if: $1 + r_h > \frac{(1 + r_f)f_1}{e_0}$

CIA- an example

Suppose:

interest rate in the US, r_h is 7% and that in the UK, r_f is 3%

spot price, e_0 = \$1.20/£ (direct quote) and one-month forward price, f_1 = \$1.25/£ (direct quote)

Do you see any CIA opportunity in this example?

Return from investing in the U.S. (home): $$1 + r_h = 1 + \frac{0.07}{12} = 1.0058$$

Return from investing in the UK: $$\frac{(1 + r_f)f_1}{e_0} = \frac{1 + \frac{0.03}{12} * 1.25}{1.20} = 1.0442$$

Funds move from the U.S. to the UK since $$1 + r_h < \frac{(1 + r_f)f_1}{e_0}$$

CIA- an example

Steps to implement

1. Borrow in the U.S. (use US$1 million) at the domestic interest rate r_h
2. Convert the borrowed fund into £ at the spot exchange rate e_0
3. Invest £ in the UK to earn the foreign interest rate r_f
4. Simultaneously sell the future value of the £ investment in the forward market
5. At the end of investment period, convert payoff from £ investment at the forward rate f_1
6. Pay back the USD loan and take profit

(next slide provides numerical explanation)

CIA- 1-month forward rate

At time 0	
1. Borrow \$1,000,000 at 7%	Need to repay \$1m*(1+0.07/12)= \$1,005,833.33 at time t (after one month)
2. Convert \$ to £ at e_0 = \$1.20/£	\$1m/1.20=£833,333
3. Invest £ at 3%	Will receive £833,333 *(1+0.03/12)= £835,416.33 at time t (after one month)
4. Lock in forward rate \$1.25/£ to sell £835,416.33 at time t (after one month)	
At time t (after one month)	
5. Convert £ to \$ at f_1 =\$1.25/£	£835,416.33 x \$1.25 =\$1,044,270.41
6. Repay Loan and take profit:	\$1,044,270.41-\$1,005,833.33 = \$38,437.08 (profit)

IRP is enforced through CIA

- In our example, demand for £ and £ denominated money-market instruments is high, market pressures develop as follows:
  - £ is bought spot and sold forward ($e_0 \uparrow$ and $f_1 \downarrow$).
  - Inflow of fund depresses i/r (r_f) in the UK: Implication of carry trade?
  - Outflow of fund boosts i/r (r_h) in the U.S.: Implication of carry trade?

$$1 + r_h \uparrow < \frac{(1 + r_f \downarrow) f_1 \downarrow}{e_0 \uparrow}$$

- Parity eventually restored

$$1 + r_h = \frac{(1 + r_f) f_1}{e_0}$$

IRP and Uncovered interest arbitrage

What happens if the investor remains uncovered/unhedged?

- Investors borrow in countries and currencies exhibiting relatively low interest rates and convert the proceed into currencies that offer much higher interest rates, but investment proceeds is not sold forward, therefore ‘uncovered’
- Investors are willing to accept the currency risk of exchanging the high-yielding currency at the future spot exchange rate
- The strategy is profitable if exchange rates are stable
- Potential for large losses when the low-yielding currency appreciates significantly

Unbiased Forward Rate

Unbiased Forward Rate

- It is often argued that the **forward rate** "fully reflects" available information about the exchange rate expectations
- The forward rate, thus, is usually viewed as an unbiased predictor of the future spot rate.
- Consider international fisher effect (IFE) and IRP:

$$\boxed{E(e_1) = e_0 \times \frac{(1 + r_h)}{(1 + r_f)}}$$

$$\boxed{E(e_1) = e_0 \times \frac{(1 + r_h)}{(1 + r_f)}}$$

- Since the RHS of each equation is equal, forward rate is equal to the future spot rate: $\boxed{f_1 = E(e_1)}$

Unbiased Forward Rate

- Is the forward rate truly unbiased?
  - The evidence suggests not.
- The evidence suggests that there is a risk premium (RP):

$$\boxed{f_1 = E(e_1) + RP}$$

  - RP can be positive or negative.
- Overall, however, regressions of the spot rate, St, on the forward rate, f_1, routinely support unbiasedness with slope coefficient near 1
- Thus, one can use the forward rate available at time t as a proxy for the prediction of the spot rate at time t+1

Forward rate and interest rate parity

Example

The US interest rate is 2.25% (per annum); in Japan, the comparable rate is 0.1%. The spot rate for the yen is \$0.009142. If interest rate parity holds, what is the 90-day forward rate?

Answer

$$f_{90} = e_0 \frac{1 + r_h}{1 + r_f}$$

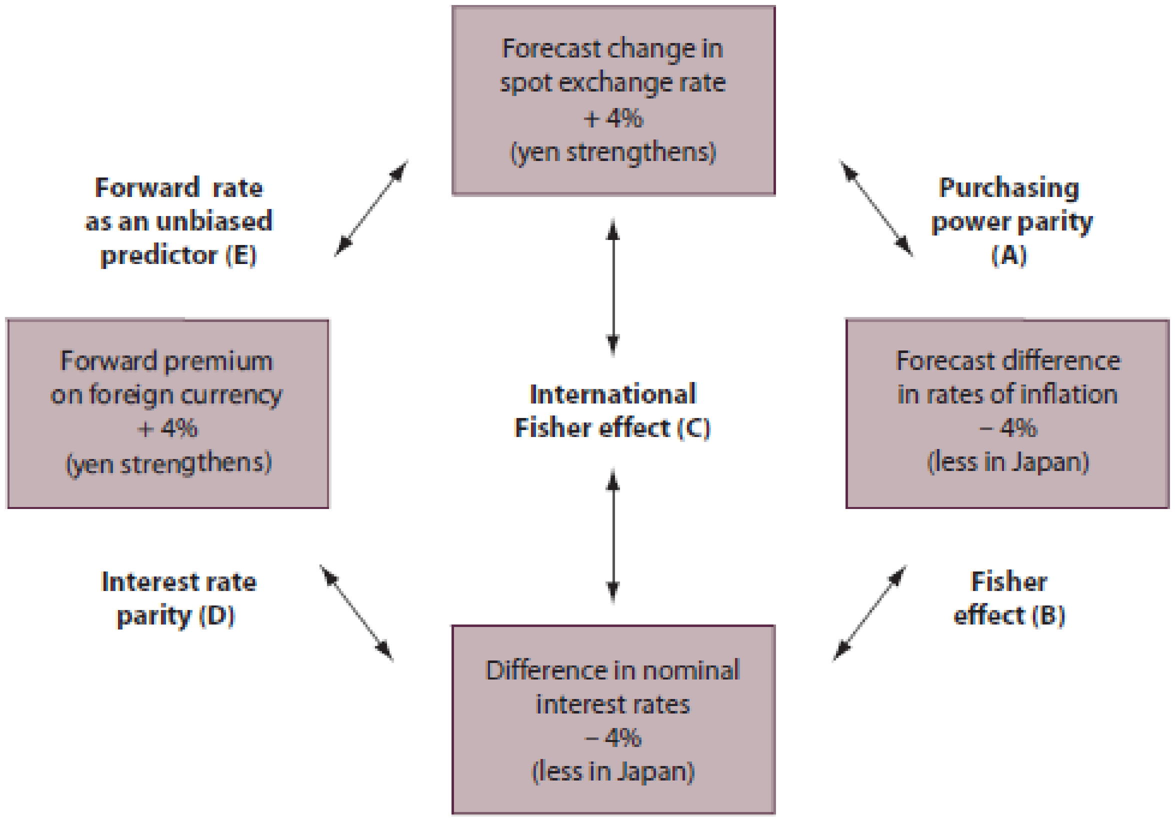
$$f_{90} = 0.009142 \times \frac{1 + \frac{0.0225}{4}}{1 + \frac{0.001}{4}}$$

$$= \$0.009191$$

Used direct quote formula assuming US\$ as the home currency.

Summary: I/N parity conditions

FIGURE 6.15 International parity conditions in equilibrium (approximate form)



Using US dollar and Yen, this figure illustrates the equilibrium under all parity relations. The forecasted inflation rates for Japan and the US are 1% and 5% respectively, a 4% differential. The nominal interest rate in the US dollar market for one-year govt bond is 8%, whereas, it is 4% in Japan, a difference of 4% over the Japanese nominal interest rate. The spot rate is ¥104/$ and the forward rate is ¥100/$. So the Japanese yen is selling at 4% forward premium, which is exactly same as the difference in nominal interest rate of 4%.

(Eiteman, page 189)